Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```python
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as 
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```python
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_wl_2h_D2_energy', 'r_NH3_wl_2h_D2_energy_rel', 'r_temperature_wl_2h_D2_energy', 'r_temperature_wl_2h_D2_energy_rel', 'r_umidity_wl_2h_D2_energy', 'r_umidity_wl_2h_D2_energy_rel']

1.2 Metadata

Nombre de lignes df	7793
Nombre de lignes df clean	7793
Nombre de features	47
% de NaN	0.0
Taille dataset entraînement	5455
Taille dataset test	2337
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

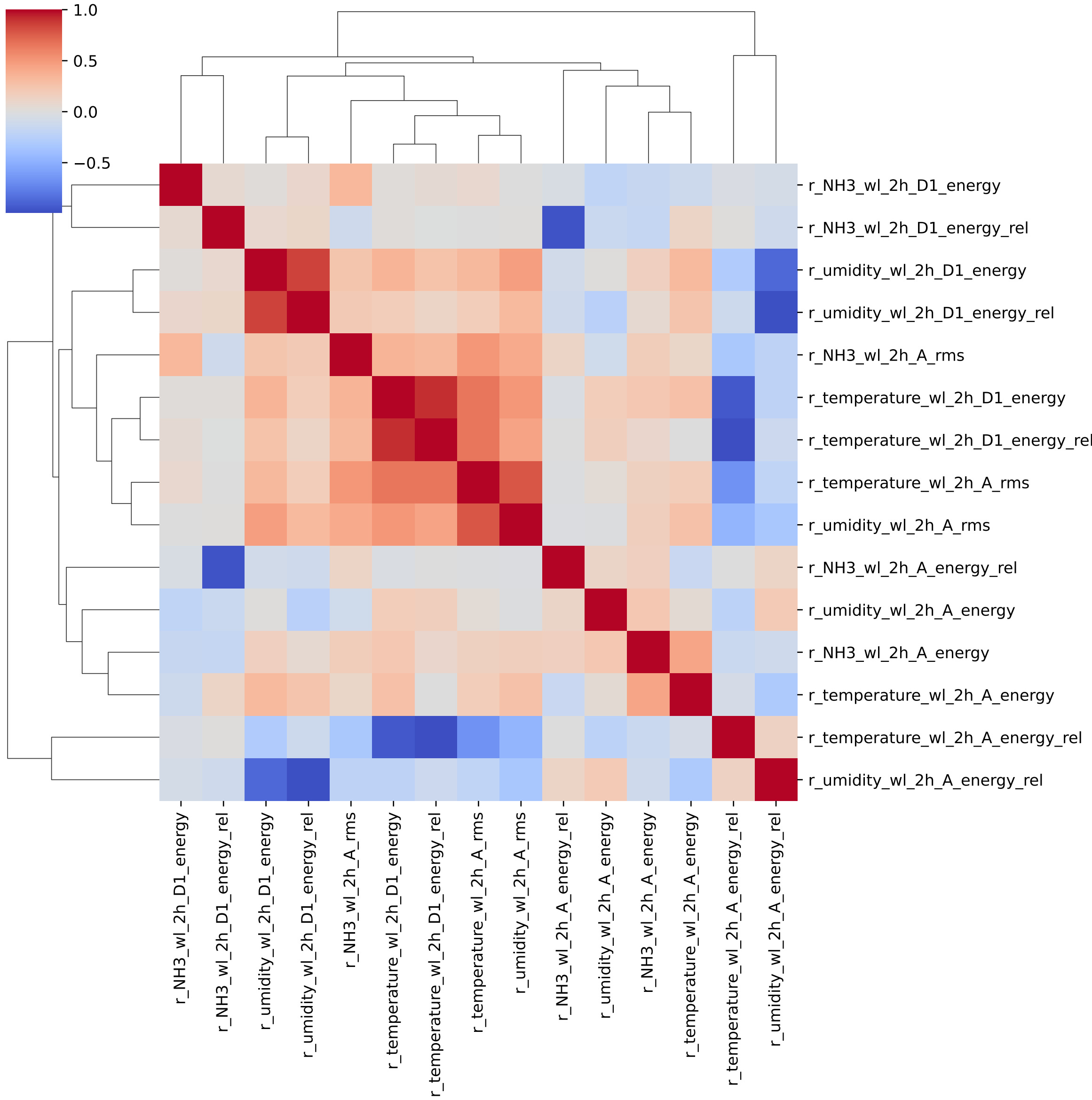
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_A_energy_rel, r_NH3_wl_2h_D1_energy_rel ; **cluster 10:** r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_D1_energy ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ; **cluster 16:** r_NH3_wl_8h_D1_energy ; **cluster 17:** r_NH3_wl_8h_D2_energy ; **cluster 18:** r_NH3_wl_8h_D3_energy ; **cluster 19:** r_NH3_wl_8h_A_energy ; **cluster 20:** r_NH3_wl_8h_A_rms ; **cluster 21:** r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_A_energy_rel ; **cluster 22:** r_NH3_wl_8h_D3_energy_rel ; **cluster 23:** r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D1_energy ; **cluster 24:** r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_D3_energy_rel ; **cluster 25:** r_temperature_wl_8h_A_energy, r_umidity_wl_8h_A_energy ; **cluster 26:** r_temperature_wl_8h_A_rms ; **cluster 27:** r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D1_energy_rel ; **cluster 28:** r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D2_energy ; **cluster 29:**

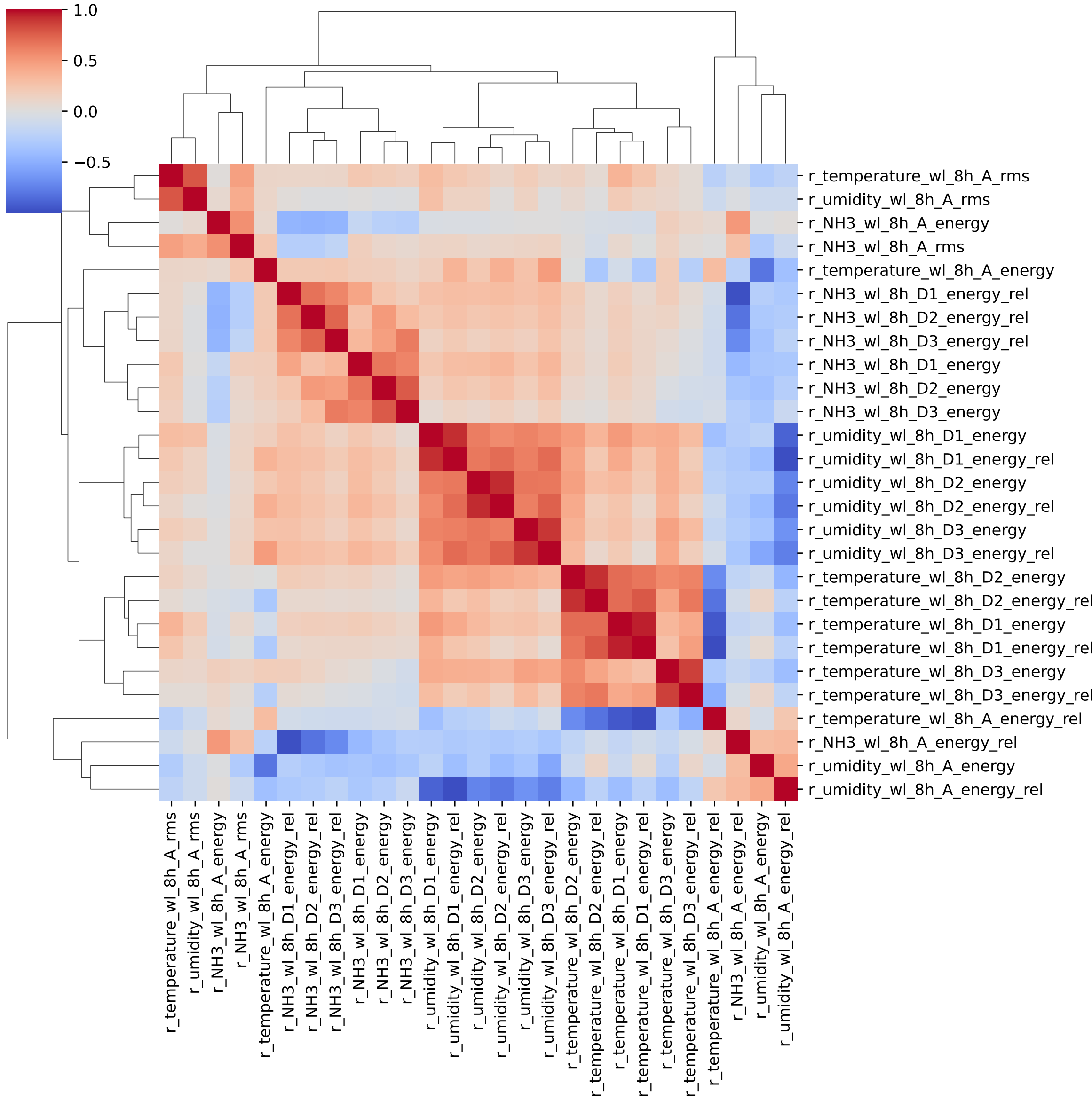
r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_D3_energy ; **cluster 30:** r_umidity_wl_8h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms


1.0
0.5
0.0
−0.5
r_NH3_wl_2h_D1_energy
r_NH3_wl_2h_D1_energy_rel
r_umidity_wl_2h_D1_energy
r_umidity_wl_2h_D1_energy_rel
r_NH3_wl_2h_A_rms
r_temperature_wl_2h_D1_energy
r_temperature_wl_2h_D1_energy_rel
r_temperature_wl_2h_A_rms
r_umidity_wl_2h_A_rms
r_NH3_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy
r_NH3_wl_2h_A_energy
r_temperature_wl_2h_A_energy
r_temperature_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy_rel



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms

Len(Features_X): 30

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

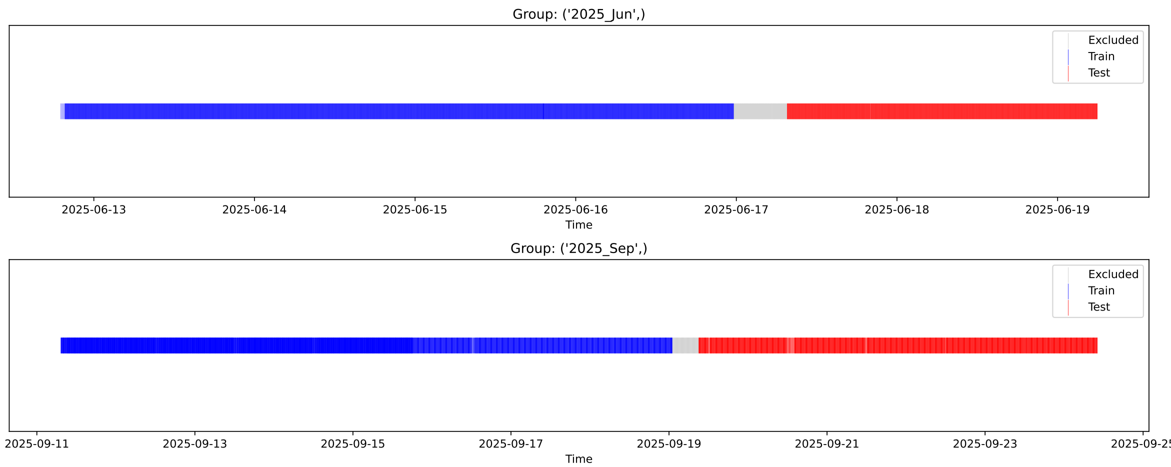
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

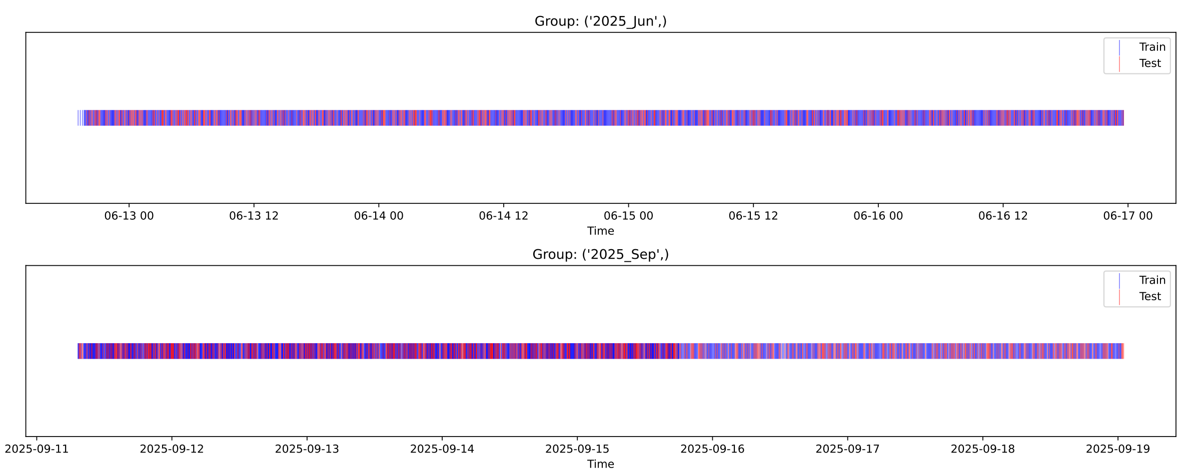
External Split: Train: 5204 samples Test: 2337 samples



3.3.2 Internal CV Folds

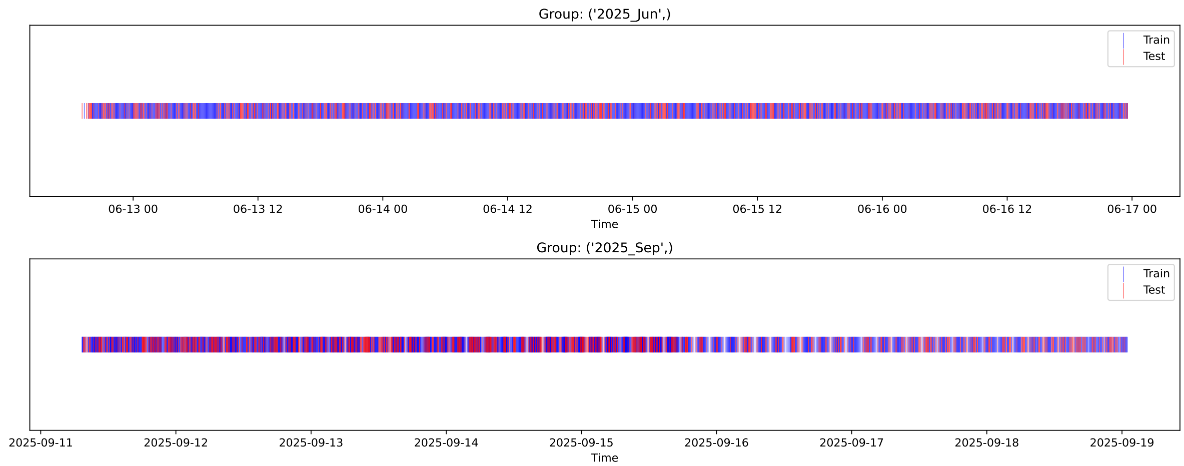
Fold 1/5

Train: 3642 samples Test: 1562 samples Excluded: 0 samples Window exclusion: 0h



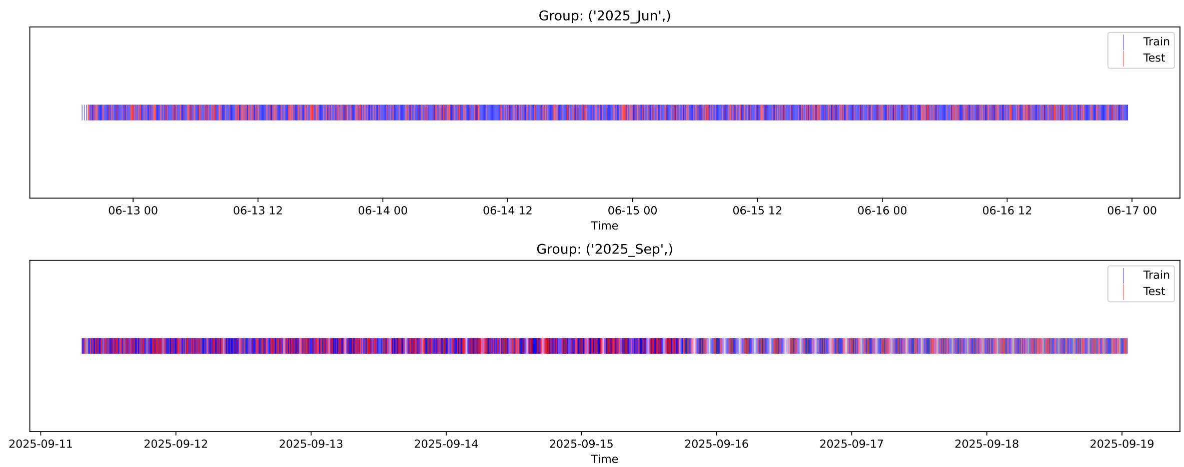
Fold 2/5

Train: 3642 samples Test: 1562 samples Excluded: 0 samples Window exclusion: 0h



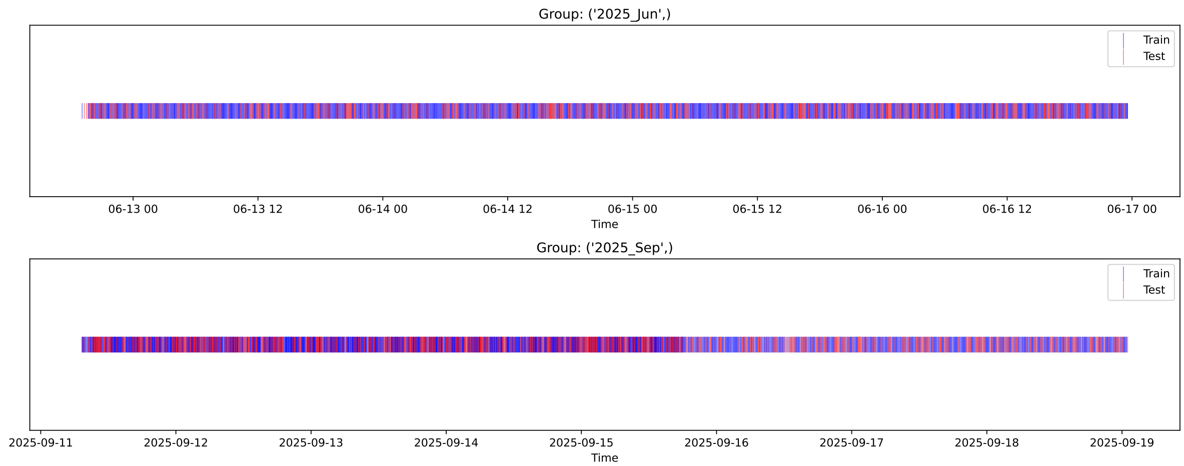
Fold 3/5

Train: 3642 samples Test: 1562 samples Excluded: 0 samples Window exclusion: 0h



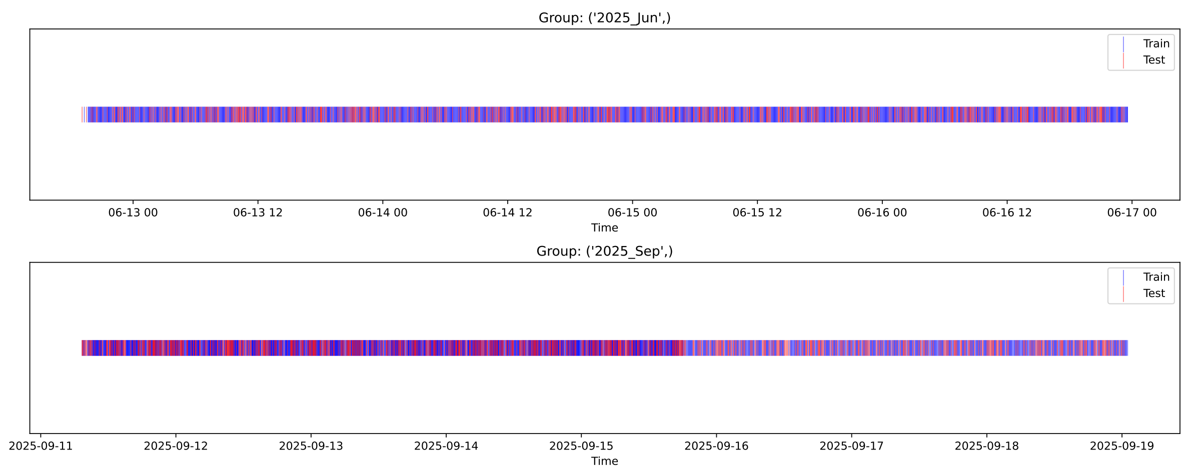
Fold 4/5

Train: 3642 samples Test: 1562 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3642 samples Test: 1562 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.367, std=0.823, min=0.959, max=8.682 y_test: mean=2.331, std=0.776, min=1.071, max=13.845

Fold 0 — y_test: mean=2.37, std=0.87, y_train: mean=2.37, std=0.80

Fold 1 — y_test: mean=2.39, std=0.85, y_train: mean=2.36, std=0.81

Fold 2 — y_test: mean=2.36, std=0.82, y_train: mean=2.37, std=0.82

Fold 3 — y_test: mean=2.37, std=0.83, y_train: mean=2.37, std=0.82

Fold 4 — y_test: mean=2.39, std=0.80, y_train: mean=2.36, std=0.83

train scores CV: [0.63311048 0.63361015 0.63420905 0.62736826 0.61131258]

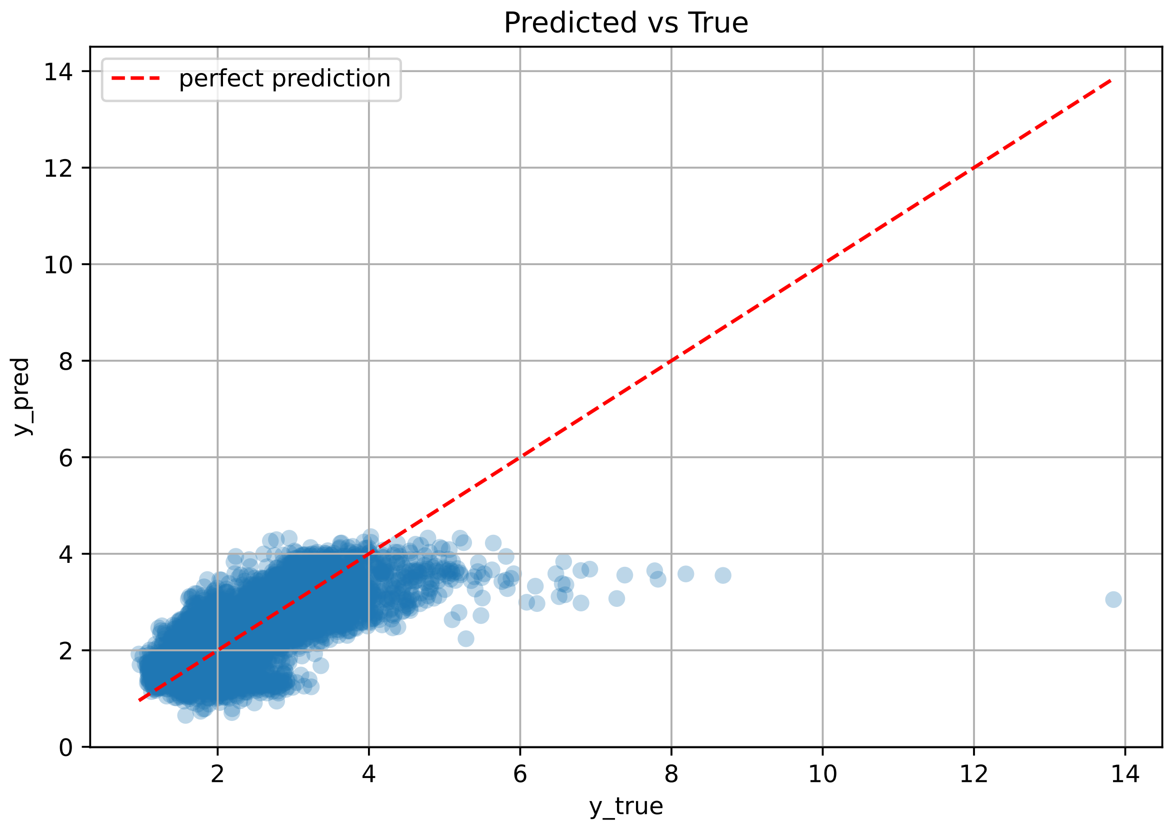
test scores CV: [0.59858774 0.59832024 0.59280284 0.60950758 0.6495509]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.6279	0.6098	0.0206	0.3128	0.3036	-0.3061	0.498	1.0298

Coefficients:

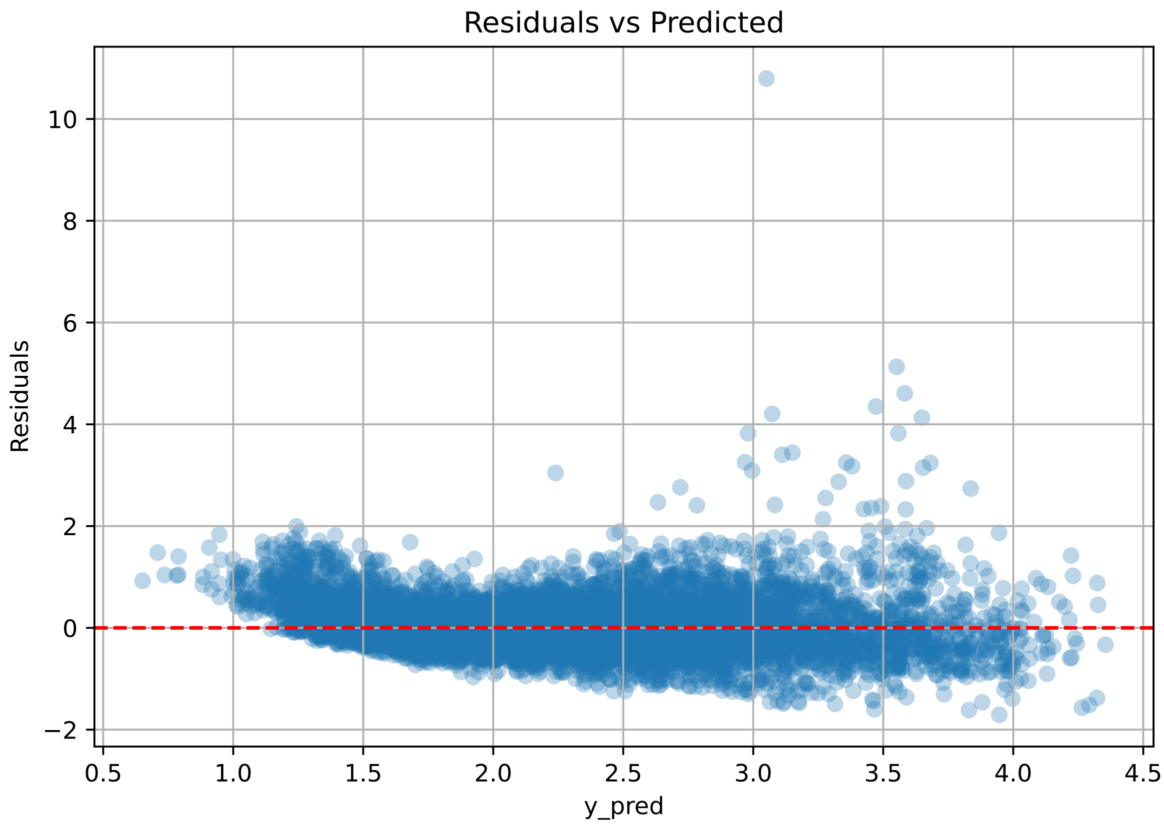
coef_r_CO2 = -0.0455, coef_r_NH3 = 0.0629, coef_r_NH3_wl_2h_A_energy = -0.1375, coef_r_NH3_wl_2h_A_energy_rel = 0.0348, coef_r_NH3_wl_2h_A_rms = -0.0112, coef_r_NH3_wl_2h_D1_energy = 0.0108, coef_r_NH3_wl_8h_A_energy = -0.0675, coef_r_NH3_wl_8h_A_rms = 0.0391, coef_r_NH3_wl_8h_D1_energy = 0.0375, coef_r_NH3_wl_8h_D2_energy = 0.1157, coef_r_NH3_wl_8h_D2_energy_rel = -0.0693, coef_r_NH3_wl_8h_D3_energy = -0.1321, coef_r_NH3_wl_8h_D3_energy_rel = 0.0059, coef_r_THI = 1.0805, coef_r_temperature = -1.1560, coef_r_temperature_wl_2h_A_energy = -0.0147, coef_r_temperature_wl_2h_A_rms = 0.1757, coef_r_temperature_wl_2h_D1_energy_rel = 0.0321, coef_r_temperature_wl_8h_A_energy = -0.1505, coef_r_temperature_wl_8h_A_energy_rel = 0.0080, coef_r_temperature_wl_8h_A_rms = 0.0777, coef_r_temperature_wl_8h_D3_energy = -0.0930, coef_r_umidity = -0.9055, coef_r_umidity_wl_2h_A_energy = 0.1842, coef_r_umidity_wl_2h_A_energy_rel = -0.0355, coef_r_umidity_wl_2h_A_rms = 0.0395, coef_r_umidity_wl_8h_A_energy_rel = 0.0366, coef_r_umidity_wl_8h_A_rms = -0.1817, coef_r_umidity_wl_8h_D2_energy_rel = -0.0236, coef_r_umidity_wl_8h_D3_energy_rel = 0.0614, intercept = 2.3671,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
0
2
4
6
8
10
12
14
2
4
6
8
10
12
14

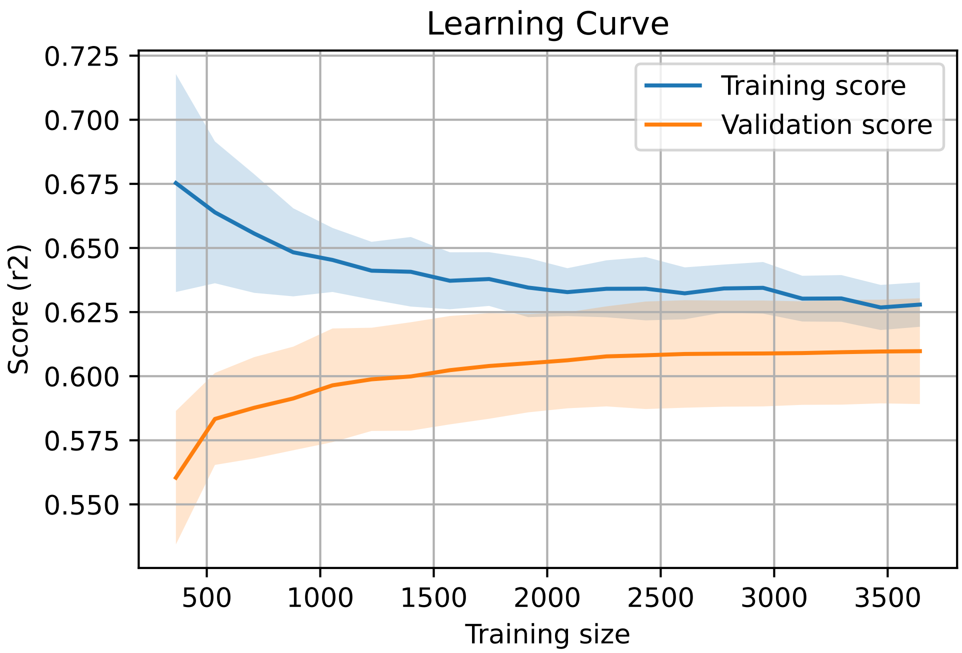
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337

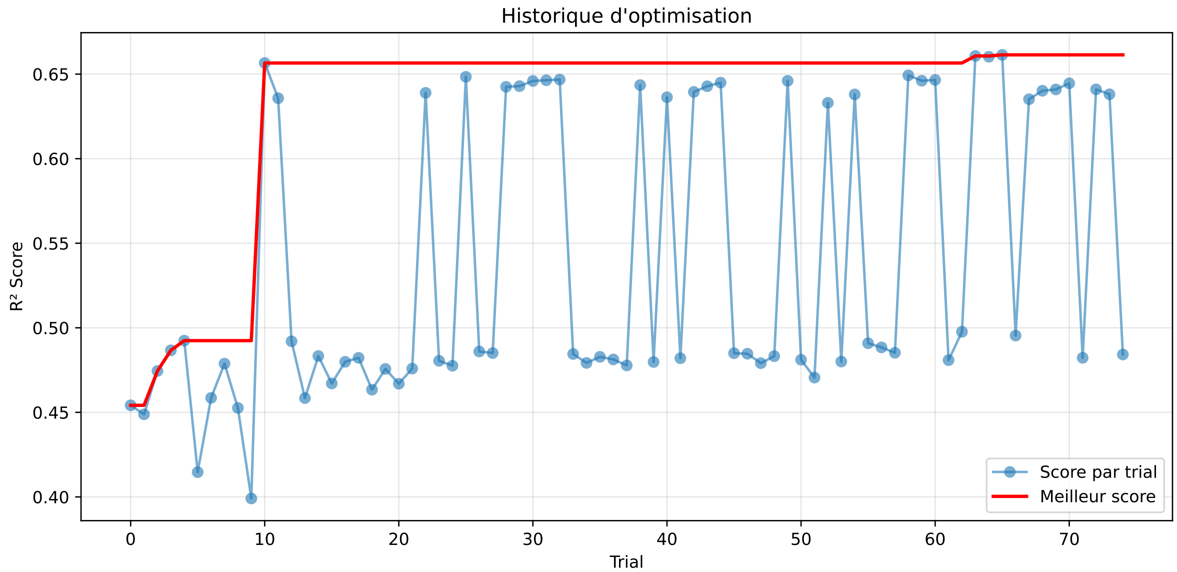


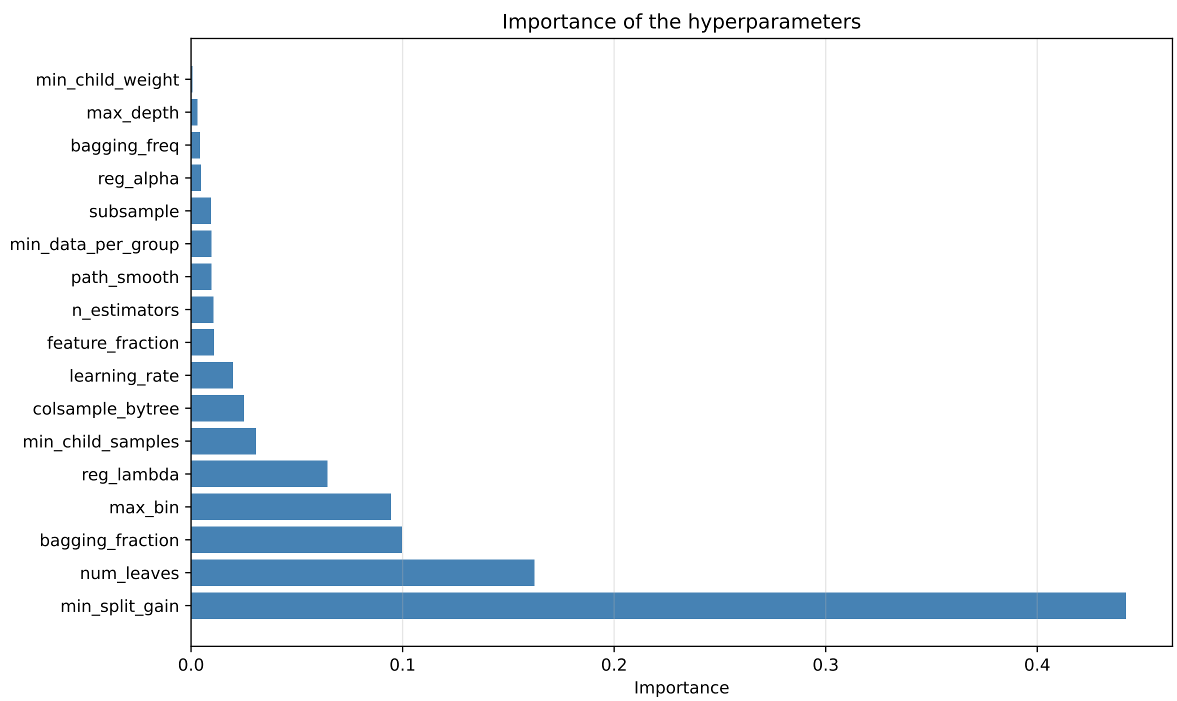
3.5 Model : LGBM

3.5.1 Gridsearch

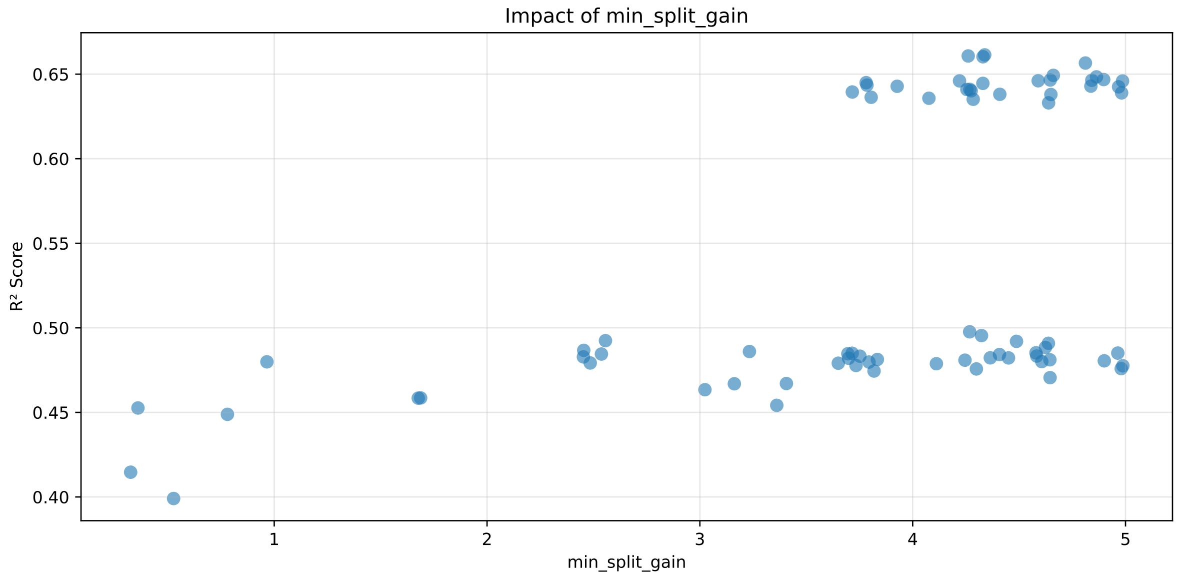
Distribution y_train vs y_test: y_train: mean=2.367, std=0.823, min=0.959, max=8.682 y_test: mean=2.331, std=0.776, min=1.071, max=13.845

==================== TOP Importance FEATURES ==================== feature importance r_umidity 111 r_CO2 107 r_NH3_wl_8h_A_energy 87 r_temperature_wl_2h_A_rms 86 r_temperature_wl_8h_A_energy 78 r_NH3_wl_8h_D1_energy 51 r_NH3 42 r_temperature 41 r_temperature_wl_2h_D1_energy_rel 39 r_NH3_wl_2h_D1_energy 33 r_temperature_wl_8h_A_energy_rel 33 r_umidity_wl_2h_A_energy 33 r_temperature_wl_2h_A_energy 32 r_umidity_wl_2h_A_energy_rel 25 r_temperature_wl_8h_D3_energy 22 r_THI 21 r_NH3_wl_2h_A_rms 19 r_NH3_wl_2h_A_energy 16 r_NH3_wl_8h_D2_energy 14 r_NH3_wl_8h_D3_energy_rel 13 r_NH3_wl_8h_D3_energy 11 r_umidity_wl_8h_A_rms 10 r_temperature_wl_8h_A_rms 9 r_umidity_wl_8h_D3_energy_rel 7 r_NH3_wl_2h_A_energy_rel 6 r_umidity_wl_2h_A_rms 6 r_NH3_wl_8h_A_rms 3 r_NH3_wl_8h_D2_energy_rel 3 r_umidity_wl_8h_A_energy_rel 3 r_umidity_wl_8h_D2_energy_rel 2

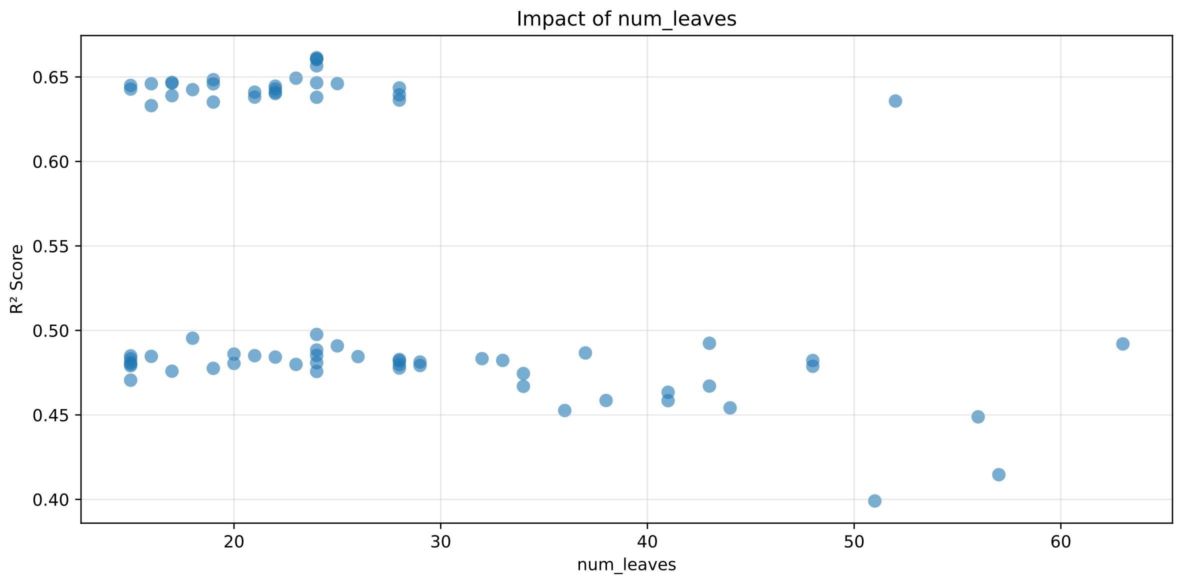




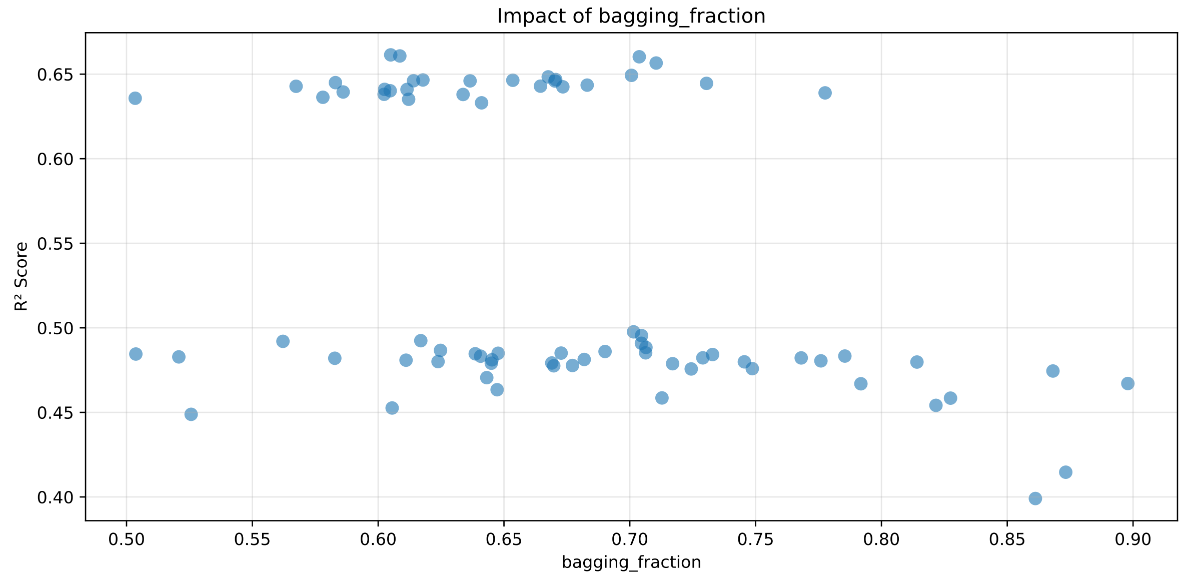
Paramètres avec >5% d'importance : ['min_split_gain', 'num_leaves', 'bagging_fraction', 'max_bin', 'reg_lambda']

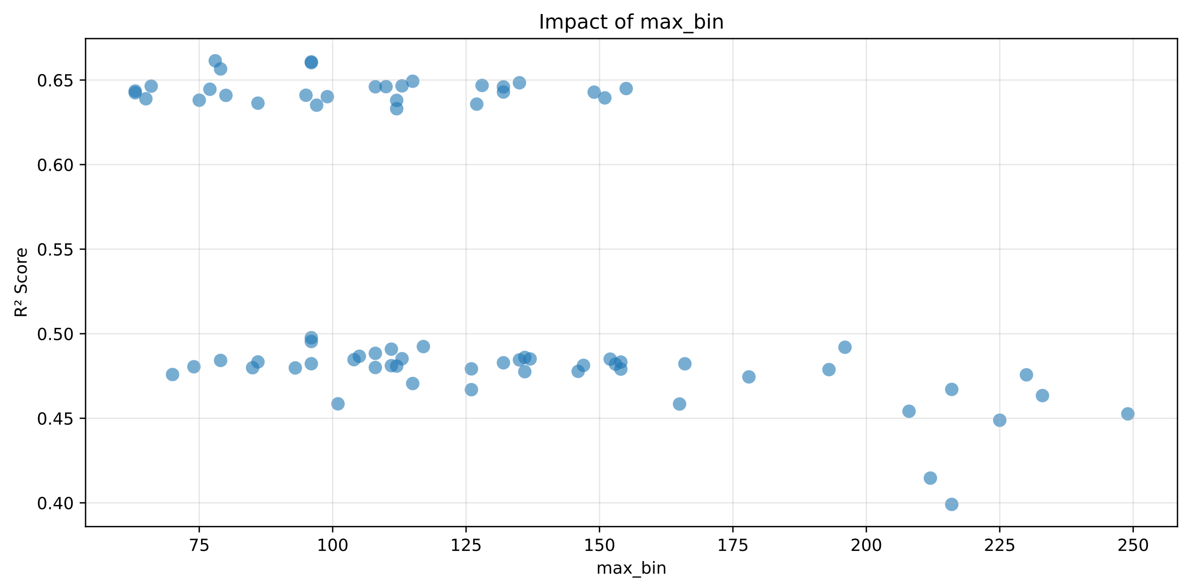


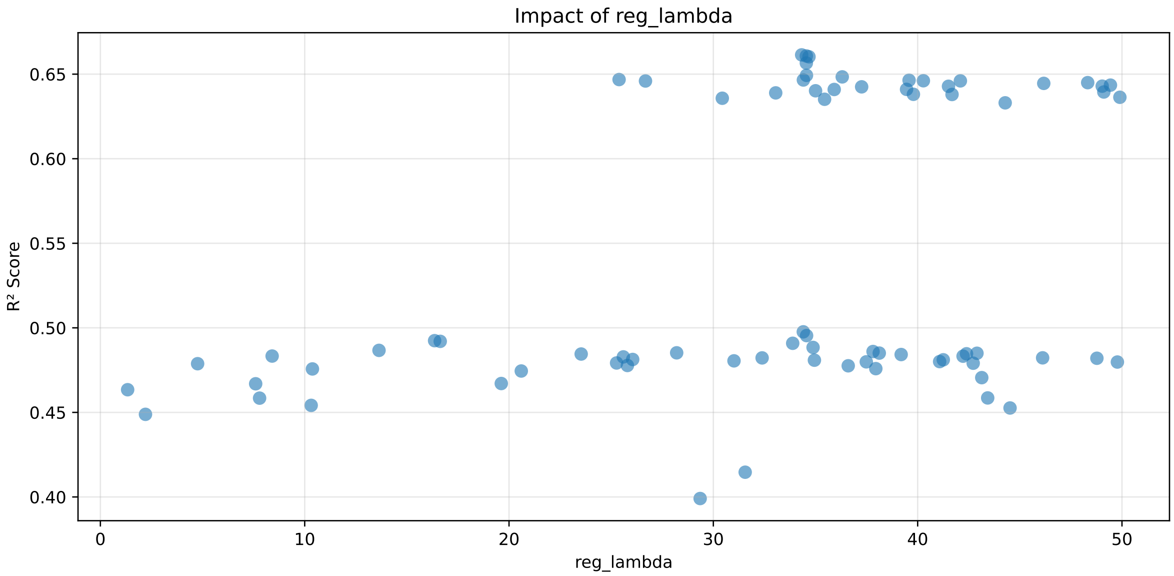
Impact of min_split_gain
R² Score
0.40
0.45
0.50
0.55
0.60
0.65
1
2
3
4
5
min_split_gain



Impact of num_leaves
R² Score
0.40
0.45
0.50
0.55
0.60
0.65
20
30
40
50
60
num_leaves


Impact of bagging_fraction
R² Score
bagging_fraction
0.65
0.60
0.55
0.50
0.45
0.40
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90


Impact of max_bin
R² Score
max_bin
0.65
0.60
0.55
0.50
0.45
0.40
75
100
125
150
175
200
225
250



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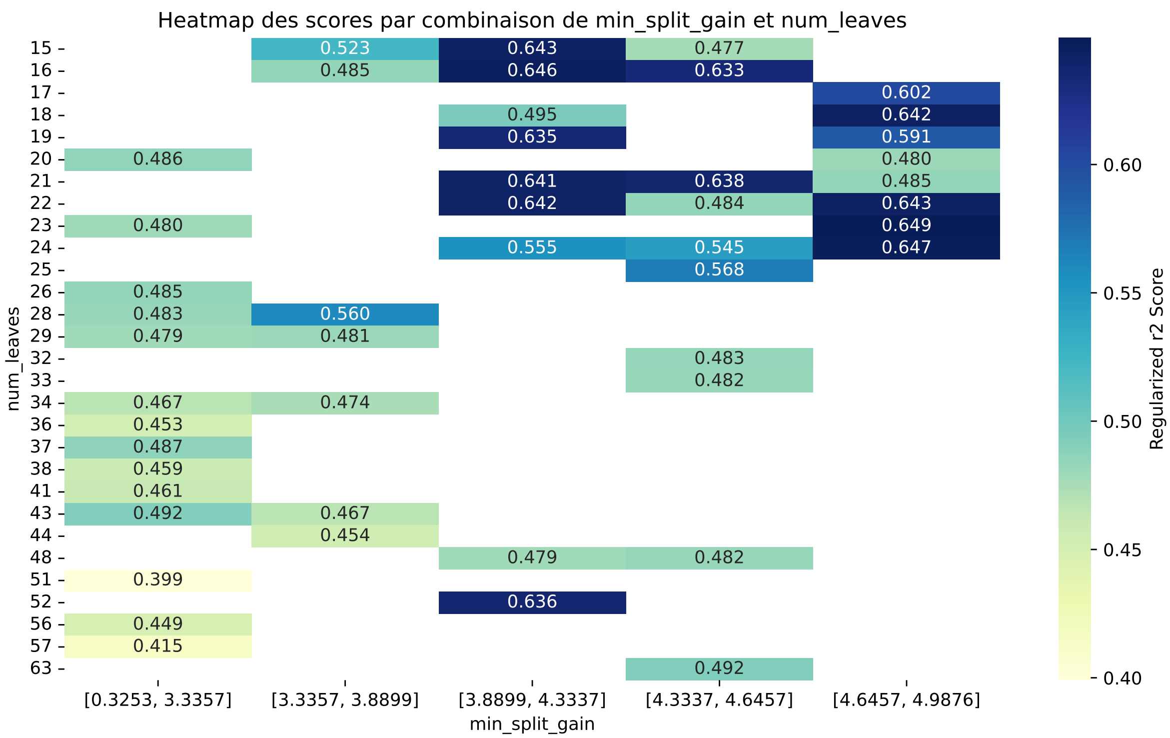
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6918	0.6614	0.5038	0.018	0.6614	0.5079	-0.1535	0.7679	1.046
2	0.6923	0.6608	0.5029	0.0183	0.6608	0.5118	-0.149	0.7745	1.0477
3	0.6924	0.6603	0.4993	0.0182	0.6603	0.4983	-0.162	0.7547	1.0487
4	0.6889	0.6566	0.4895	0.0164	0.6566	0.4944	-0.1622	0.753	1.0492
5	0.6817	0.6493	0.5036	0.0163	0.6493	0.5109	-0.1384	0.7869	1.0499
6	0.6807	0.6484	0.5032	0.0162	0.6484	0.5085	-0.1399	0.7842	1.0498
7	0.6781	0.6468	0.4995	0.0168	0.6468	0.5042	-0.1426	0.7795	1.0483
8	0.6782	0.6466	0.5019	0.0135	0.6466	0.5028	-0.1438	0.7777	1.0489
9	0.677	0.6464	0.5018	0.0177	0.6464	0.5113	-0.135	0.7911	1.0473
10	0.6763	0.6461	0.4998	0.0158	0.6461	0.5163	-0.1298	0.799	1.0468
---	---	---	---	---	---	---	---	---	---
75	0.8102	0.7417	0.5143	0.0201	0.3991	0.5126	-0.2291	0.6911	1.0924

Hyperparameters:

Rank 1: {'reg___n_estimators': 800, 'reg___max_depth': 6, 'reg___learning_rate': 0.029716569275548686, 'reg___num_leaves': 24, 'reg___subsample': 0.7136345814137156, 'reg___colsample_bytree': 0.8592355873620957, 'reg___min_child_samples': 20, 'reg___reg_alpha': 4.609835249290976, 'reg___reg_lambda': 34.32267605144712, 'reg___min_split_gain': 4.338460698824724, 'reg___path_smooth': 3.994865973642536, 'reg___min_child_weight': 0.06869994762226171, 'reg___feature_fraction': 0.6748283089073124, 'reg___bagging_fraction': 0.6049873879153187, 'reg___bagging_freq': 1, 'reg___max_bin': 78, 'reg___min_data_per_group': 199}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 6, 'reg___learning_rate': 0.060063774836364499, 'reg___num_leaves': 24, 'reg___subsample': 0.5997023123578079, 'reg___colsample_bytree': 0.8149302472243063, 'reg___min_child_samples': 22, 'reg___reg_alpha': 5.711479772543532, 'reg___reg_lambda': 34.55186293773799, 'reg___min_split_gain': 4.260858187675785, 'reg___path_smooth': 2.4346245932808506, 'reg___min_child_weight': 0.16322582425504095, 'reg___feature_fraction': 0.6727673890261682, 'reg___bagging_fraction': 0.6086080407792593, 'reg___bagging_freq': 1, 'reg___max_bin': 96, 'reg___min_data_per_group': 188}

Rank 3: {'reg___n_estimators': 800, 'reg___max_depth': 6, 'reg___learning_rate': 0.05903628886247723, 'reg___num_leaves': 24, 'reg___subsample': 0.7097566559393068, 'reg___colsample_bytree': 0.855921521332331, 'reg___min_child_samples': 22, 'reg___reg_alpha': 8.351957777619607, 'reg___reg_lambda': 34.68128155148215, 'reg___min_split_gain': 4.3304800366158895, 'reg___path_smooth': 2.368386201692428, 'reg___min_child_weight': 0.1615883050446997, 'reg___feature_fraction': 0.7954070341667907, 'reg___bagging_fraction': 0.7037641565562117, 'reg___bagging_freq': 1, 'reg___max_bin': 96, 'reg___min_data_per_group': 182}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.03252309030676203, 'reg__num_leaves': 24, 'reg__subsample': 0.6064227265049901, 'reg__colsample_bytree': 0.862440238923748, 'reg__min_child_samples': 21, 'reg__reg_alpha': 3.4161057952593965, 'reg__reg_lambda': 34.55171187928401, 'reg__min_split_gain': 4.811633004765362, 'reg__path_smooth': 3.6761111803132263, 'reg__min_child_weight': 0.029769940990377216, 'reg__feature_fraction': 0.89486331246355, 'reg__bagging_fraction': 0.7104792207955768, 'reg__bagging_freq': 1, 'reg__max_bin': 79, 'reg__min_data_per_group': 154}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.015861296099080754, 'reg__num_leaves': 23, 'reg__subsample': 0.6027396509004457, 'reg__colsample_bytree': 0.8521112346306351, 'reg__min_child_samples': 25, 'reg__reg_alpha': 9.000873704676643, 'reg__reg_lambda': 34.55787194023973, 'reg__min_split_gain': 4.660870904389053, 'reg__path_smooth': 2.7218555128621027, 'reg__min_child_weight': 0.07068159392917495, 'reg__feature_fraction': 0.6766539349043303, 'reg__bagging_fraction': 0.7006652118420023, 'reg__bagging_freq': 1, 'reg__max_bin': 115, 'reg__min_data_per_group': 82}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.02639550297153088, 'reg__num_leaves': 19, 'reg__subsample': 0.6283018823641956, 'reg__colsample_bytree': 0.7355501948925487, 'reg__min_child_samples': 35, 'reg__reg_alpha': 7.369028091727934, 'reg__reg_lambda': 36.30853568985603, 'reg__min_split_gain': 4.863498422828848, 'reg__path_smooth': 3.182374811150089, 'reg__min_child_weight': 0.0011566219969005394, 'reg__feature_fraction': 0.6957145815674507, 'reg__bagging_fraction': 0.6675877767564242, 'reg__bagging_freq': 1, 'reg__max_bin': 135, 'reg__min_data_per_group': 107}

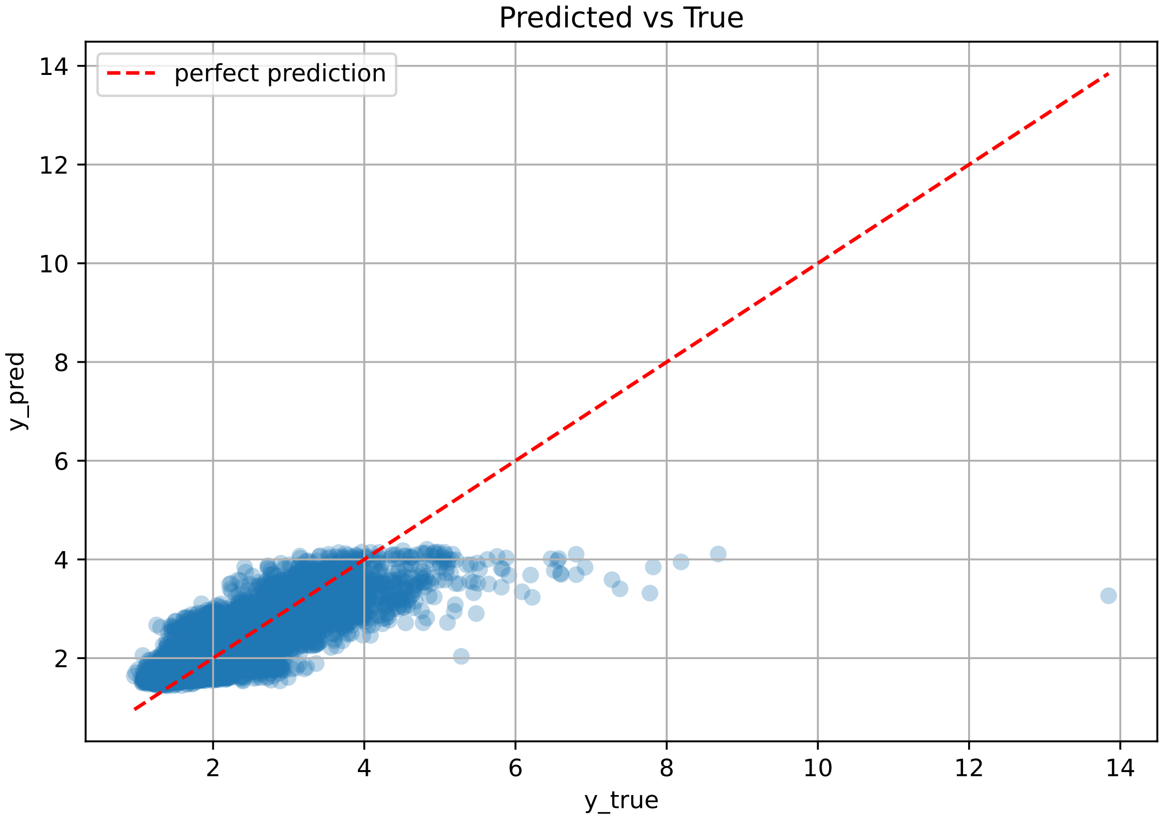
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.034291432989132474, 'reg__num_leaves': 17, 'reg__subsample': 0.6701833680043187, 'reg__colsample_bytree': 0.738349432788341, 'reg__min_child_samples': 18, 'reg__reg_alpha': 7.402882985709375, 'reg__reg_lambda': 25.38673639497943, 'reg__min_split_gain': 4.897502379506666, 'reg__path_smooth': 3.3805495075965912, 'reg__min_child_weight': 0.001341824219447002, 'reg__feature_fraction': 0.6708558967641087, 'reg__bagging_fraction': 0.6705509585626321, 'reg__bagging_freq': 1, 'reg__max_bin': 128, 'reg__min_data_per_group': 173}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.04085251955213484, 'reg__num_leaves': 24, 'reg__subsample': 0.5999759234501165, 'reg__colsample_bytree': 0.8113661689750231, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.449585846297203, 'reg__reg_lambda': 34.40663262768304, 'reg__min_split_gain': 4.646318295586776, 'reg__path_smooth': 2.7984523620425446, 'reg__min_child_weight': 0.0021902988494142566, 'reg__feature_fraction': 0.8037065507117893, 'reg__bagging_fraction': 0.6177864284477157, 'reg__bagging_freq': 3, 'reg__max_bin': 113, 'reg__min_data_per_group': 200}

Rank 9: {'reg___n_estimators': 600, 'reg___max_depth': 6, 'reg___learning_rate': 0.05987656671419554, 'reg___num_leaves': 17, 'reg___subsample': 0.6621505395396858, 'reg___colsample_bytree': 0.7426091047019934, 'reg___min_child_samples': 34, 'reg___reg_alpha': 8.08944975734854, 'reg___reg_lambda': 39.57949697935576, 'reg___min_split_gain': 4.841585309381926, 'reg___path_smooth': 3.3805483994825045, 'reg___min_child_weight': 8.550474071287756, 'reg___feature_fraction': 0.6980512902031173, 'reg___bagging_fraction': 0.6535081724716737, 'reg___bagging_freq': 1, 'reg___max_bin': 66, 'reg___min_data_per_group': 176}

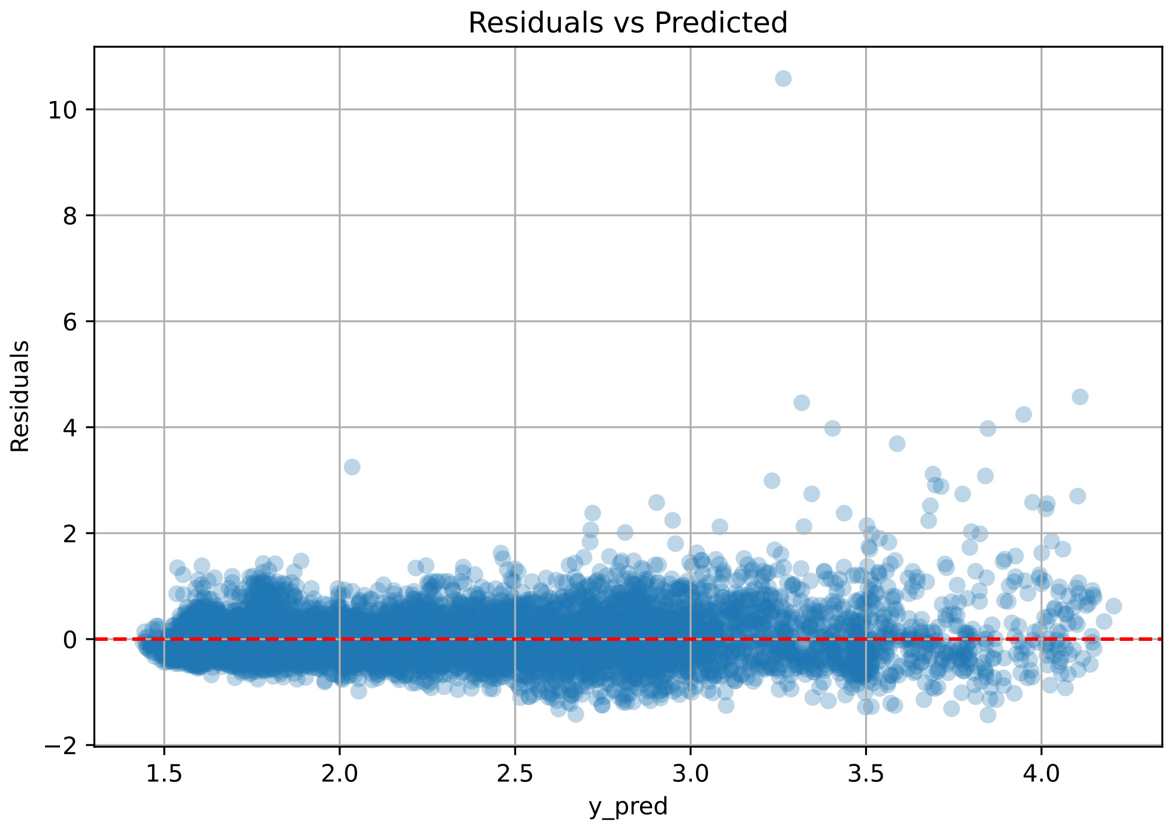
Rank 10: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.05949974546656608, 'reg___num_leaves': 25, 'reg___subsample': 0.5993822510121548, 'reg___colsample_bytree': 0.8177877711287805, 'reg___min_child_samples': 22, 'reg___reg_alpha': 4.0906661792625965, 'reg___reg_lambda': 40.2797669832001, 'reg___min_split_gain': 4.58955805335785, 'reg___path_smooth': 4.392901714845174, 'reg___min_child_weight': 0.0026524556370069187, 'reg___feature_fraction': 0.6734714172954485, 'reg___bagging_fraction': 0.6140600176744981, 'reg___bagging_freq': 3, 'reg___max_bin': 110, 'reg___min_data_per_group': 185}

3.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

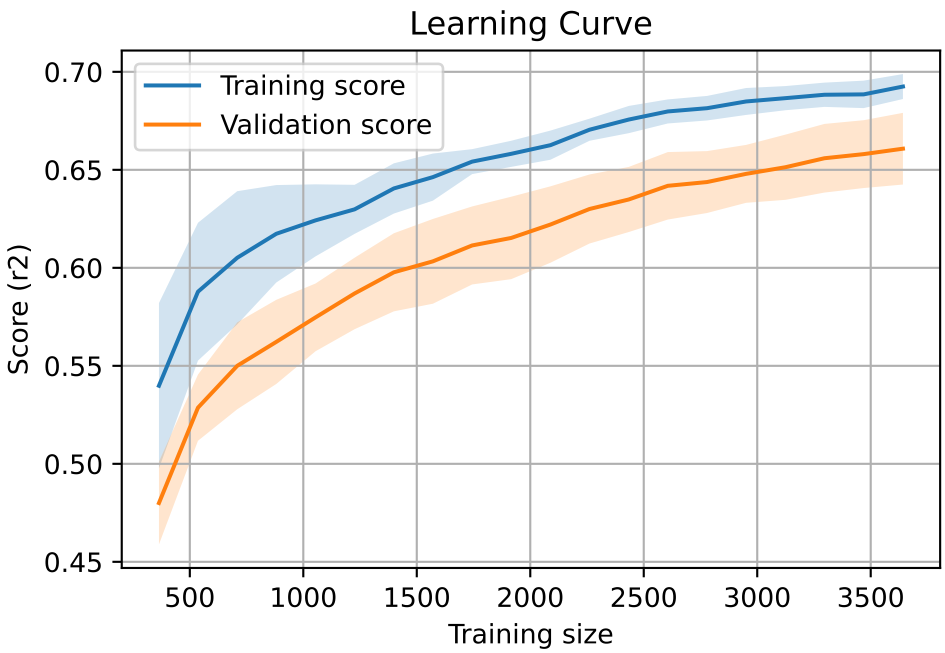
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337

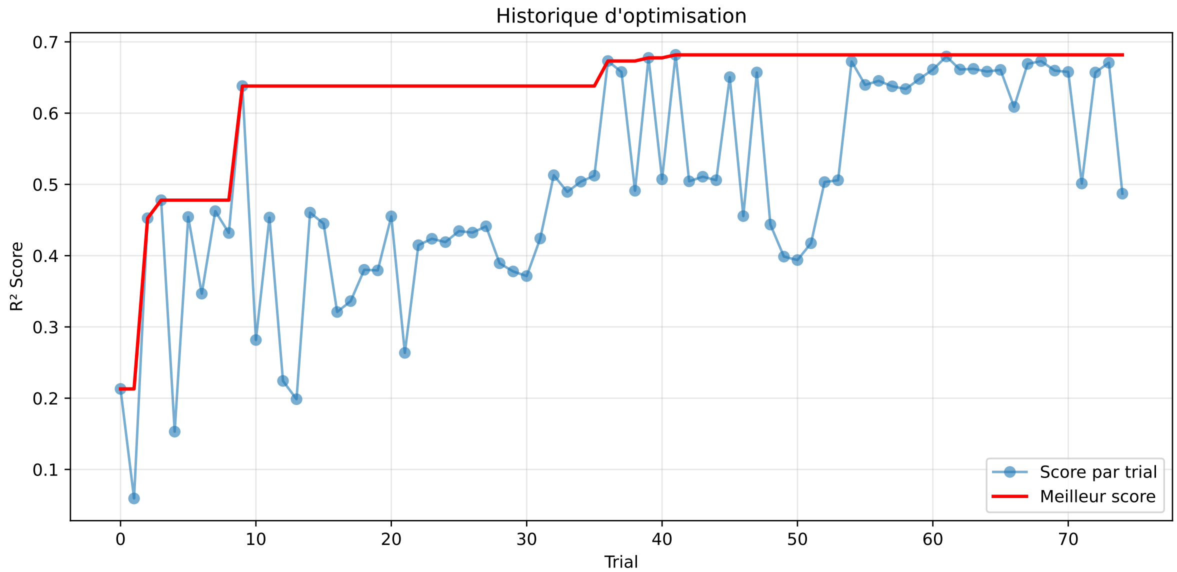


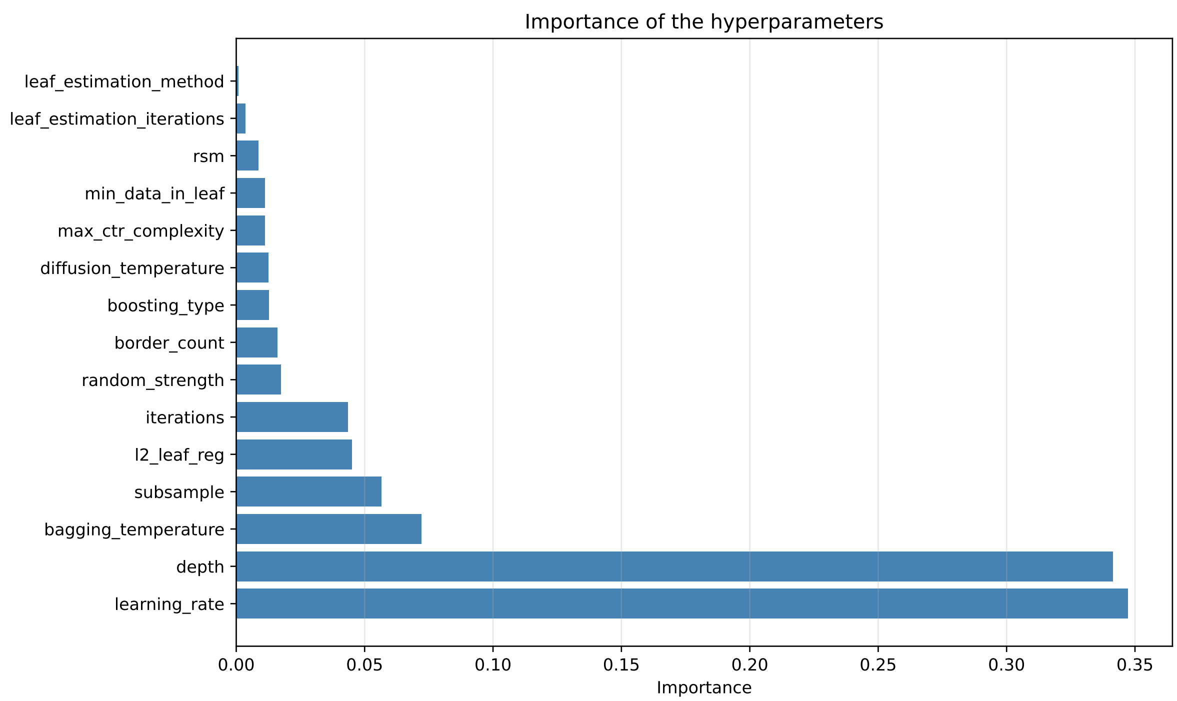
3.6 Model : CatBoost

3.6.1 Gridsearch

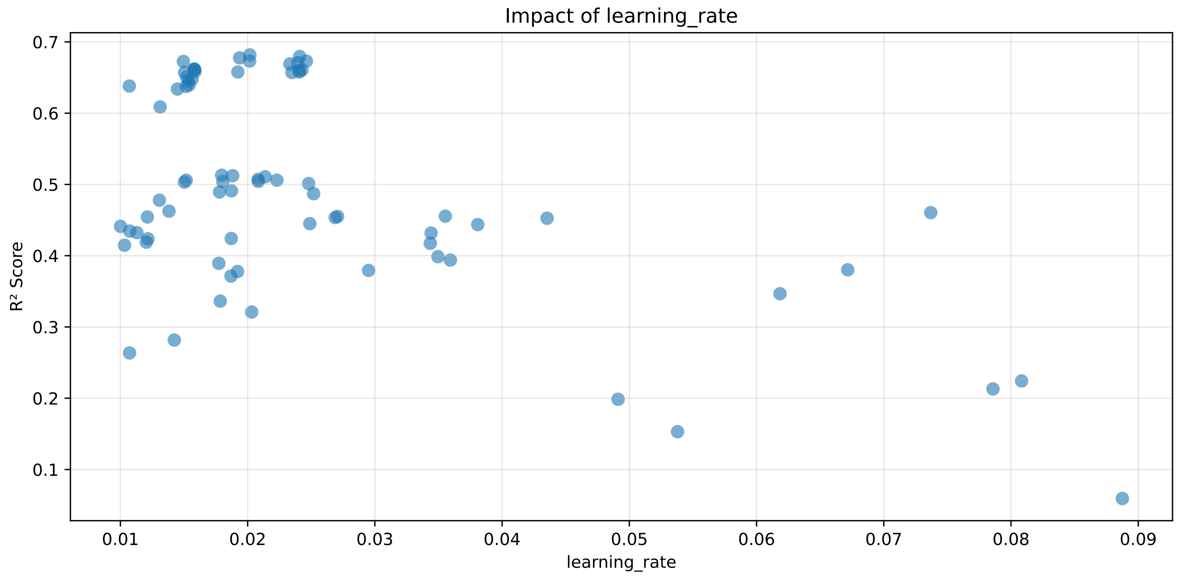
Distribution y_train vs y_test: y_train: mean=2.367, std=0.823, min=0.959, max=8.682 y_test: mean=2.331, std=0.776, min=1.071, max=13.845

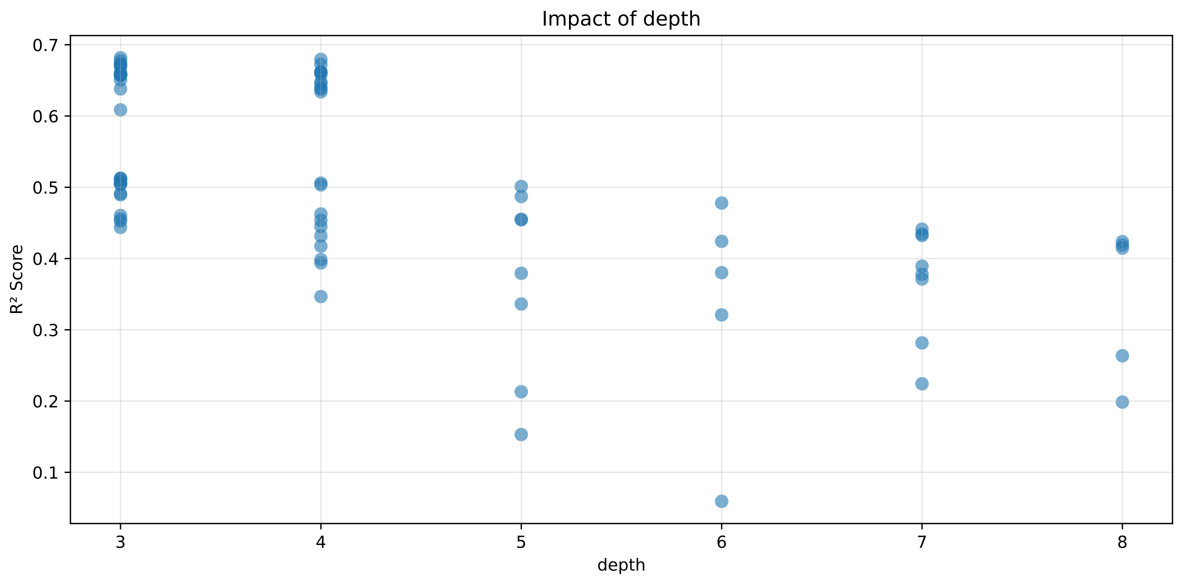
==================== TOP Importance FEATURES ==================== feature importance r_umidity 24.046105 r_temperature_wl_2h_A_rms 15.119796 r_temperature_wl_8h_A_energy_rel 5.268305 r_temperature_wl_2h_D1_energy_rel 5.143958 r_umidity_wl_2h_A_energy_rel 4.666736 r_temperature_wl_8h_A_energy 4.339145 r_CO2 4.019368 r_temperature 3.591689 r_NH3 3.495641 r_NH3_wl_8h_D1_energy 2.840678 r_NH3_wl_8h_D2_energy 2.701304 r_NH3_wl_8h_A_energy 2.660160 r_umidity_wl_2h_A_energy 1.950207 r_THI 1.875526 r_temperature_wl_2h_A_energy 1.772415 r_NH3_wl_2h_A_rms 1.730798 r_NH3_wl_8h_D3_energy 1.699626 r_NH3_wl_2h_D1_energy 1.653437 r_umidity_wl_2h_A_rms 1.587849 r_temperature_wl_8h_D3_energy 1.442913 r_NH3_wl_2h_A_energy 1.392162 r_NH3_wl_8h_D3_energy_rel 1.269405 r_NH3_wl_8h_D2_energy_rel 1.208418 r_umidity_wl_8h_A_energy_rel 0.862997 r_umidity_wl_8h_A_rms 0.854649 r_NH3_wl_8h_A_rms 0.662170 r_NH3_wl_2h_A_energy_rel 0.592842 r_umidity_wl_8h_D2_energy_rel 0.589109 r_umidity_wl_8h_D3_energy_rel 0.562089 r_temperature_wl_8h_A_rms 0.400503

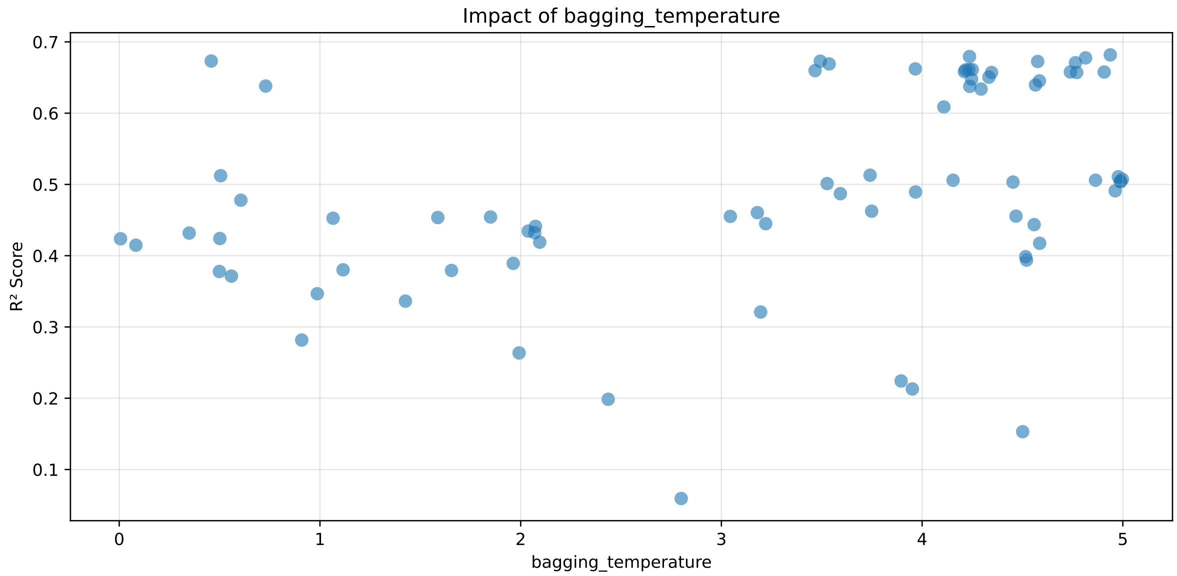


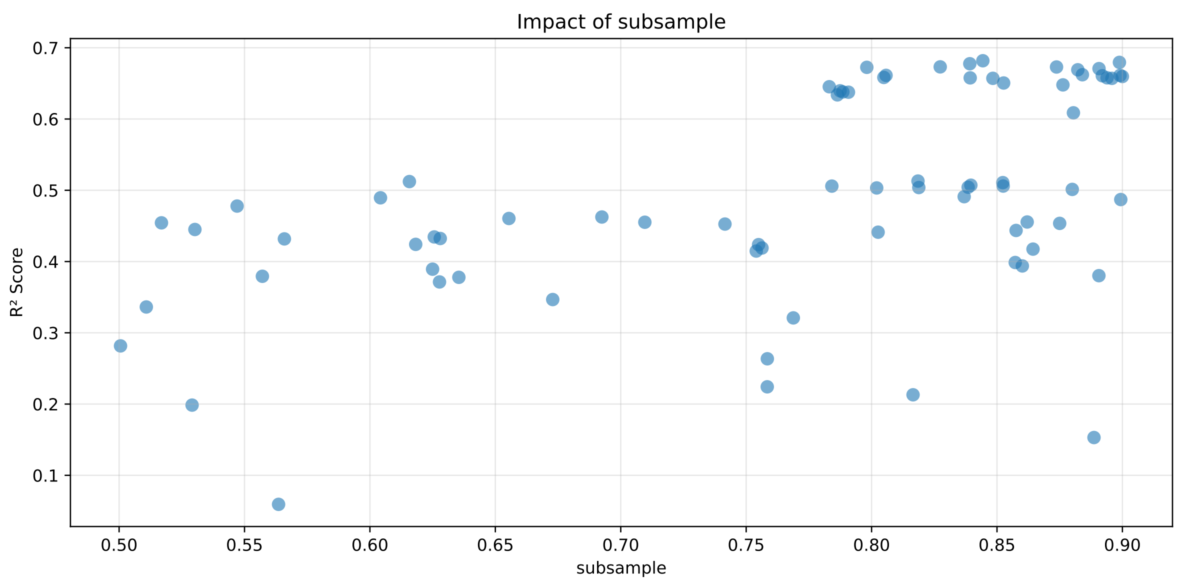


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'bagging_temperature', 'subsample']


Impact of learning_rate
R² Score
learning_rate
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09

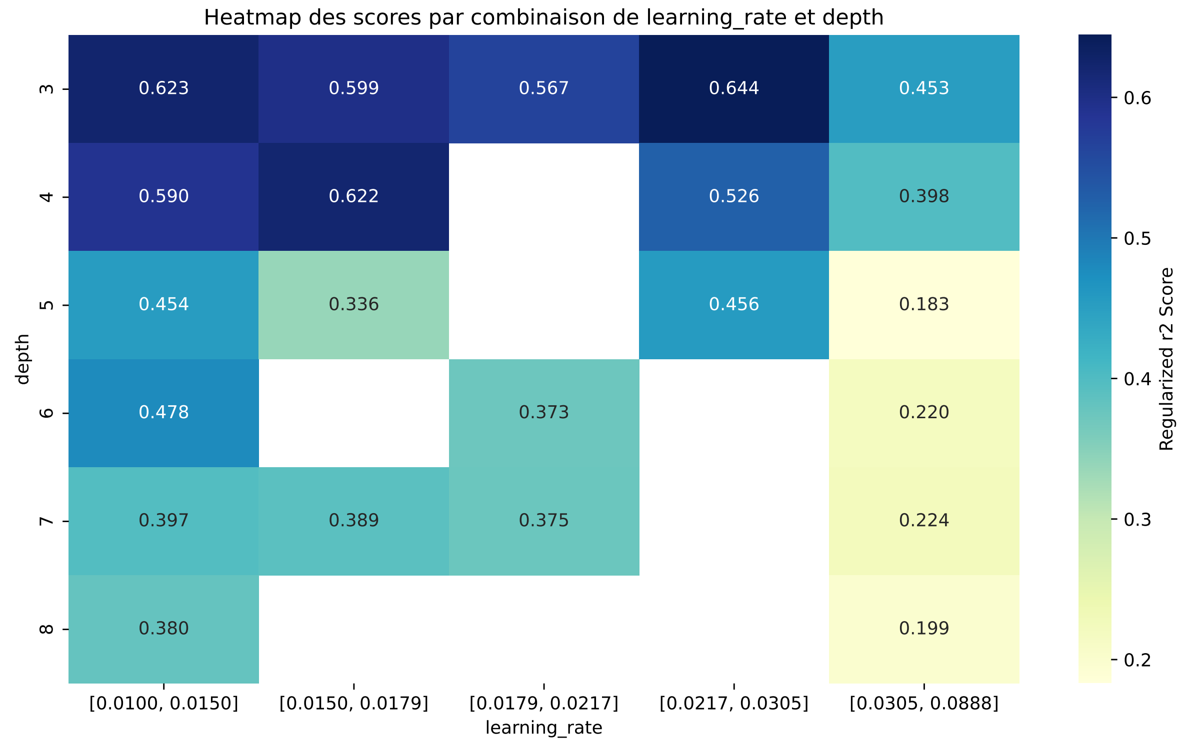

Impact of depth
R² Score
depth
0.1
0.2
0.3
0.4
0.5
0.6
0.7
3
4
5
6
7
8






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7147	0.6818	0.5094	0.0182	0.6818	0.5124	-0.1693	0.7516	1.0483
2	0.7131	0.6795	0.5184	0.0177	0.6795	0.5154	-0.1641	0.7585	1.0495
3	0.7094	0.6776	0.5039	0.0163	0.6776	0.5049	-0.1727	0.7452	1.047
4	0.7039	0.6732	0.5071	0.0161	0.6732	0.5123	-0.1609	0.761	1.0457
5	0.7051	0.673	0.5026	0.0167	0.673	0.4968	-0.1762	0.7382	1.0476
6	0.7057	0.6725	0.5199	0.0164	0.6725	0.5239	-0.1487	0.7789	1.0492
7	0.7011	0.6707	0.5128	0.0151	0.6707	0.5137	-0.1571	0.7658	1.0452
8	0.6987	0.6691	0.5021	0.0173	0.6691	0.498	-0.1711	0.7443	1.0443
9	0.6926	0.6621	0.5123	0.0164	0.6621	0.5182	-0.144	0.7826	1.046
10	0.6924	0.6613	0.515	0.0175	0.6613	0.5125	-0.1487	0.7751	1.0471
---	---	---	---	---	---	---	---	---	---
75	0.9755	0.8228	0.521	0.0182	0.0593	0.543	-0.2797	0.66	1.1856

Hyperparameters:

Rank 1: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.020168677311316927, 'reg___l2_leaf_reg': 12.792545466985171, 'reg___bagging_temperature': 4.937472344413193, 'reg___random_strength': 4.030523730159674, 'reg___subsample': 0.8443574002293304, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6706202447749925, 'reg___border_count': 86, 'reg___leaf_estimation_method': 'Newton',

’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 6795.297233759822}

Rank 2: {’reg__iterations’: 400, ’reg__depth’: 4, ’reg__learning_rate’: 0.02408872392402344, ’reg__l2_leaf_reg’: 34.558374209031044, ’reg__bagging_temperature’: 4.236171801312011, ’reg__random_strength’: 4.317130741810088, ’reg__subsample’: 0.8988116643419745, ’reg__min_data_in_leaf’: 44, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’: 0.664346566885608, ’reg__border_count’: 70, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 4884.132730939522}

Rank 3: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.01936807161441199, ’reg__l2_leaf_reg’: 33.15800637539332, ’reg__bagging_temperature’: 4.81433829635048, ’reg__random_strength’: 2.879739489496972, ’reg__subsample’: 0.8391566222337067, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 4, ’reg__rsm’: 0.6402030131007981, ’reg__border_count’: 62, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 4382.234822844996}

Rank 4: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.020153709948713995, ’reg__l2_leaf_reg’: 32.85843430882763, ’reg__bagging_temperature’: 0.4583738720052777, ’reg__random_strength’: 4.26755201878405, ’reg__subsample’: 0.8273637722382183, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 5, ’reg__rsm’: 0.632861877254738, ’reg__border_count’: 62, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 6466.071628435432}

Rank 5: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.02461722400774788, ’reg__l2_leaf_reg’: 47.98377396409974, ’reg__bagging_temperature’: 3.4927137433138093, ’reg__random_strength’: 4.345516293139346, ’reg__subsample’: 0.8737349190622125, ’reg__min_data_in_leaf’: 42, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.6586603918147711, ’reg__border_count’: 55, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 2911.686281276392}

Rank 6: {’reg__iterations’: 500, ’reg__depth’: 4, ’reg__learning_rate’: 0.014946445470331911, ’reg__l2_leaf_reg’: 43.135771885756995, ’reg__bagging_temperature’: 4.575760777684075, ’reg__random_strength’: 4.528714958351408, ’reg__subsample’: 0.7980995532496495, ’reg__min_data_in_leaf’: 35, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.608326784522955, ’reg__border_count’: 69, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 3195.427131288176}

Rank 7: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.023936531404337297, ’reg__l2_leaf_reg’: 40.0332682258844, ’reg__bagging_temperature’: 4.763851499279089, ’reg__random_strength’: 4.360346028701641, ’reg__subsample’: 0.8906726900127121,

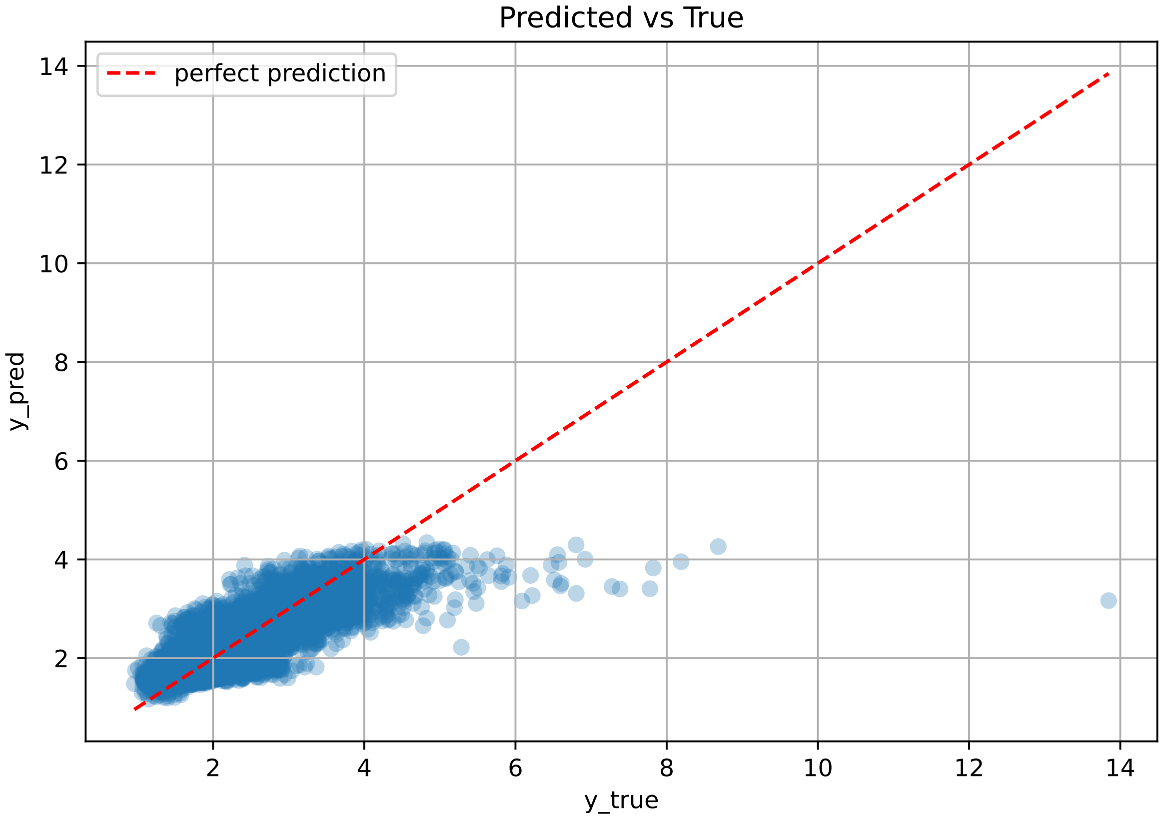
’reg___min_data_in_leaf’: 42, ’reg___leaf_estimation_iterations’: 3, ’reg___rsm’: 0.6573312650149186, ’reg___border_count’: 76, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 2881.4995781466855}

Rank 8: {’reg___iterations’: 400, ’reg___depth’: 3, ’reg___learning_rate’: 0.023321386040630845, ’reg___l2_leaf_reg’: 46.45786949845832, ’reg___bagging_temperature’: 3.5370021653064034, ’reg___random_strength’: 2.5209985283905514, ’reg___subsample’: 0.8822163822753374, ’reg___min_data_in_leaf’: 42, ’reg___leaf_estimation_iterations’: 1, ’reg___rsm’: 0.6619899049425538, ’reg___border_count’: 72, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 3485.7005414627984}

Rank 9: {’reg___iterations’: 400, ’reg___depth’: 4, ’reg___learning_rate’: 0.015797625591701083, ’reg___l2_leaf_reg’: 47.88389301705155, ’reg___bagging_temperature’: 3.9662765116309777, ’reg___random_strength’: 4.3331322209901, ’reg___subsample’: 0.8840479893045513, ’reg___min_data_in_leaf’: 42, ’reg___leaf_estimation_iterations’: 6, ’reg___rsm’: 0.6682693839740996, ’reg___border_count’: 70, ’reg___leaf_estimation_method’: ‘Newton’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 3536.4262603682782}

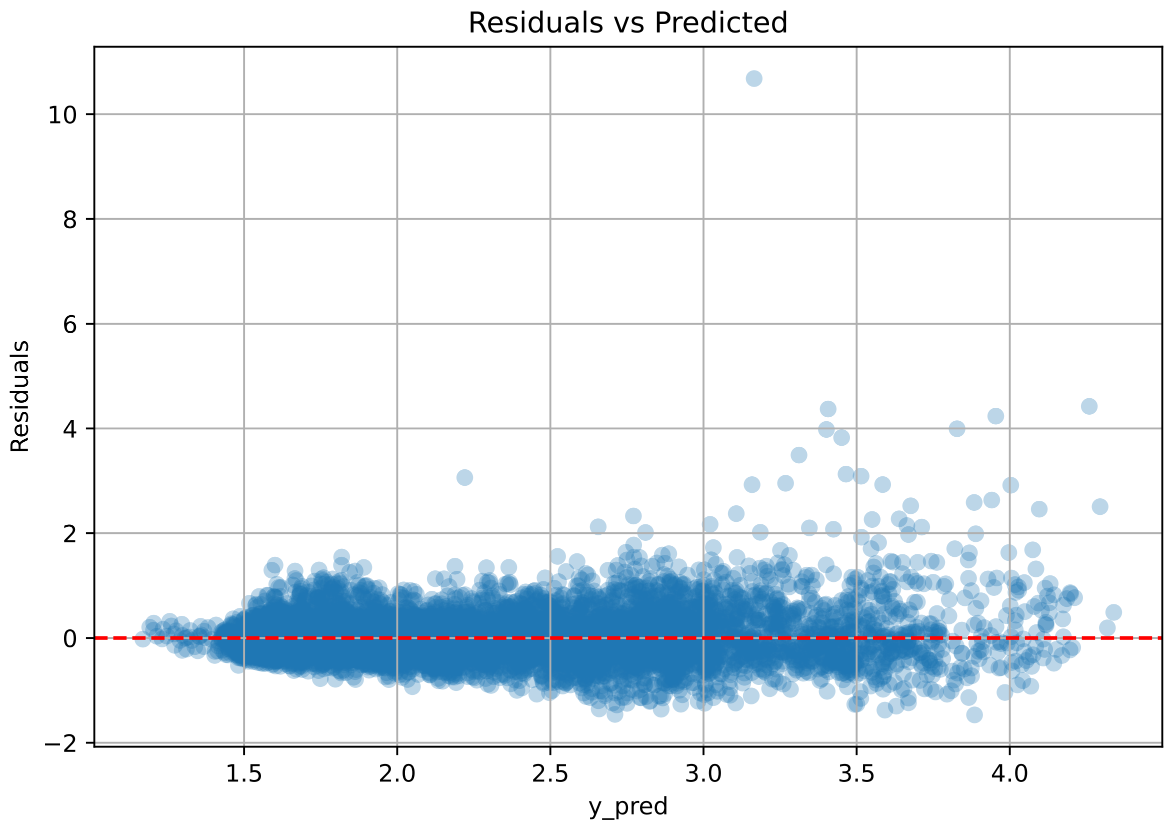
Rank 10: {’reg___iterations’: 400, ’reg___depth’: 4, ’reg___learning_rate’: 0.015828534548918954, ’reg___l2_leaf_reg’: 34.23880782443616, ’reg___bagging_temperature’: 4.232573108924241, ’reg___random_strength’: 4.393435244803804, ’reg___subsample’: 0.8057682834859554, ’reg___min_data_in_leaf’: 42, ’reg___leaf_estimation_iterations’: 8, ’reg___rsm’: 0.6588063594759642, ’reg___border_count’: 70, ’reg___leaf_estimation_method’: ‘Newton’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 3429.7252827870525}

3.6.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

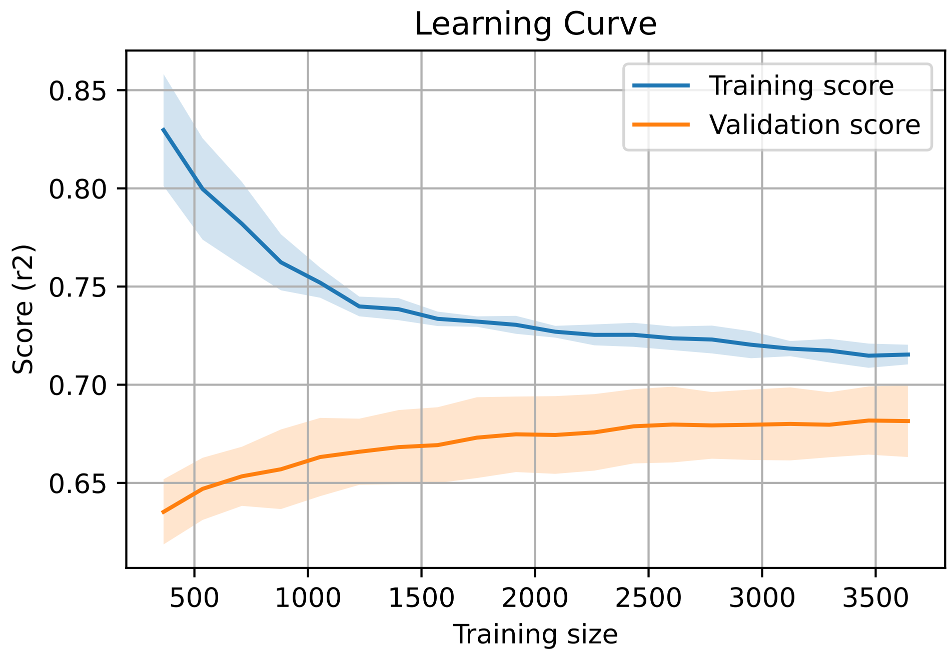
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337



3.7 Sumup-table

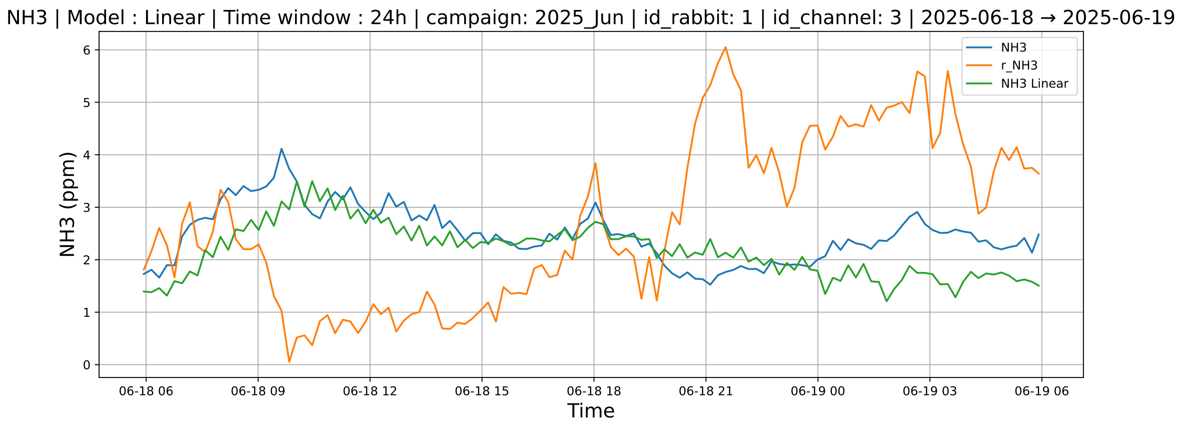
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.628	0.61	0.021	0.313	0.304	-0.306	0.498	1.03	nan
LGBM	0.692	0.661	0.018	0.504	0.508	-0.154	0.768	1.046	0.6614
CatBoost	0.715	0.682	0.018	0.509	0.512	-0.169	0.752	1.048	0.6818

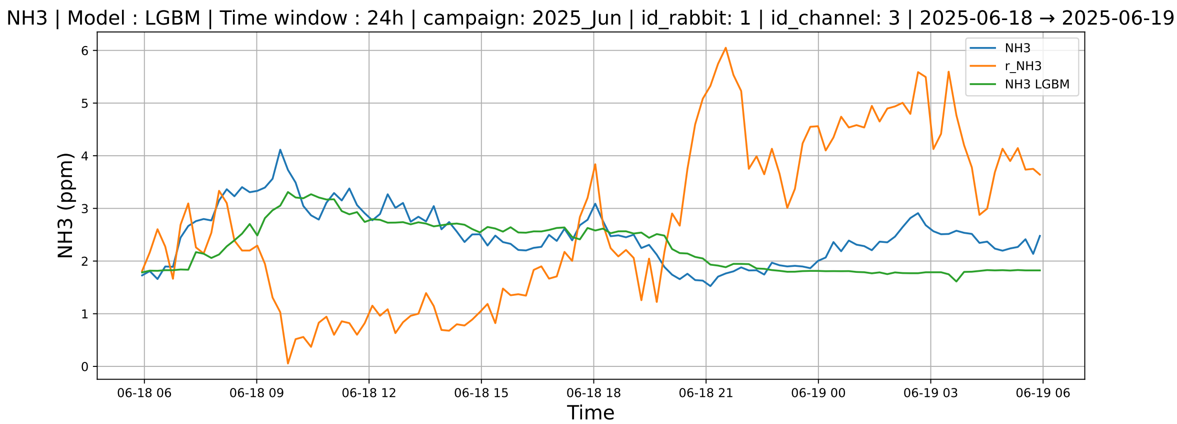
4 Plot

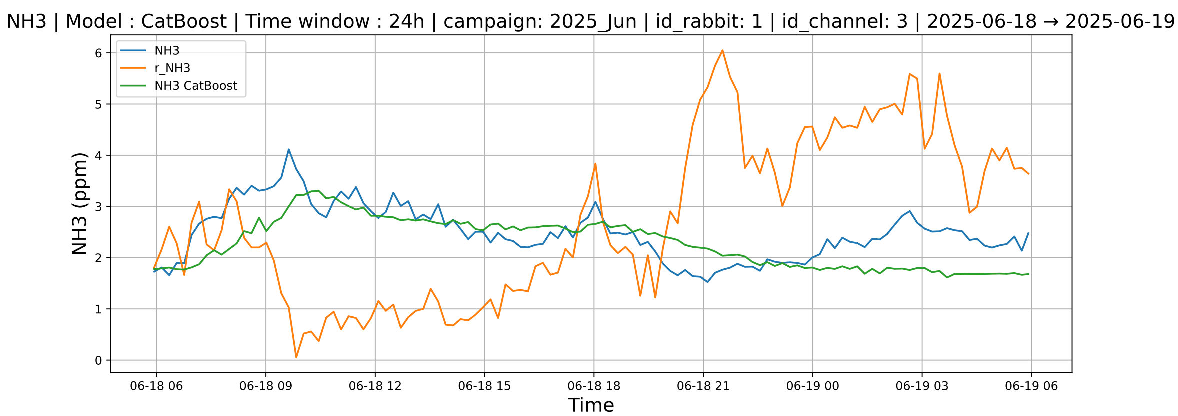
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

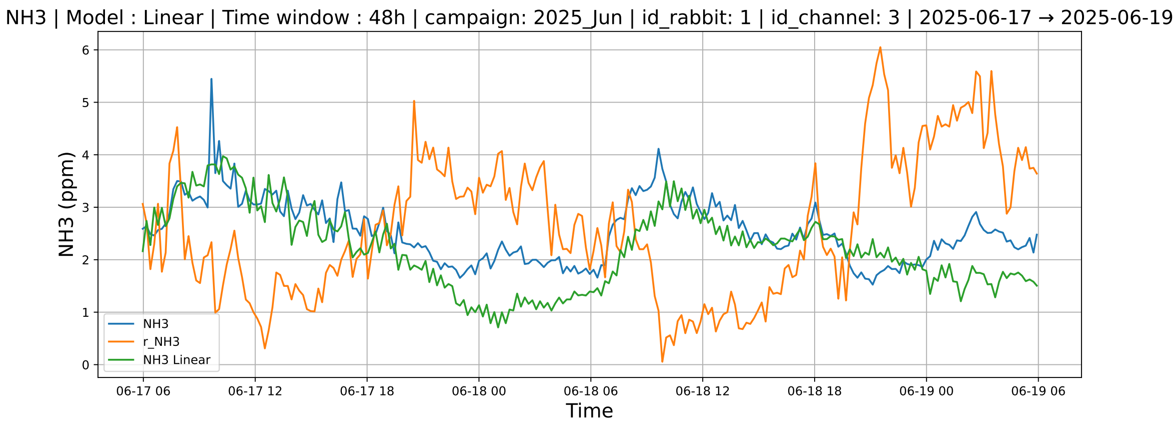
4.1.1.1 Time window : 24

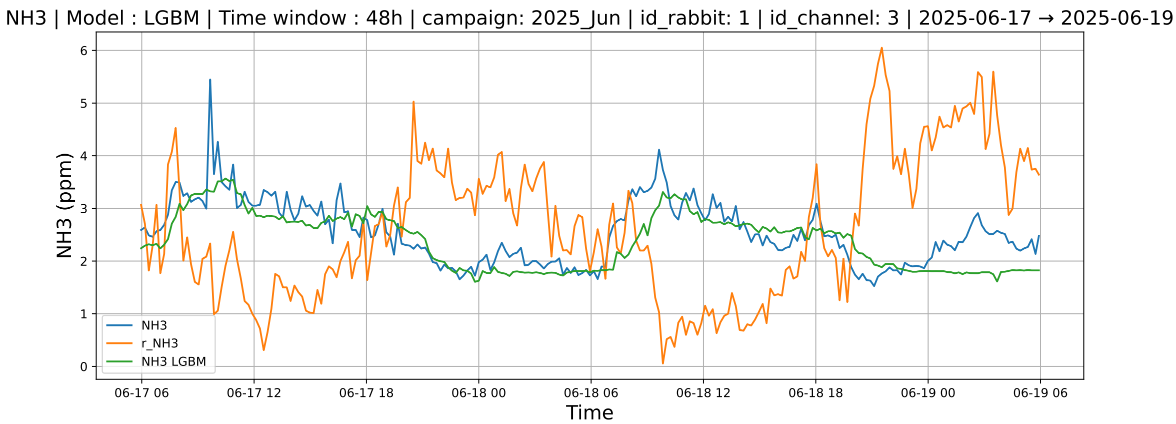


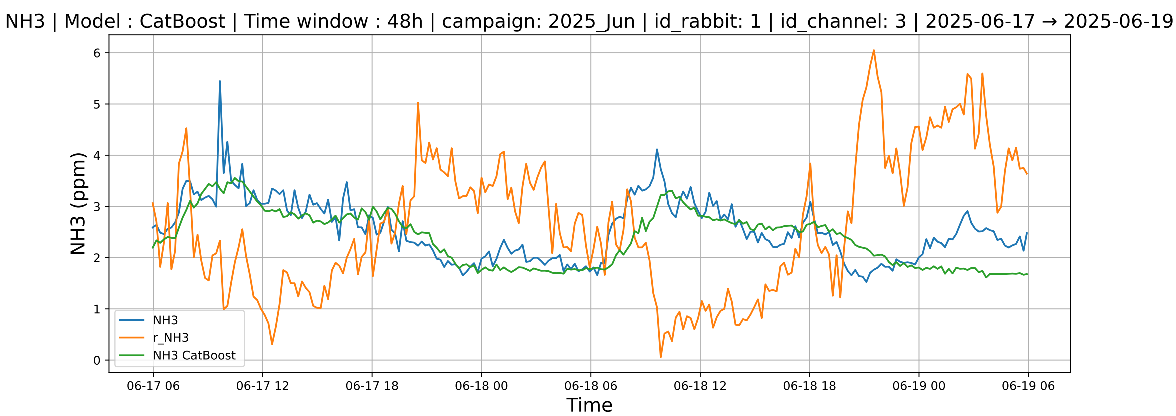




4.1.1.2 Time window : 48

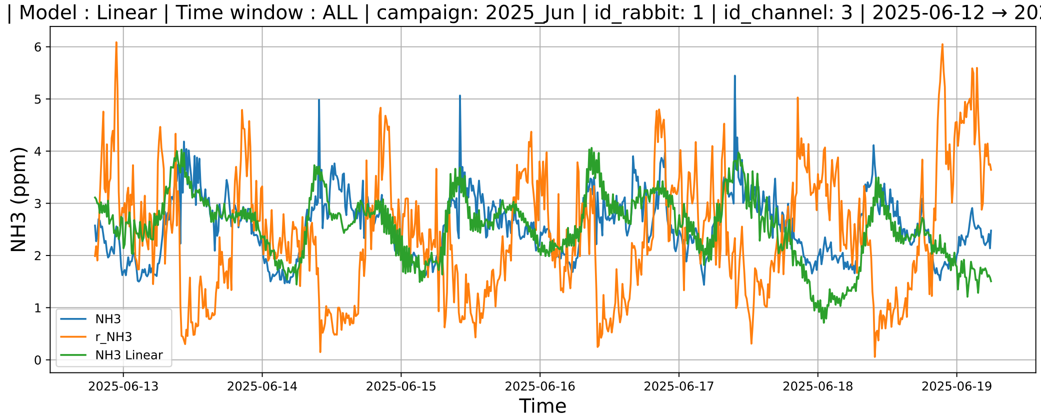




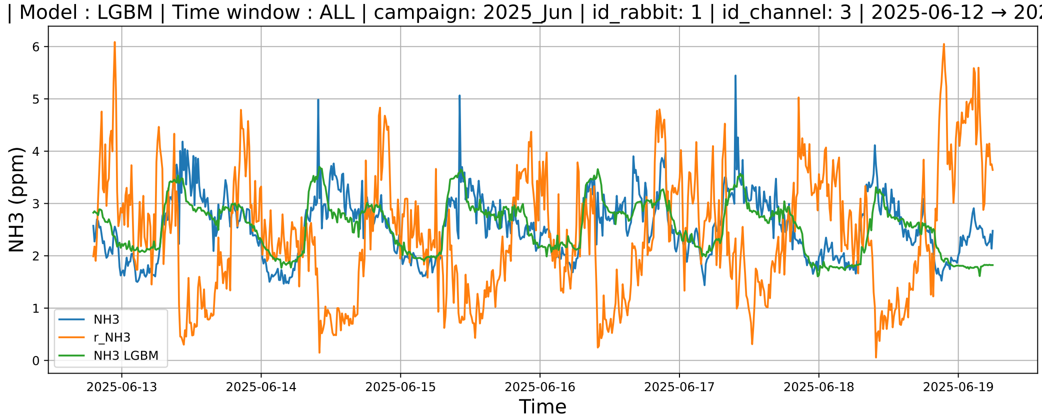


4.1.1.3 Time window : ALL

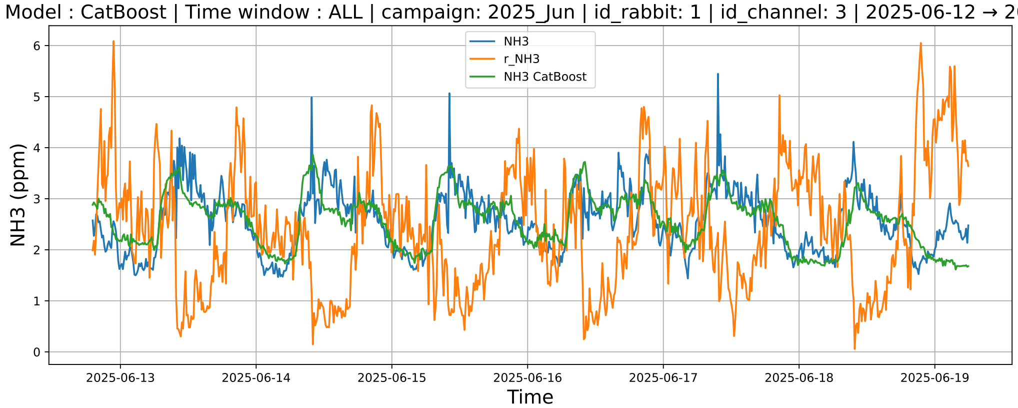
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



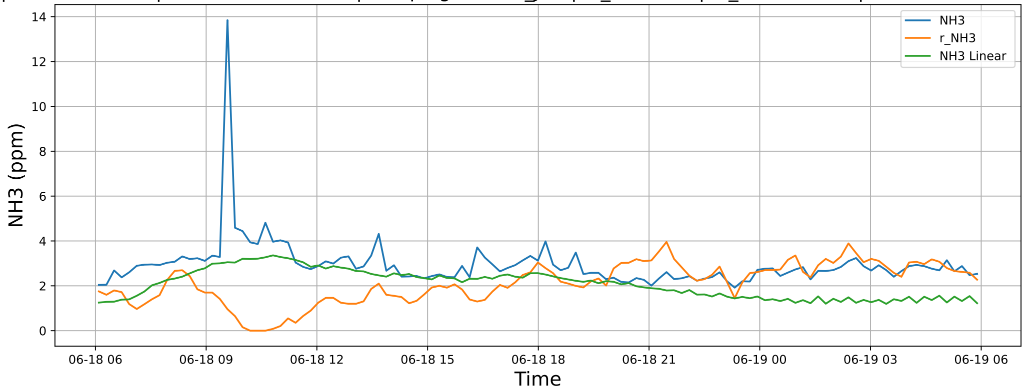
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



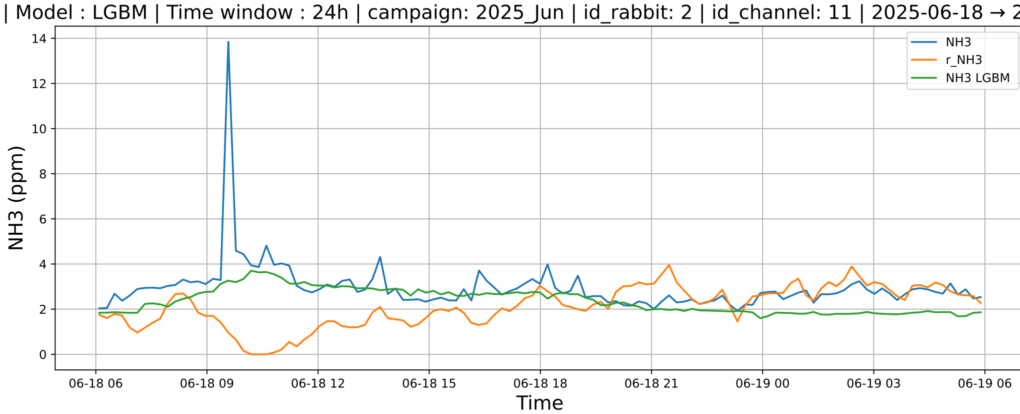
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

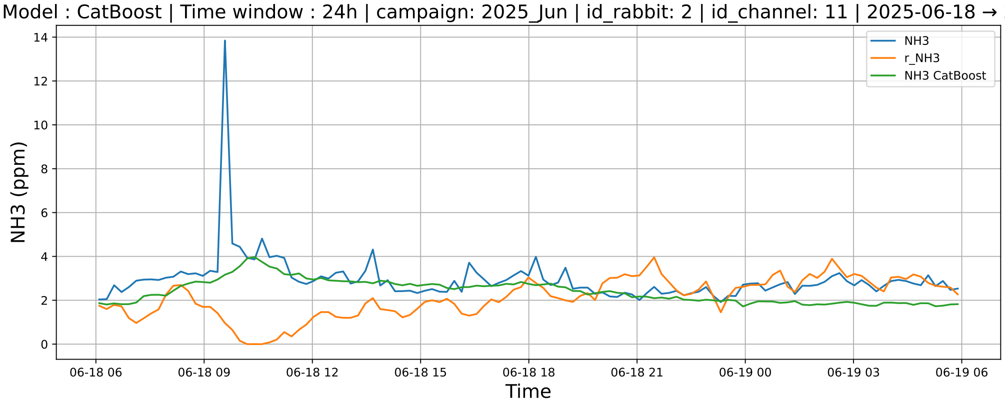
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



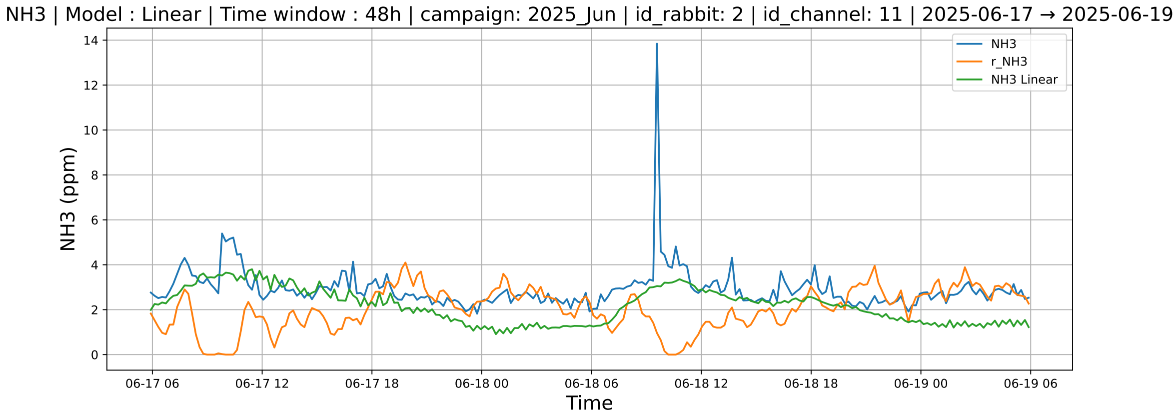
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

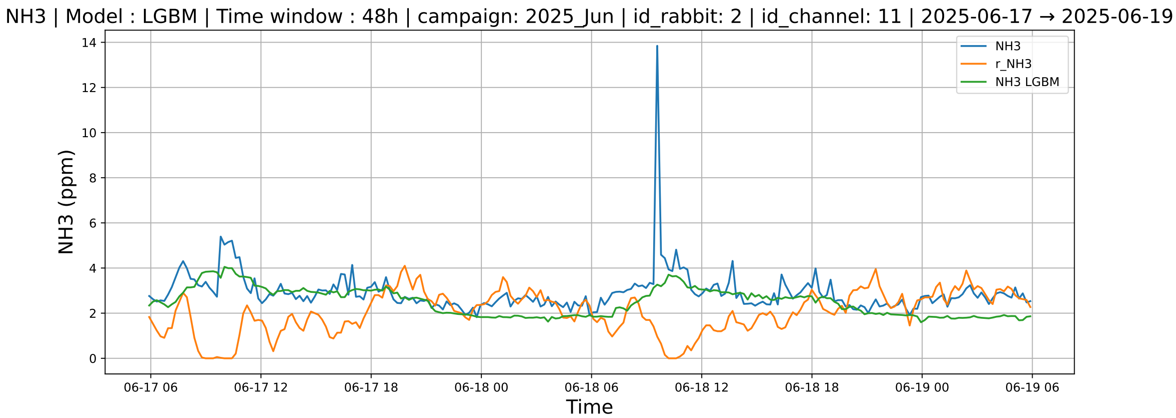


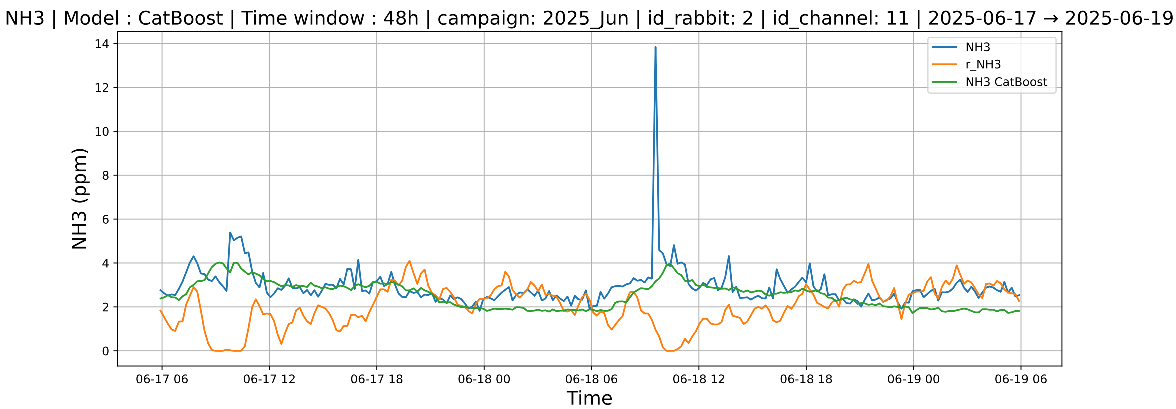
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



4.1.2.2 Time window : 48

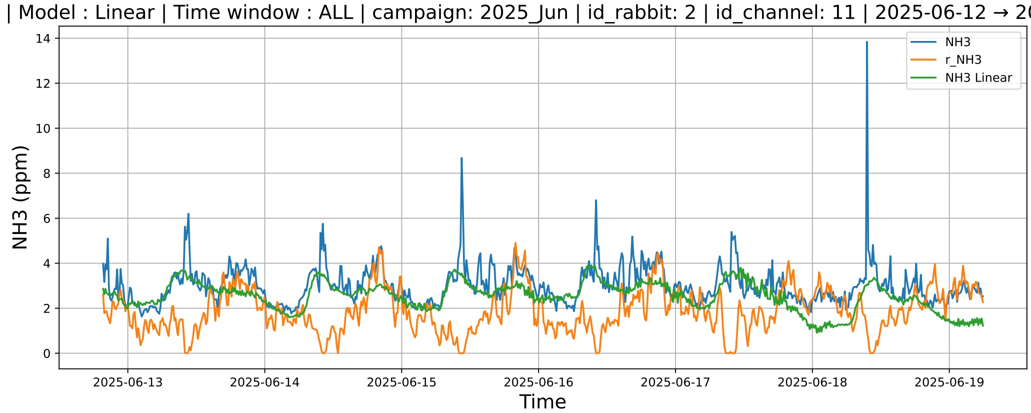




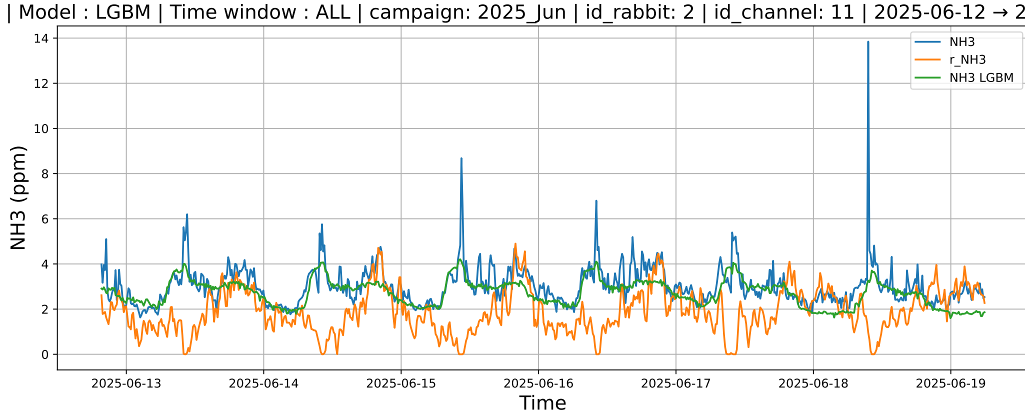


4.1.2.3 Time window : ALL

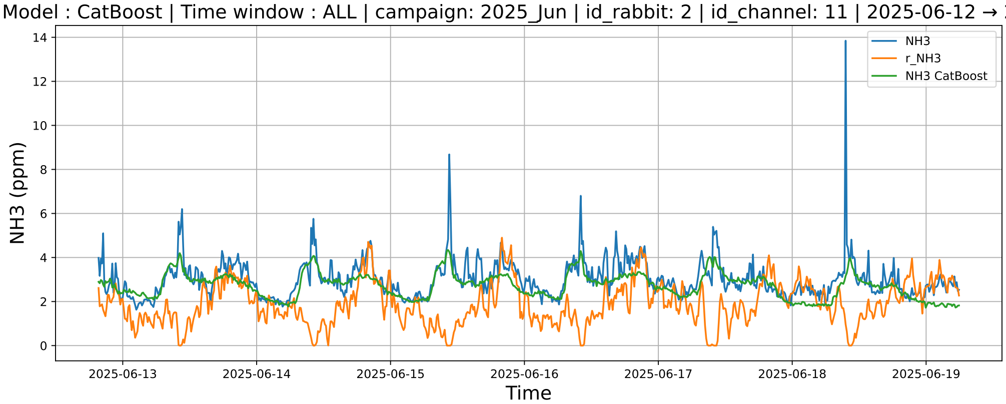
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



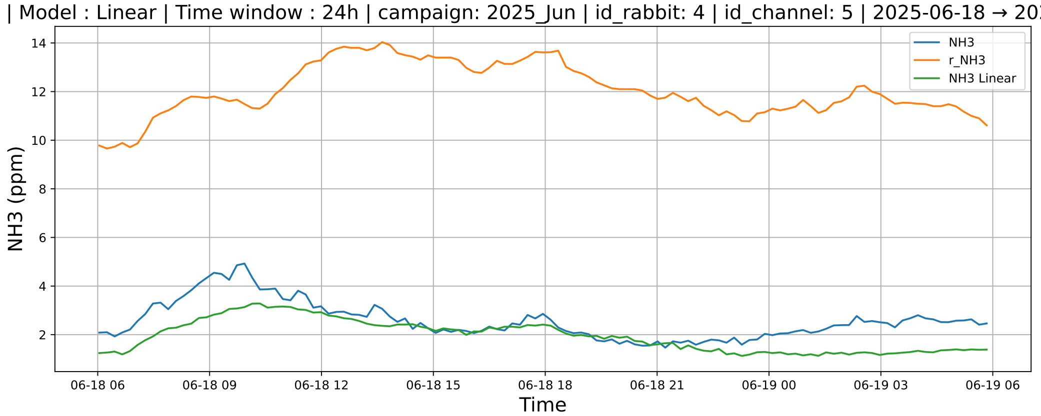
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



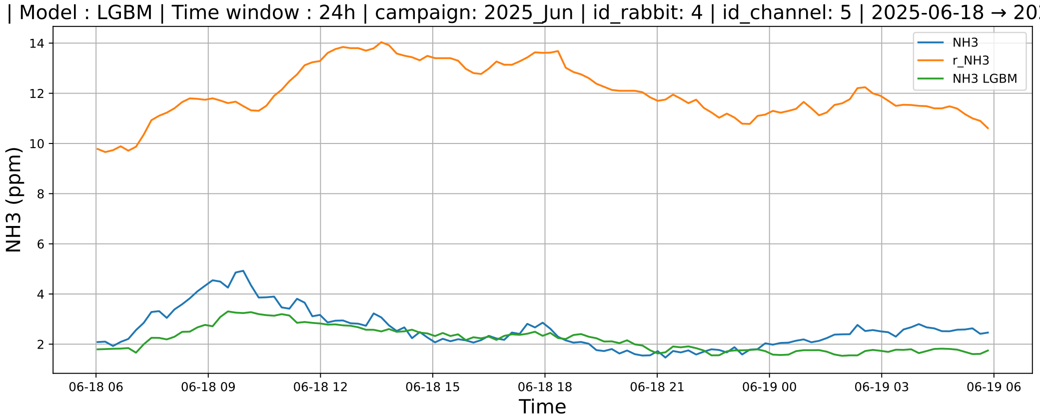
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

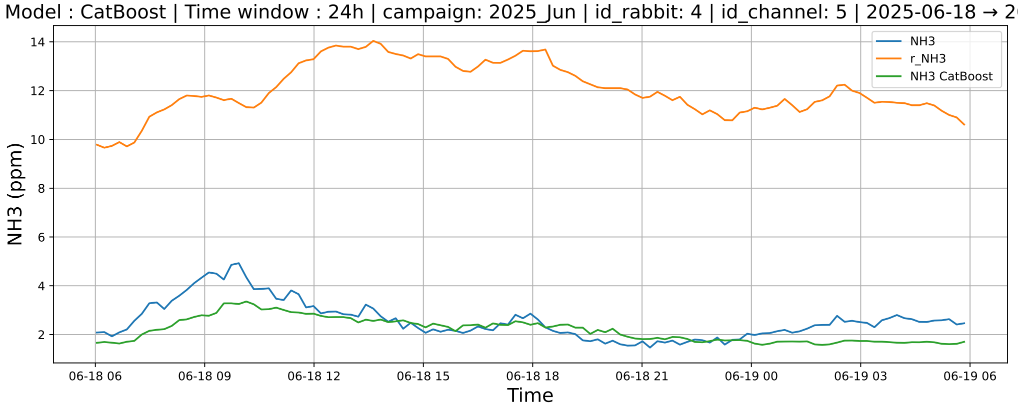
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

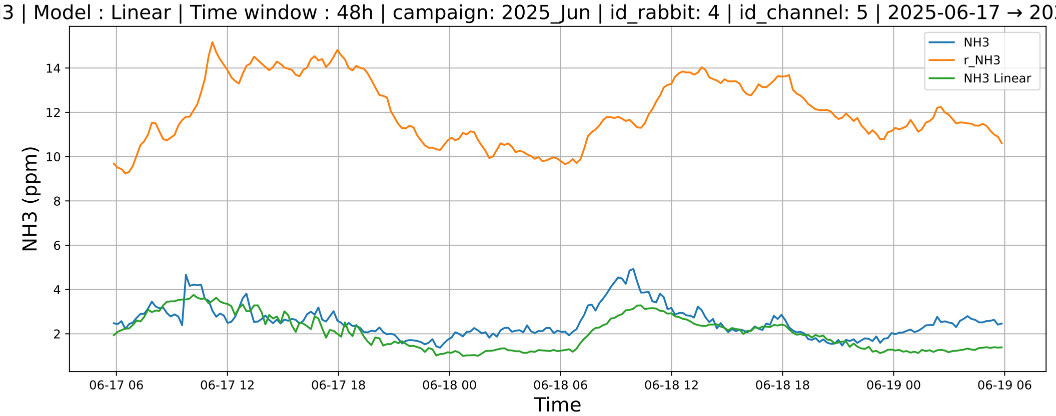


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

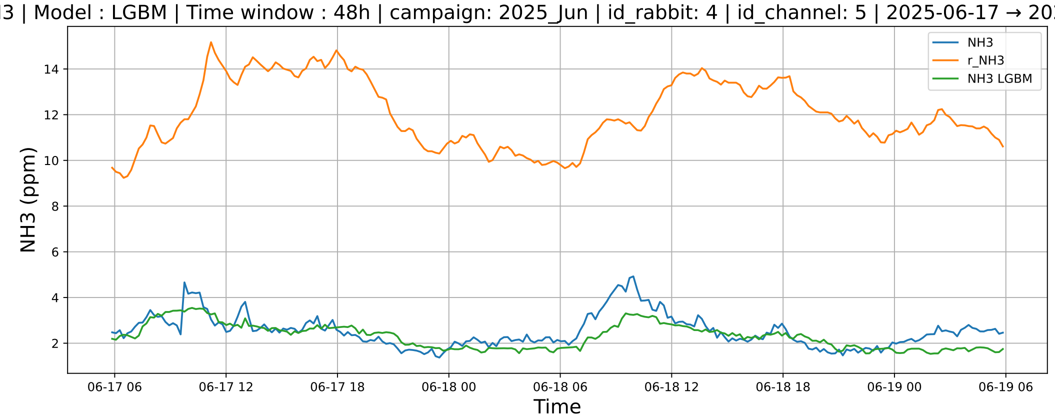


4.1.3.2 Time window : 48

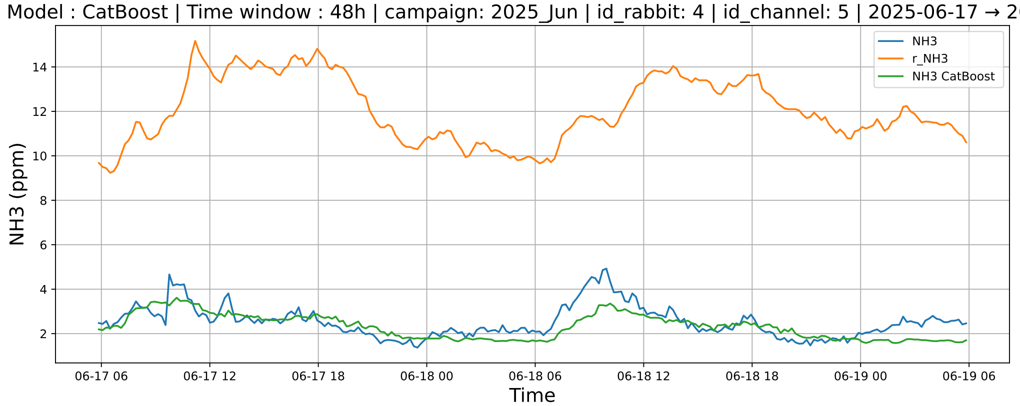
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



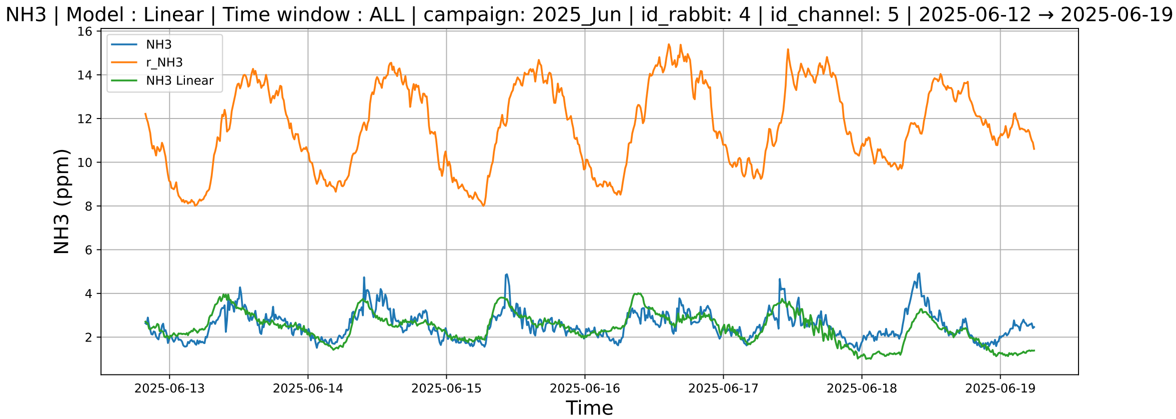
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

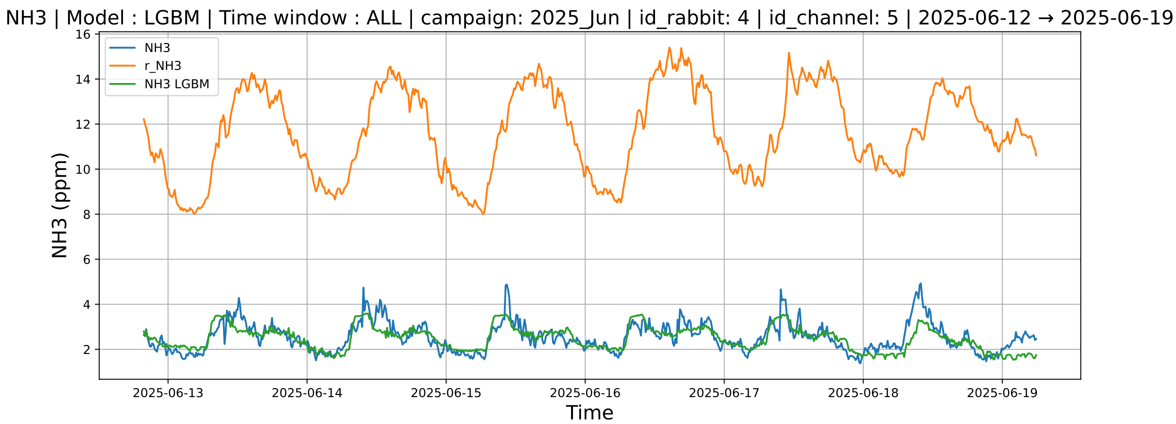


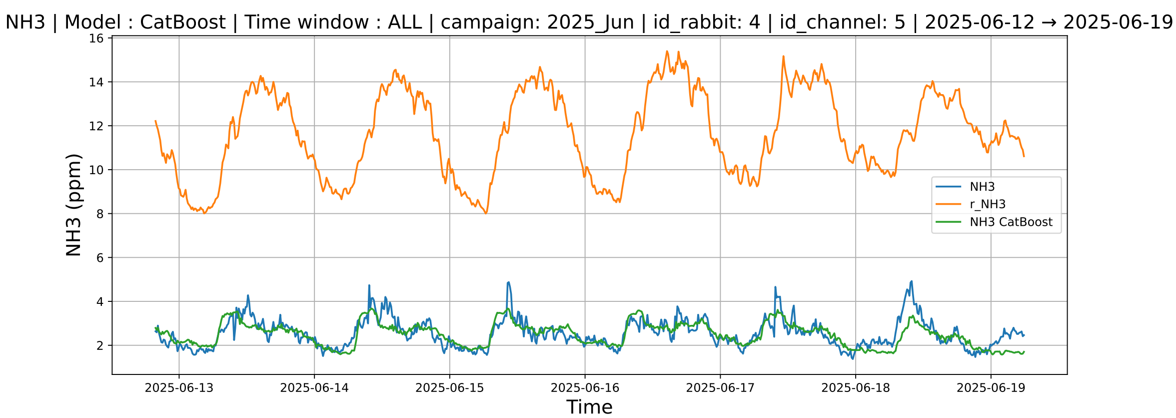
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.1.3.3 Time window : ALL

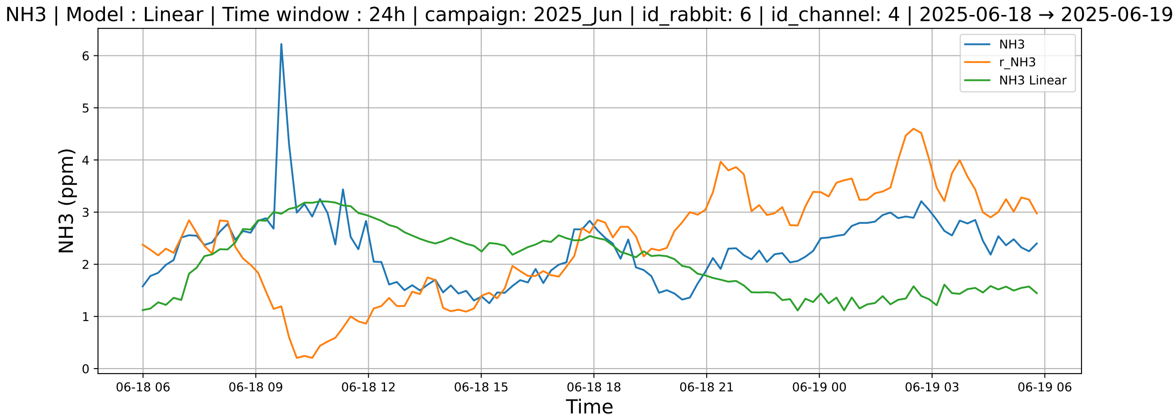


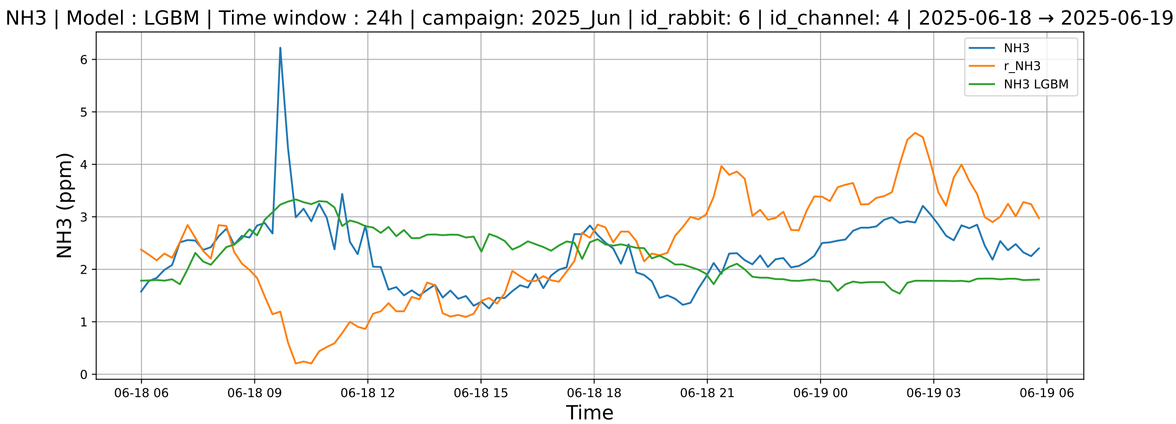


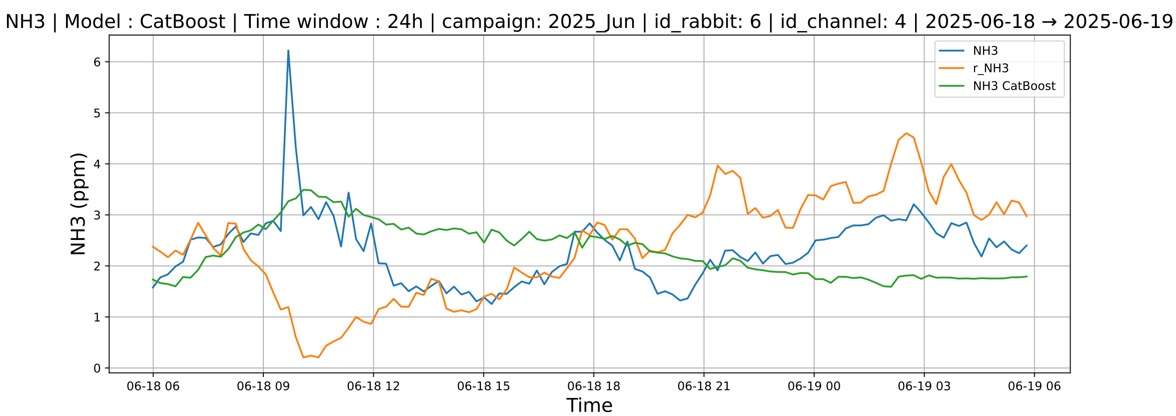


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

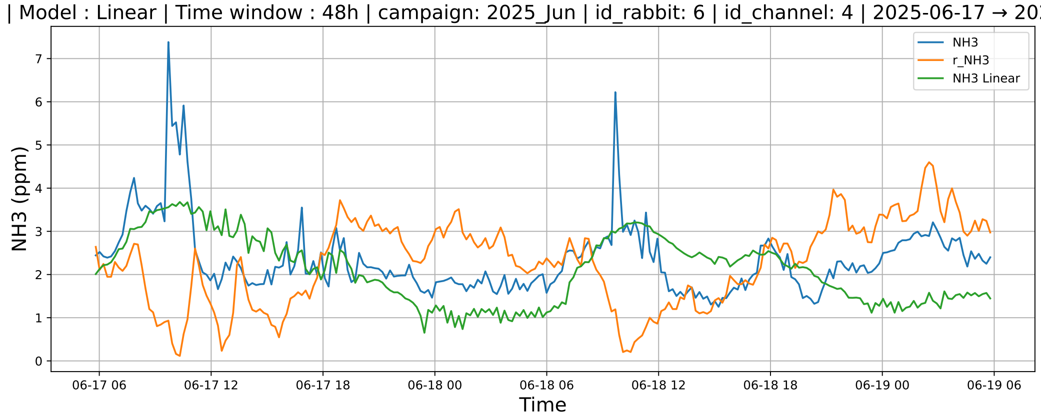




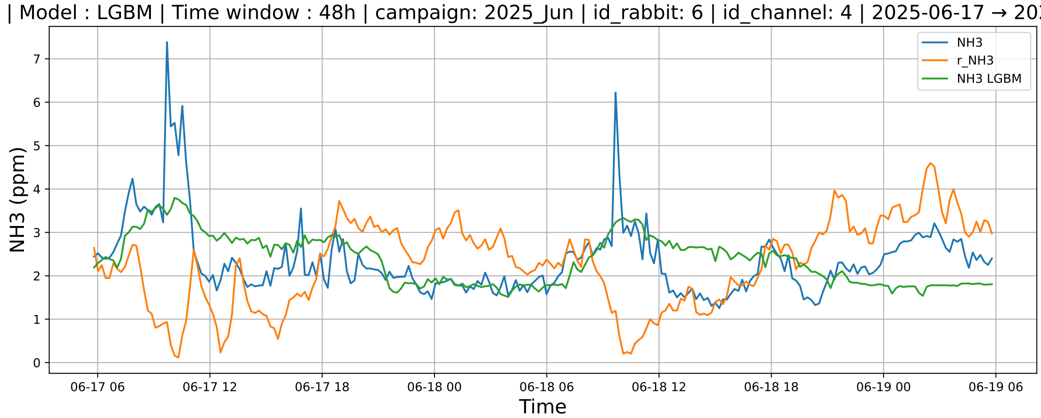


4.1.4.2 Time window : 48

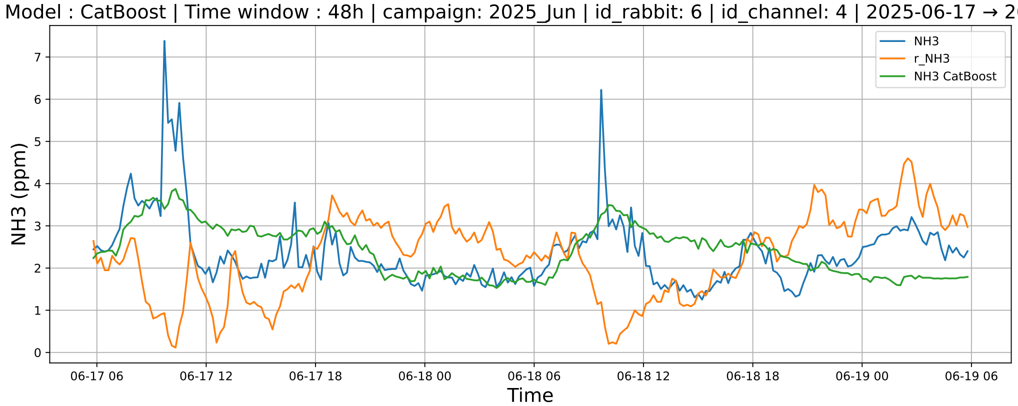
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



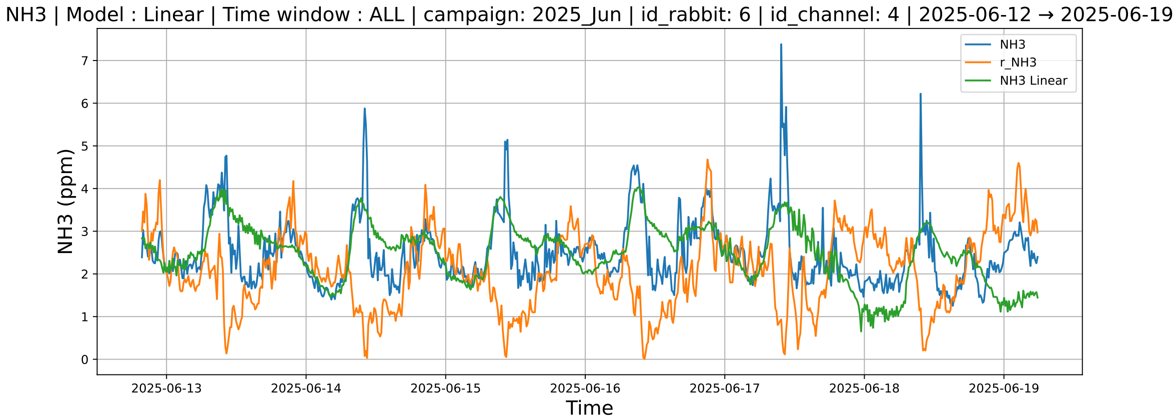
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

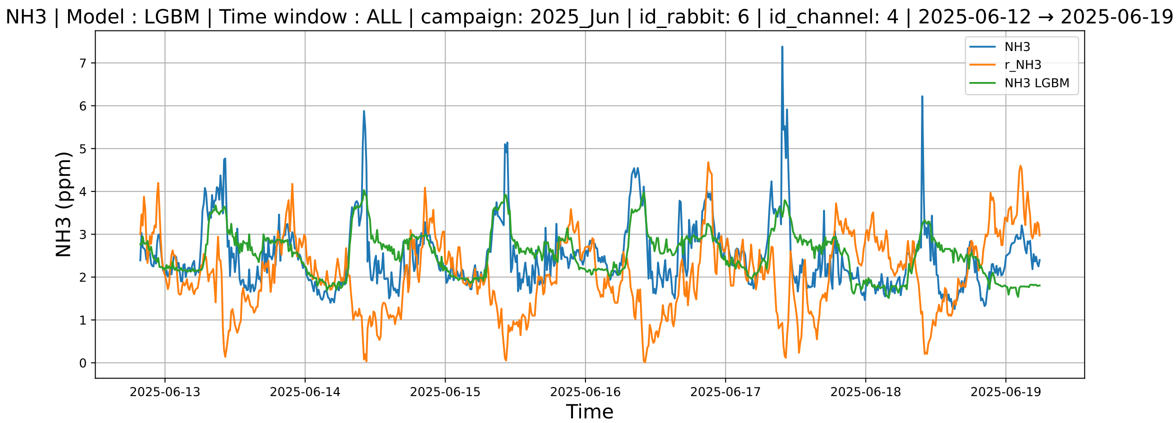


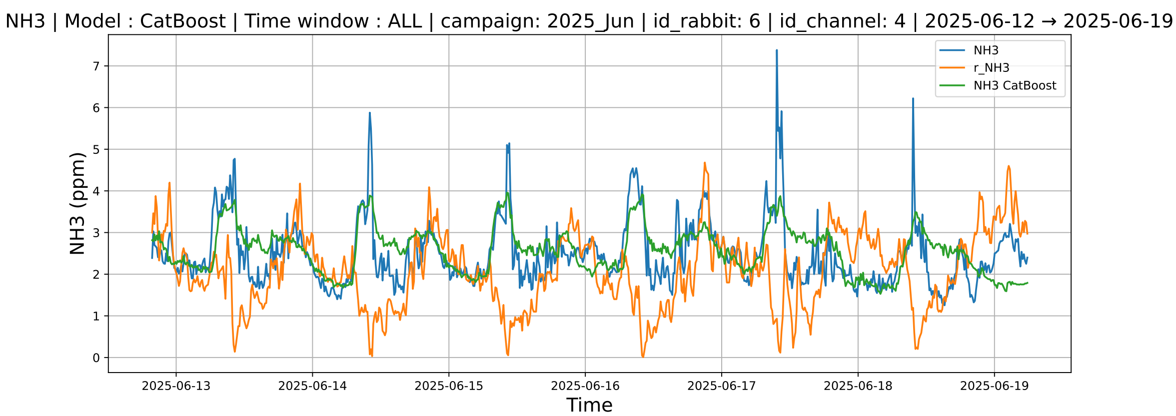
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.1.4.3 Time window : ALL



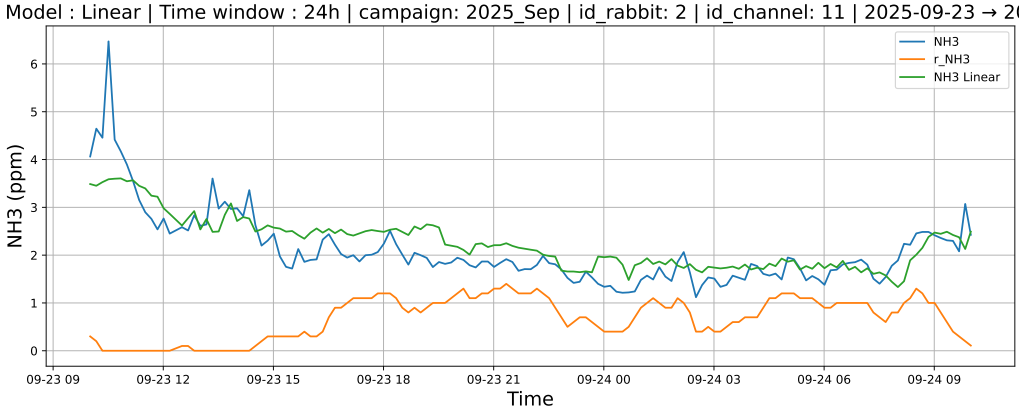




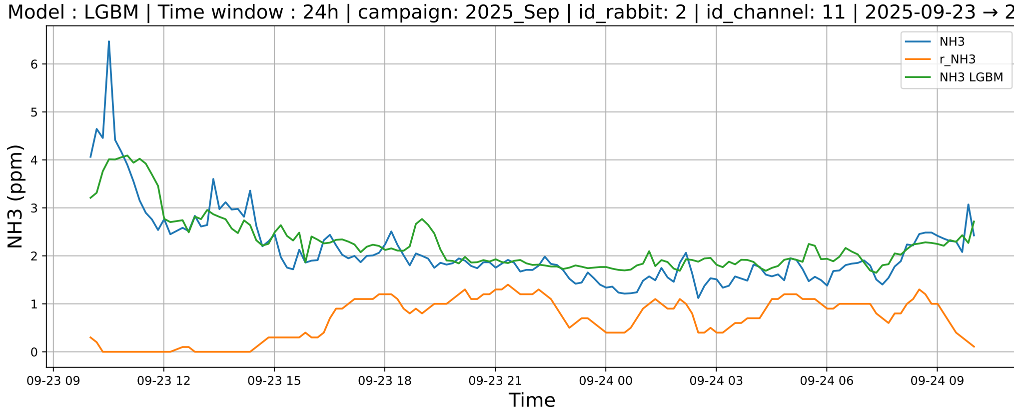
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

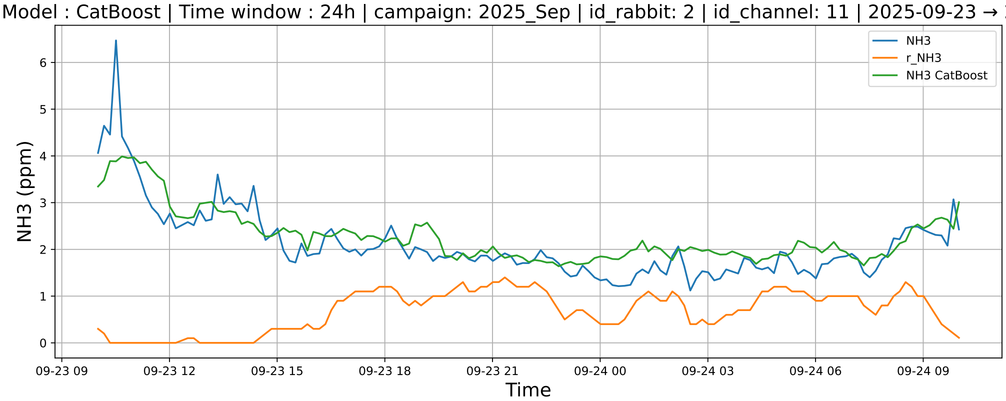
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

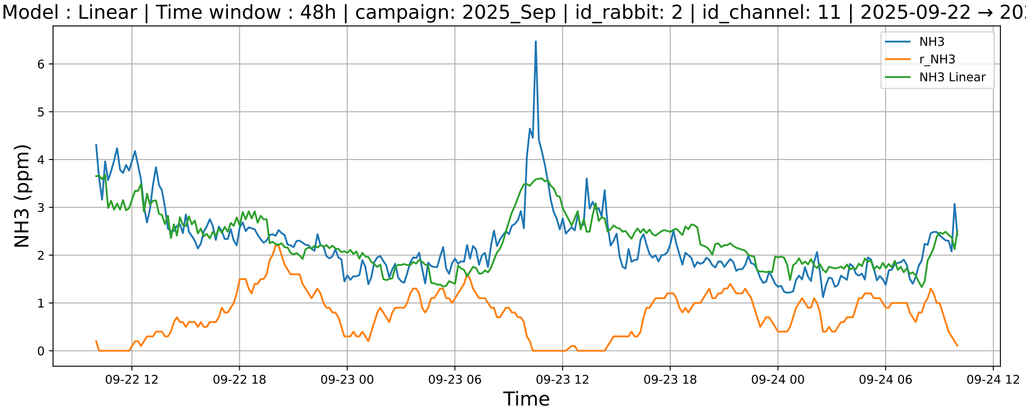


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

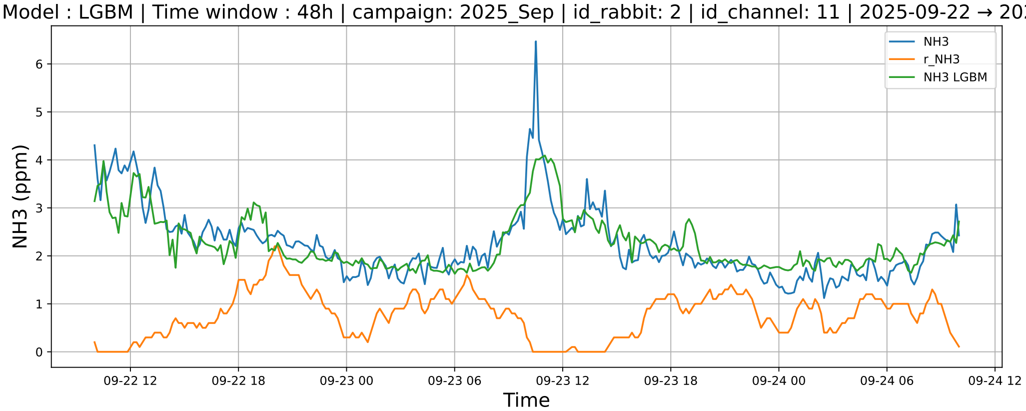


4.1.5.2 Time window : 48

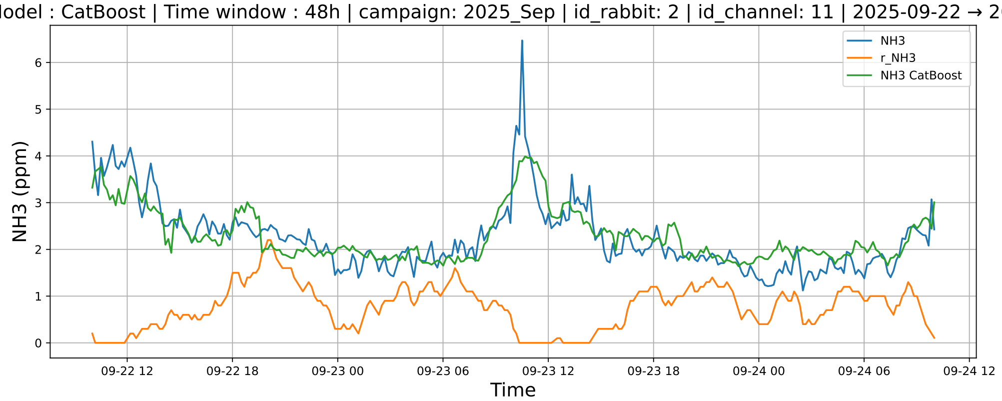
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

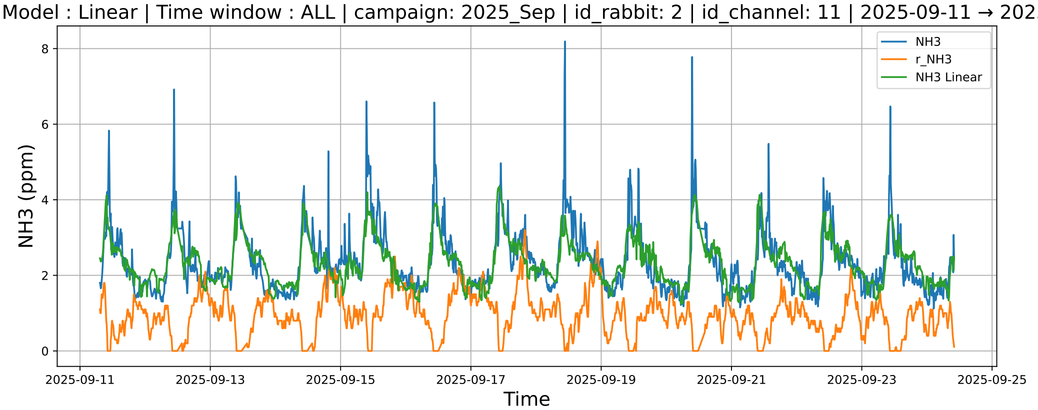


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

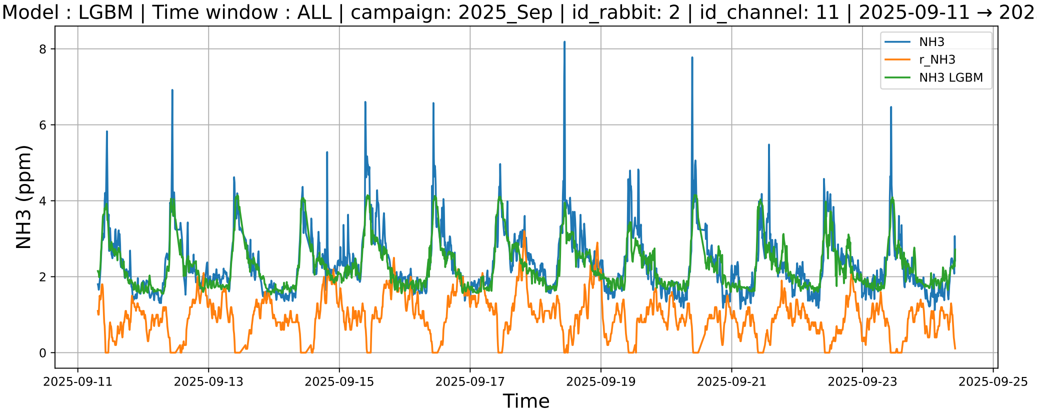


4.1.5.3 Time window : ALL

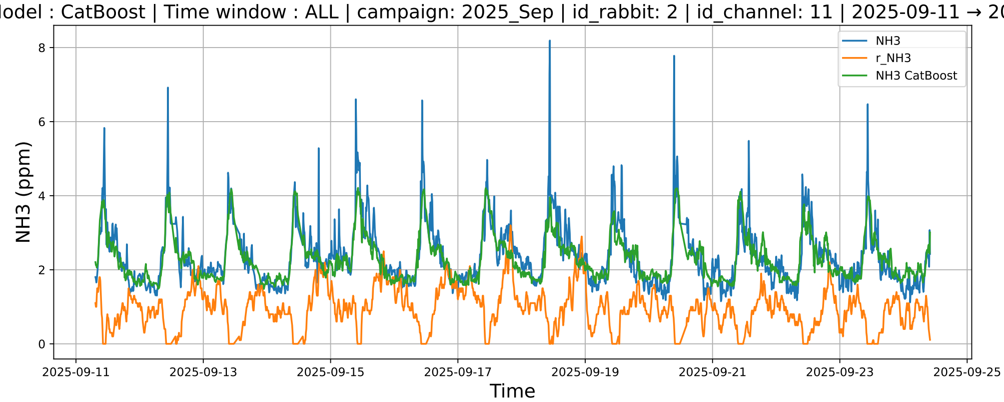
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



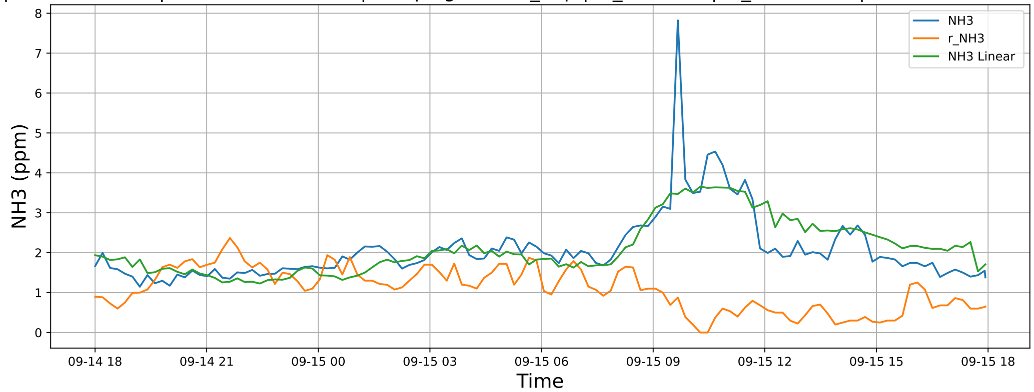
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



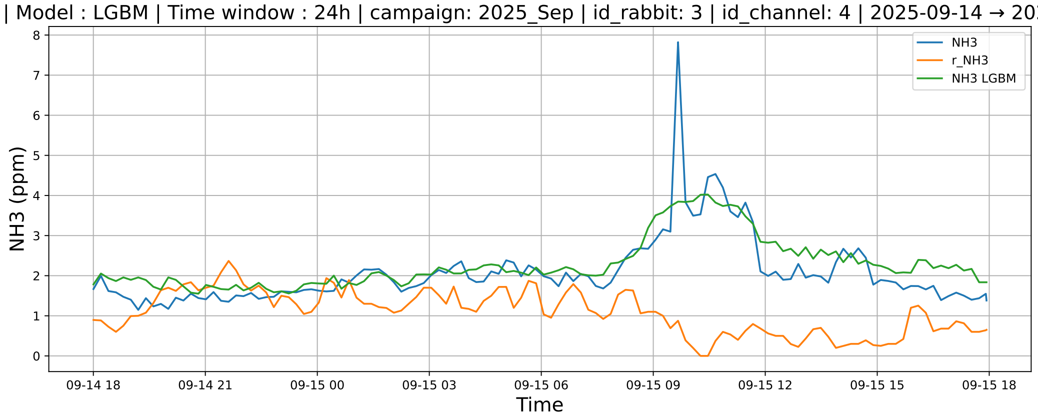
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

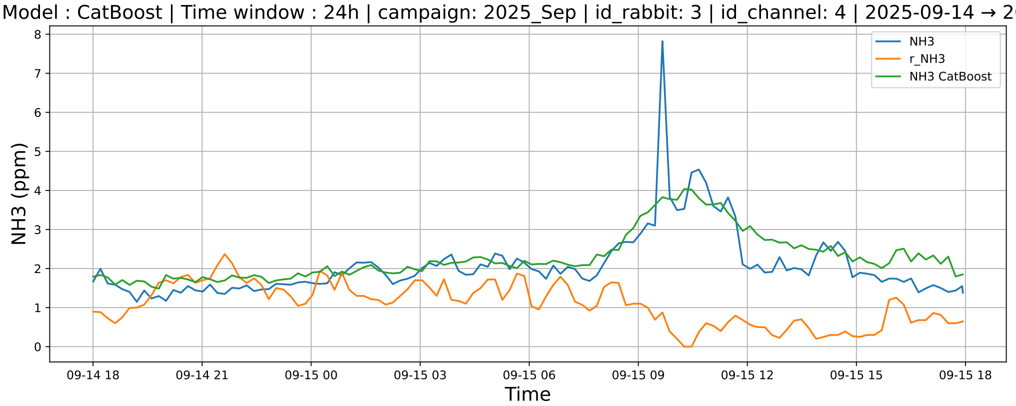
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

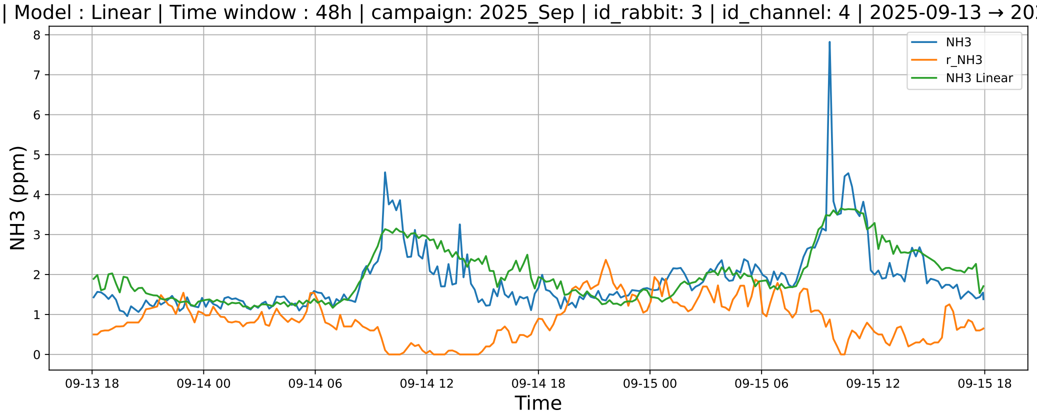


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

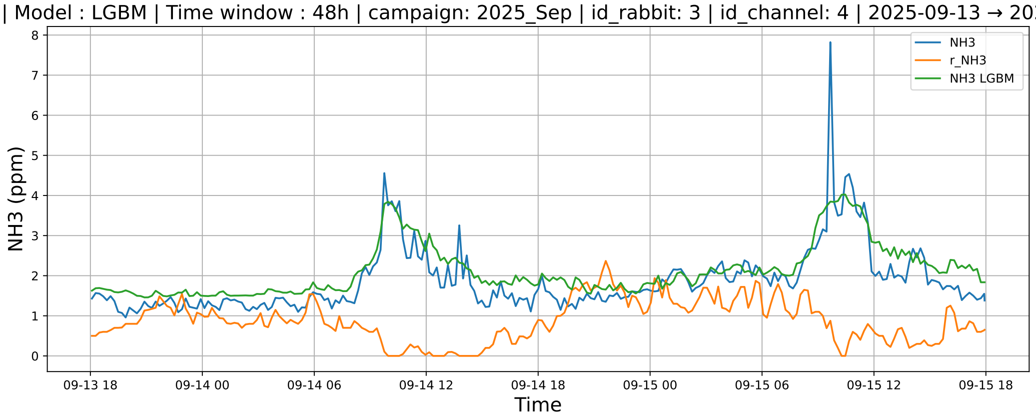


4.1.6.2 Time window : 48

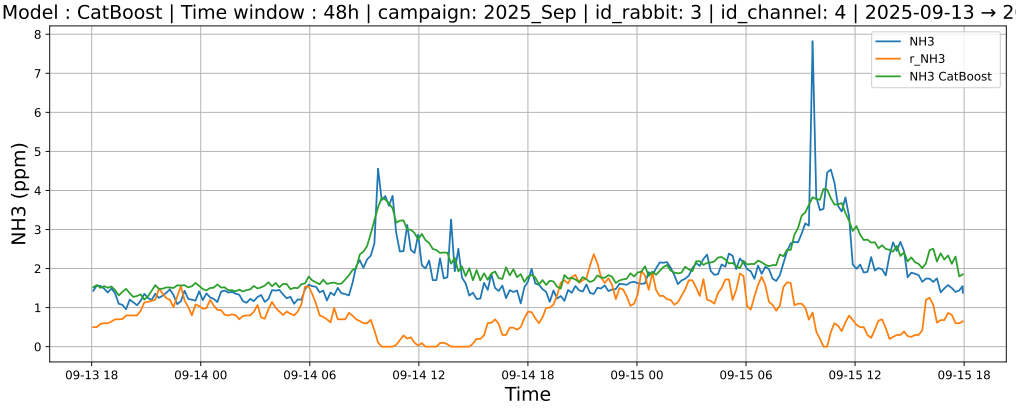
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



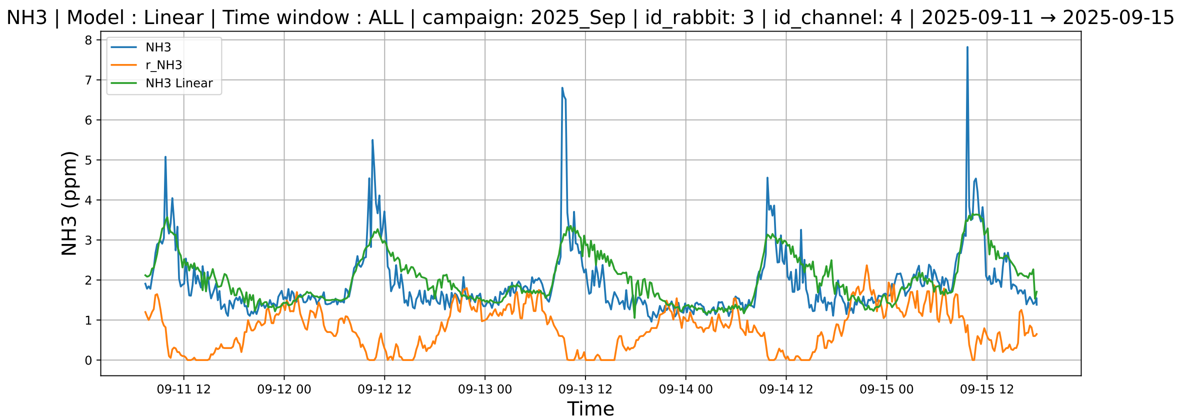
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

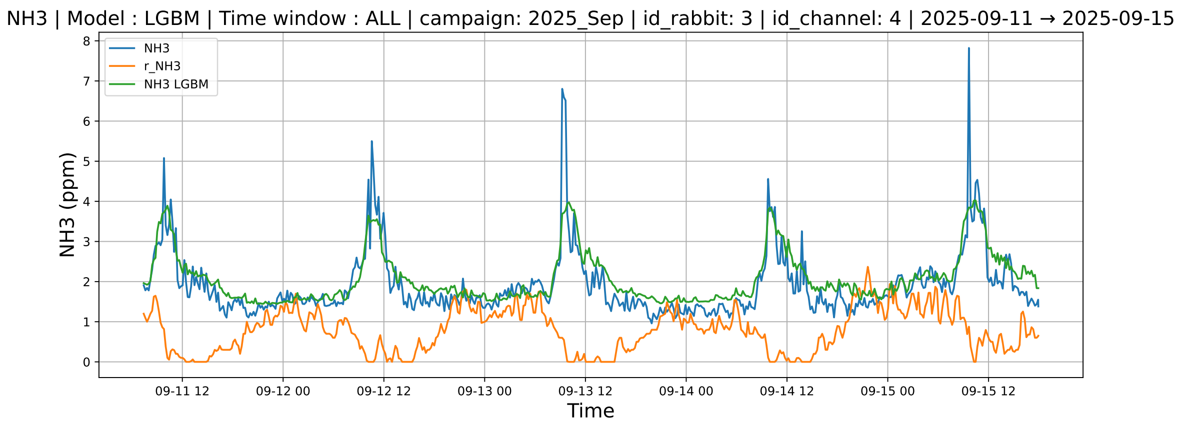


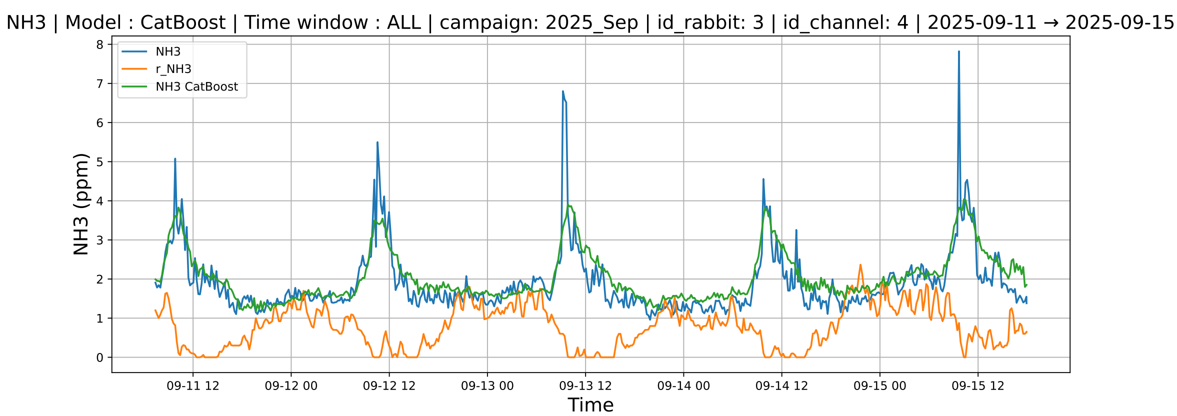
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.1.6.3 Time window : ALL



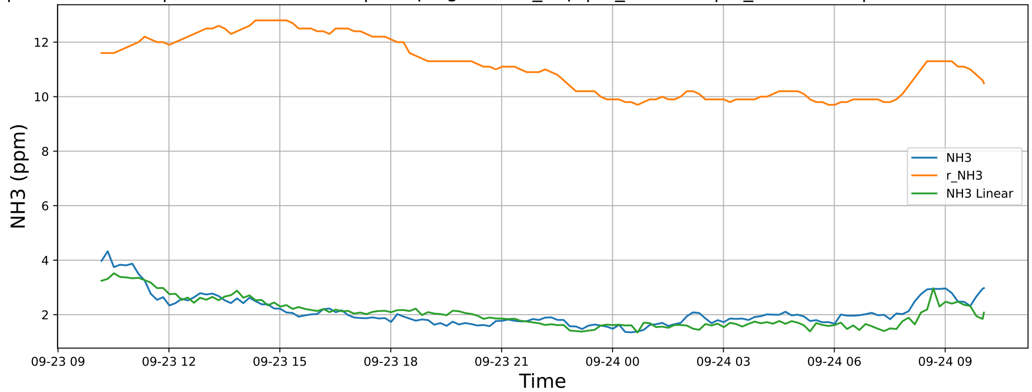




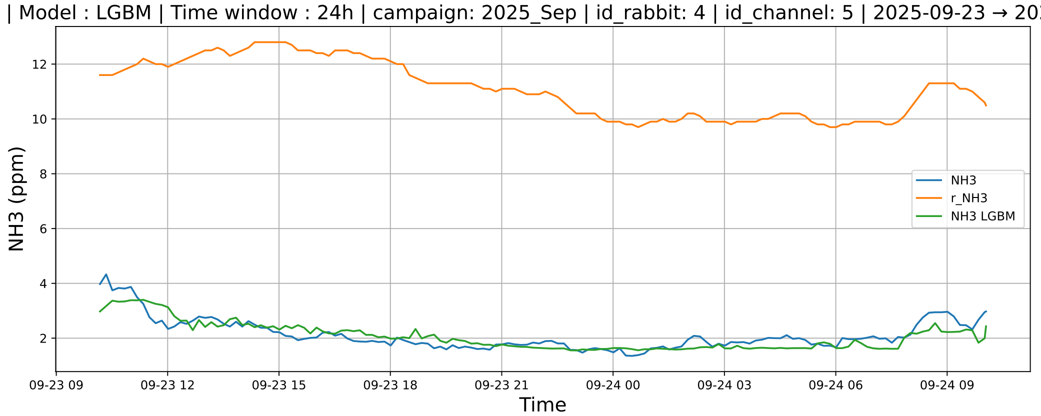
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

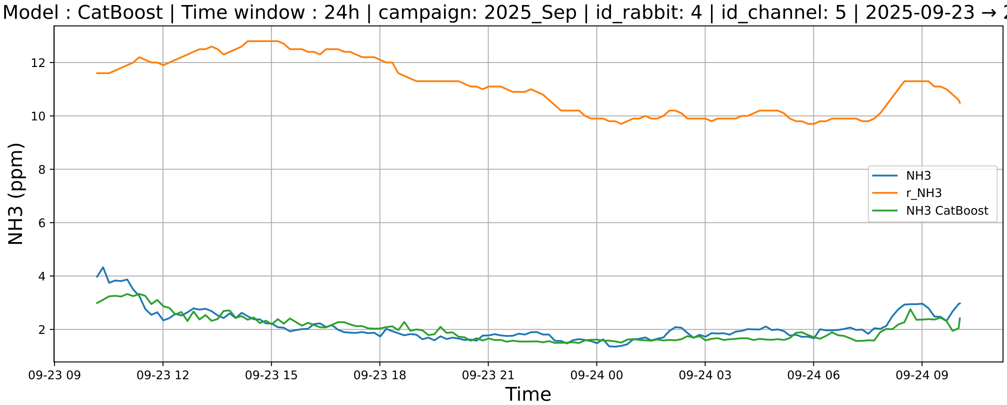
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



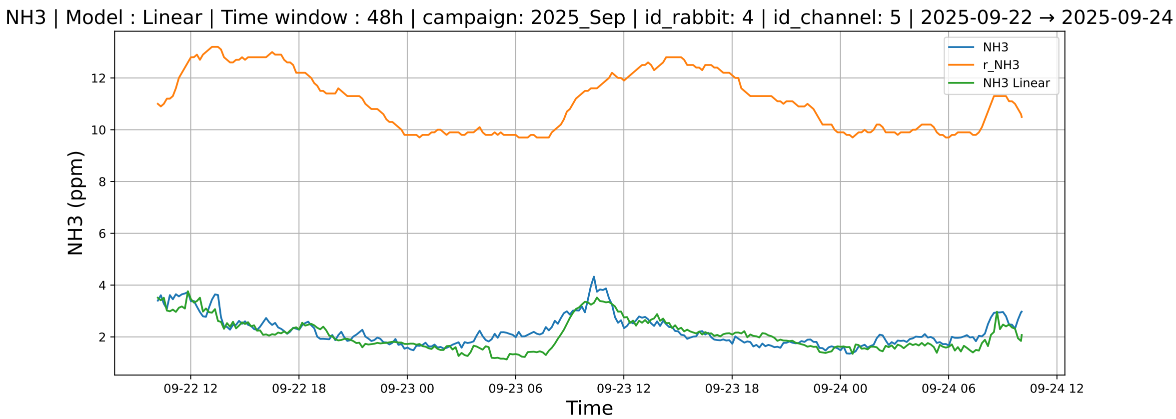
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

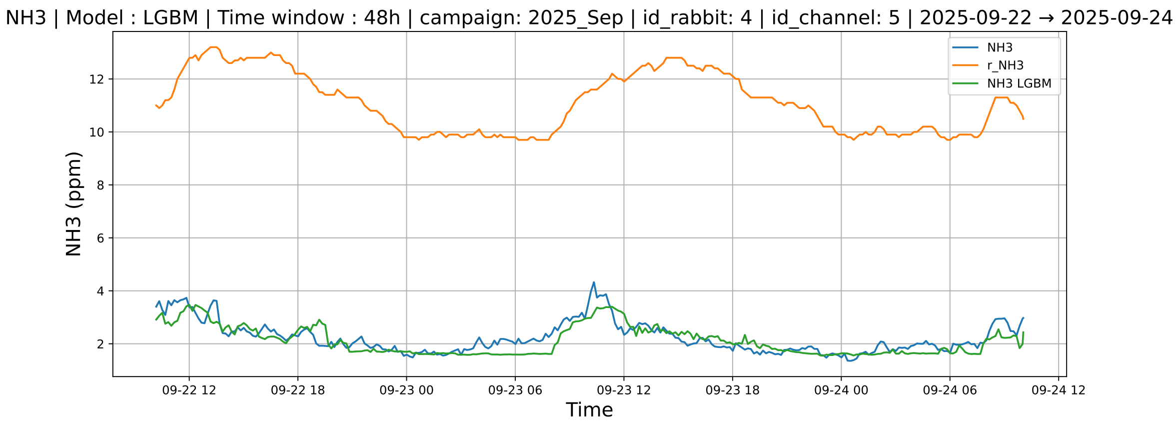


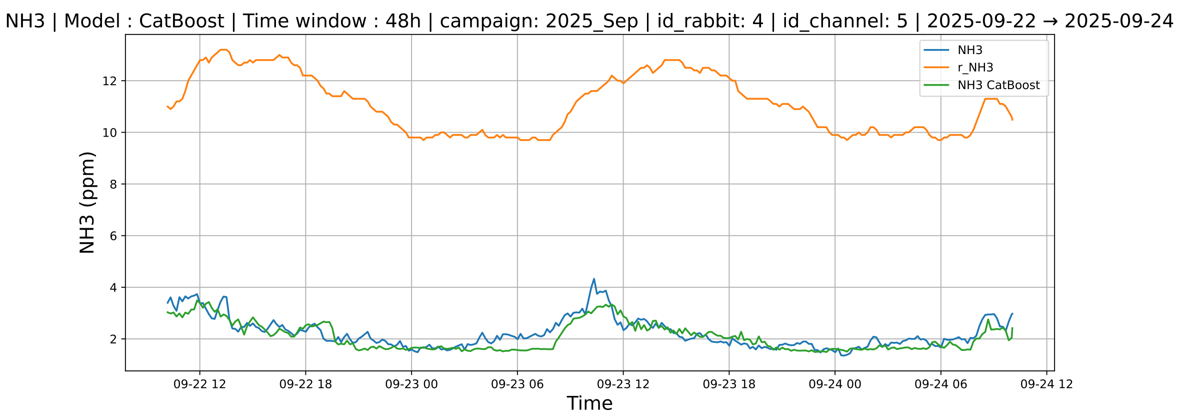
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



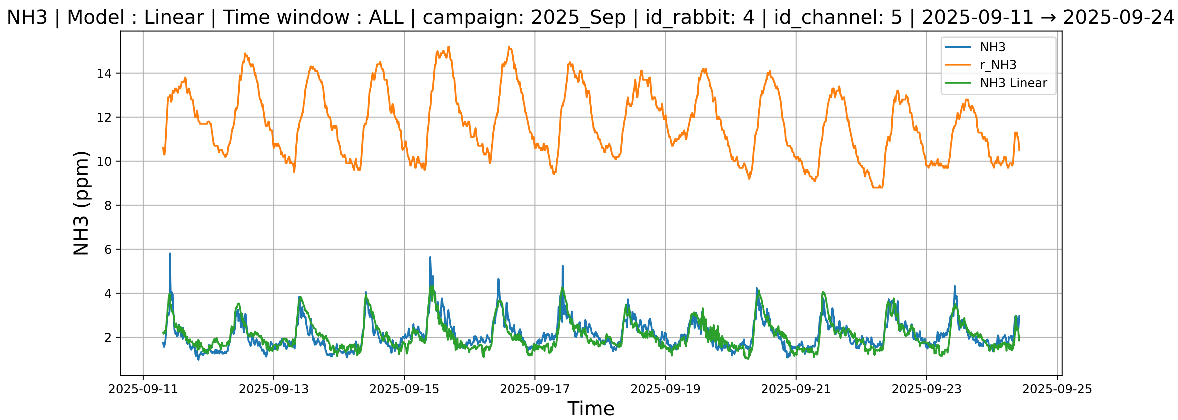
4.1.7.2 Time window : 48

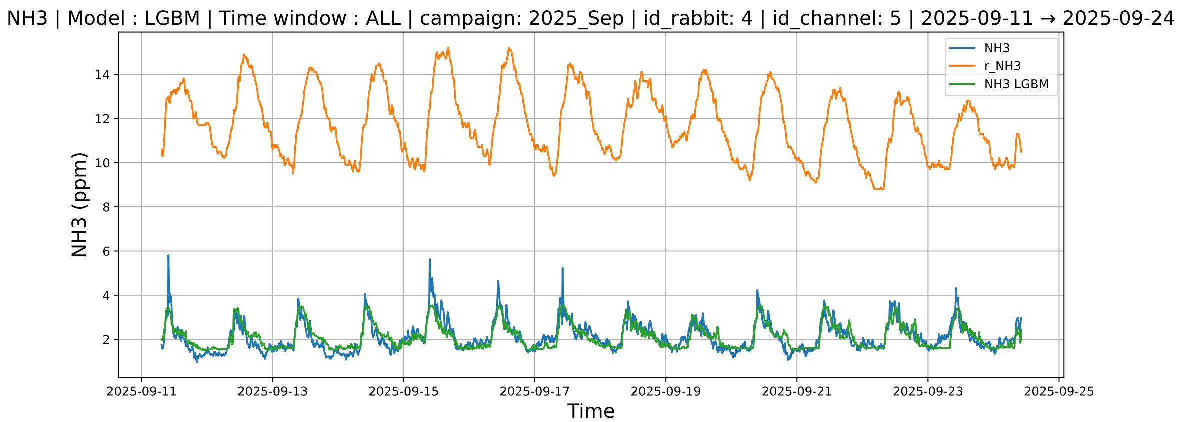


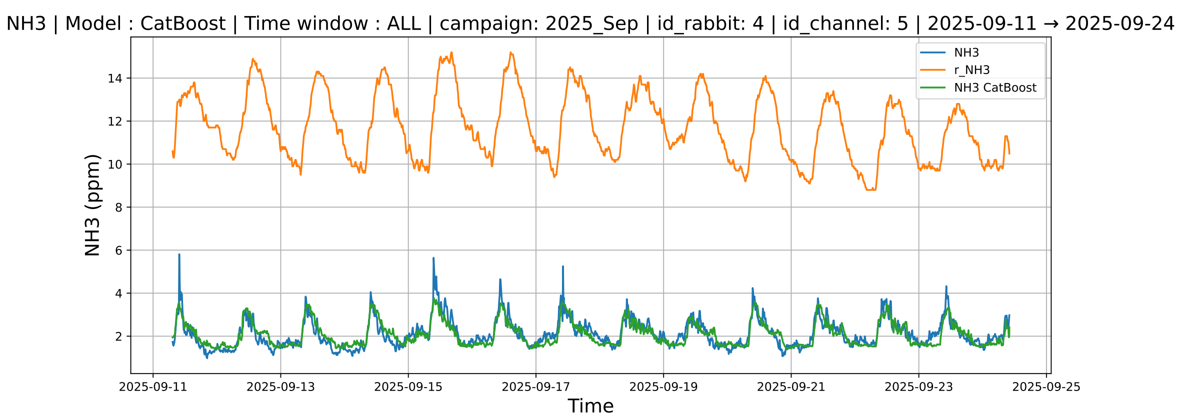




4.1.7.3 Time window : ALL

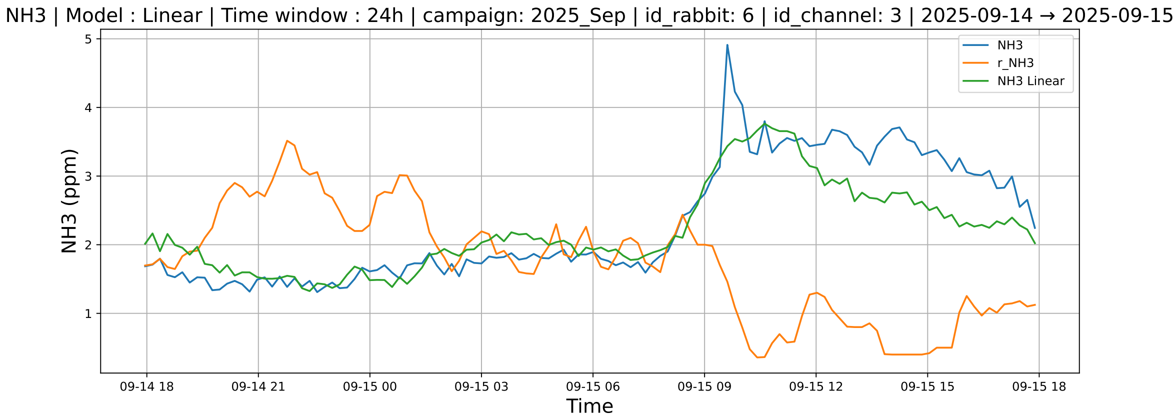


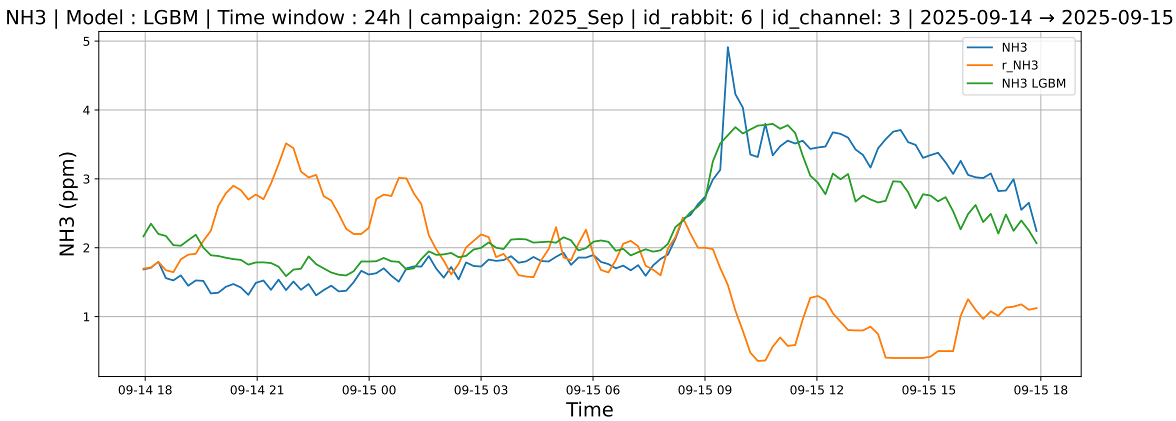


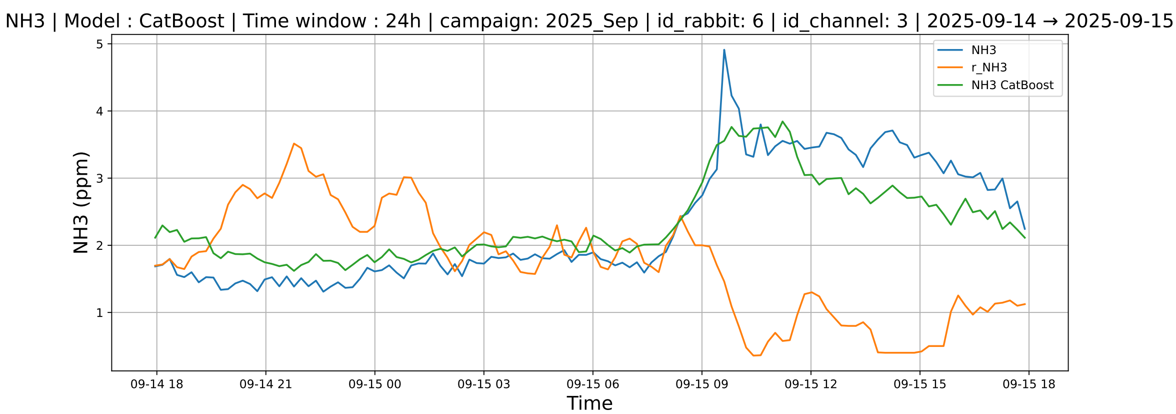


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

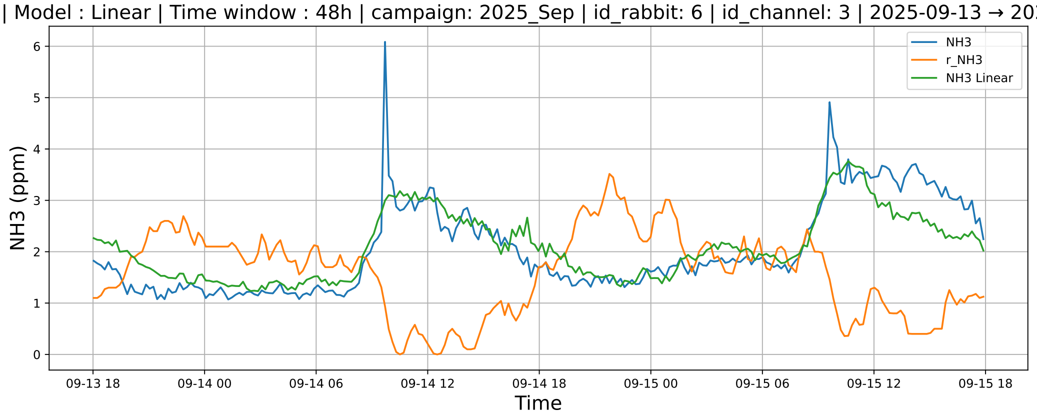




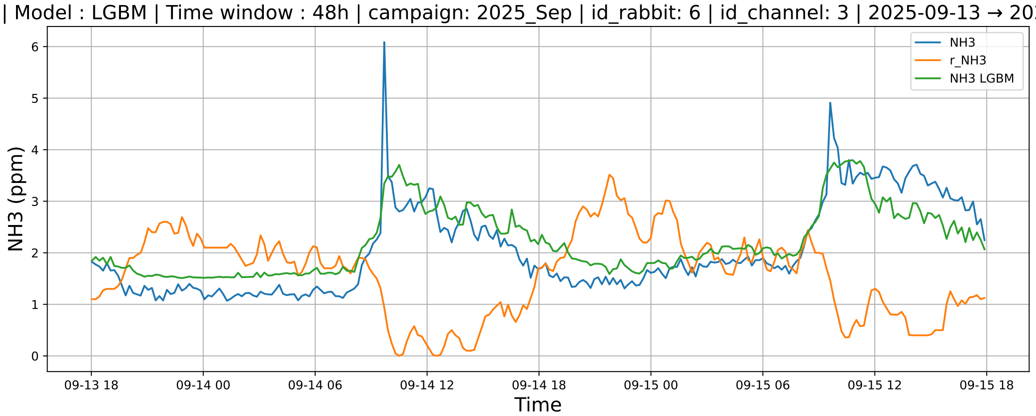


4.1.8.2 Time window : 48

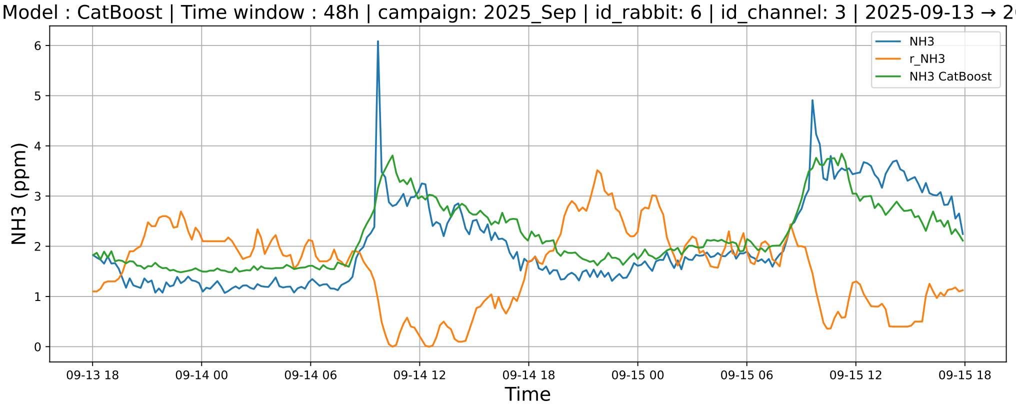
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

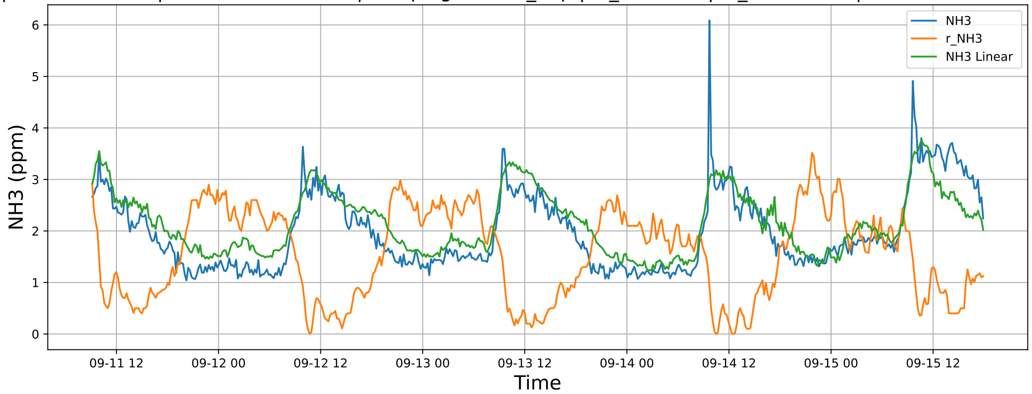


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

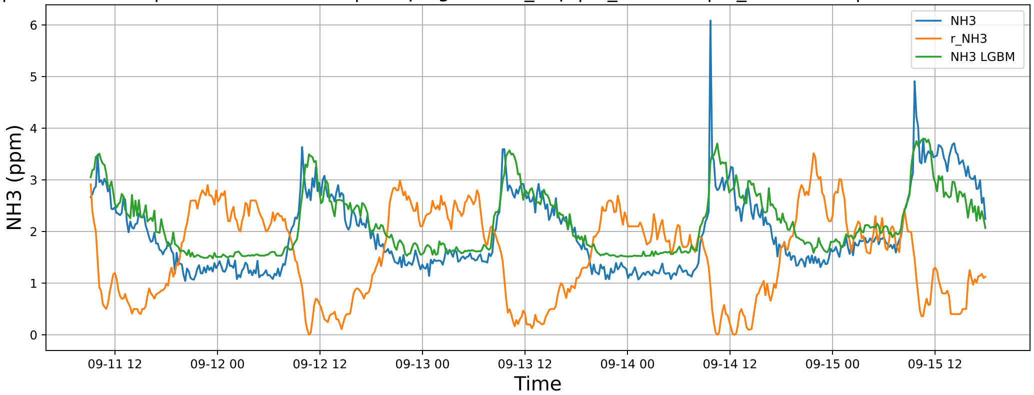


4.1.8.3 Time window : ALL

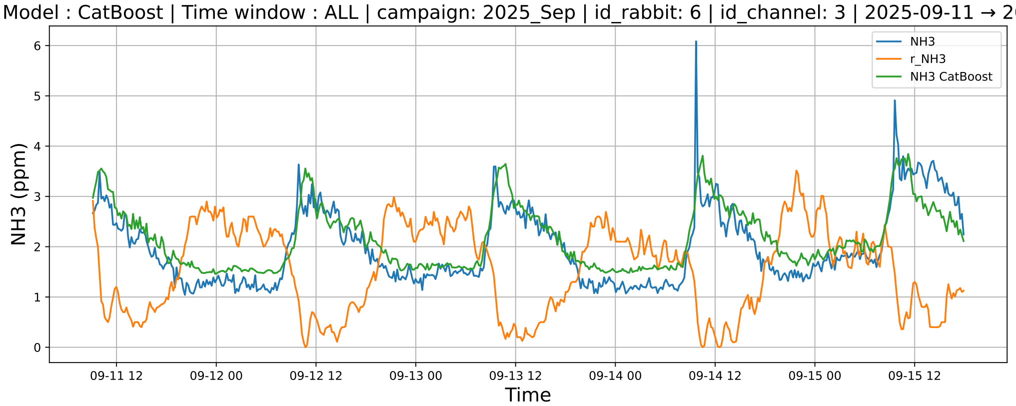
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

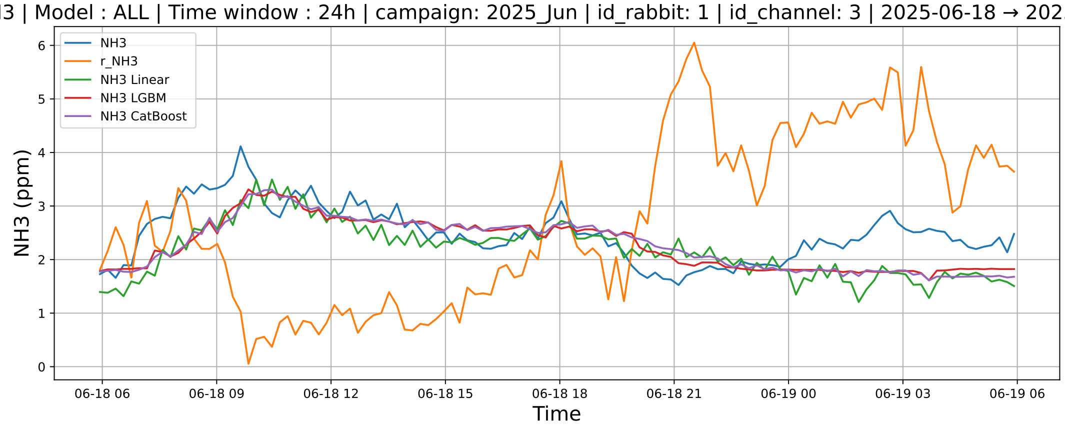


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

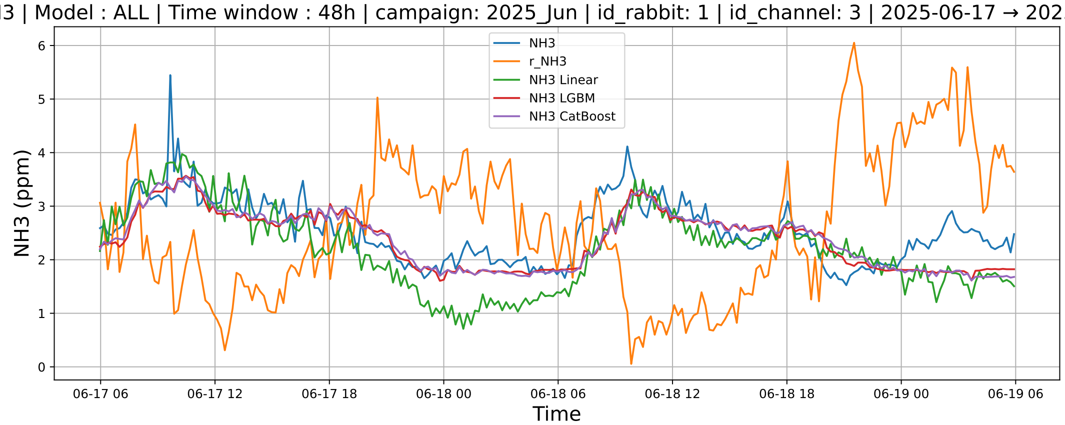
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



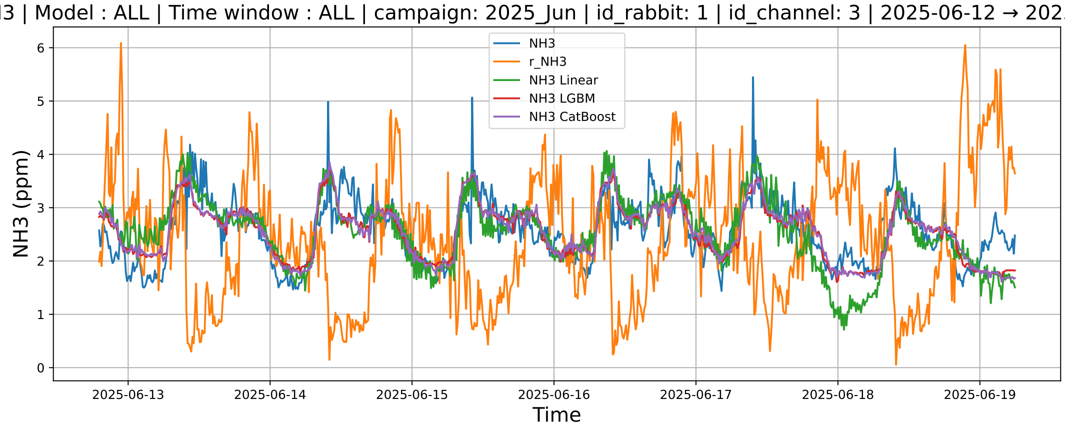
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



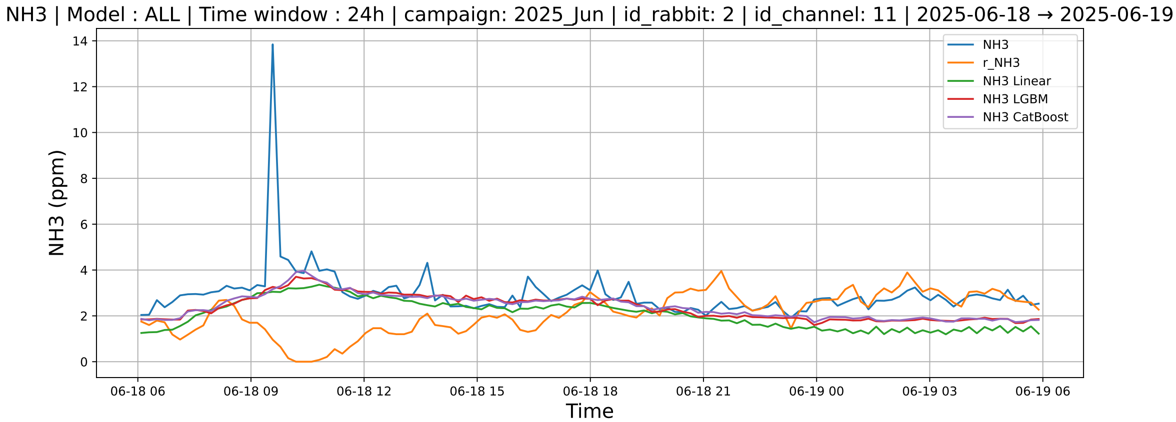
4.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

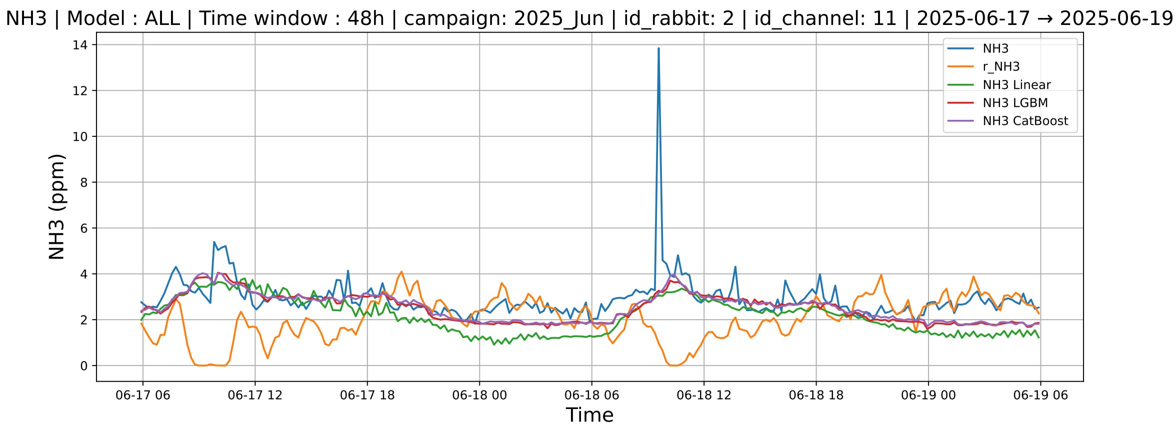


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

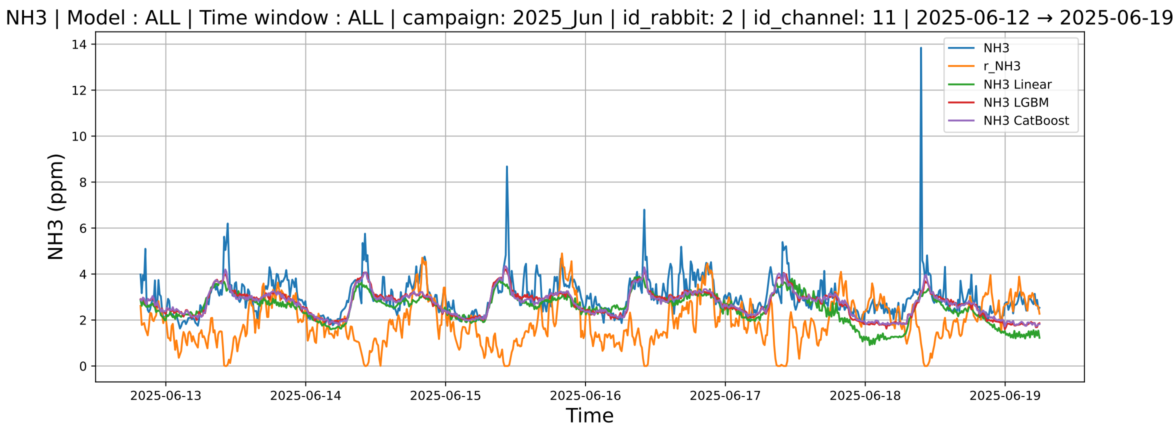
4.2.2.1 Time window : 24



4.2.2.2 Time window : 48

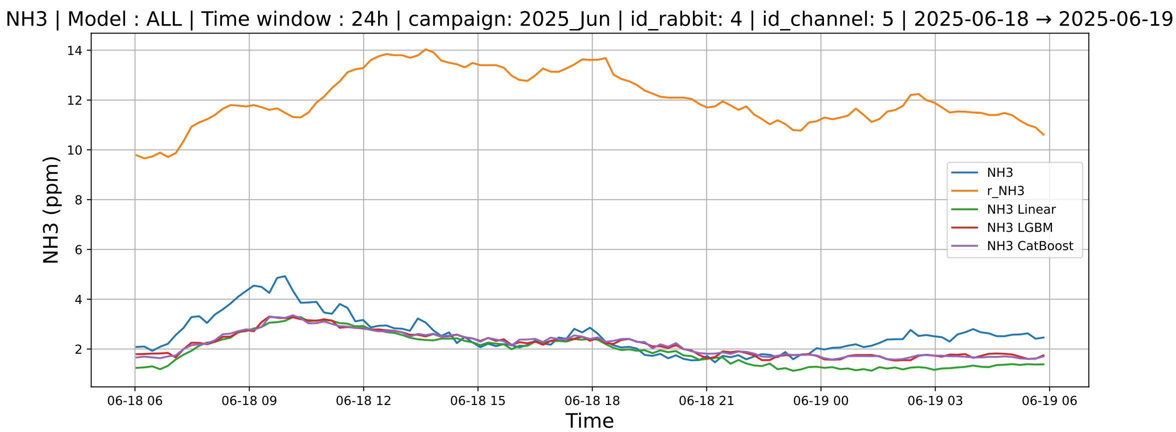


4.2.2.3 Time window : ALL

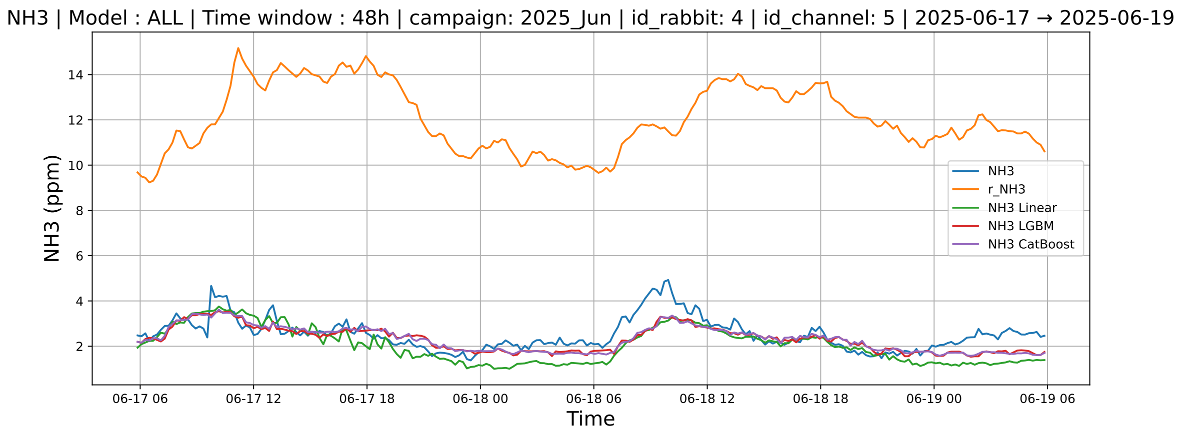


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

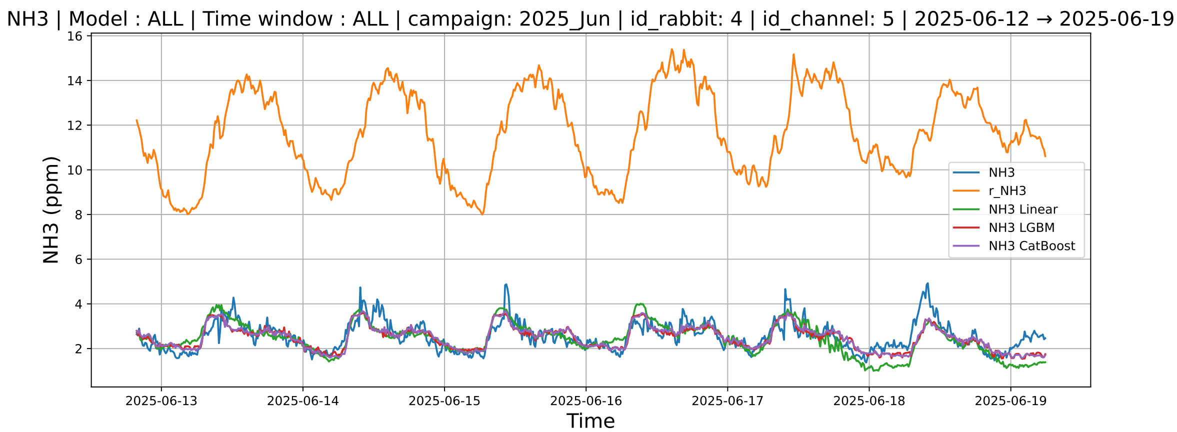
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

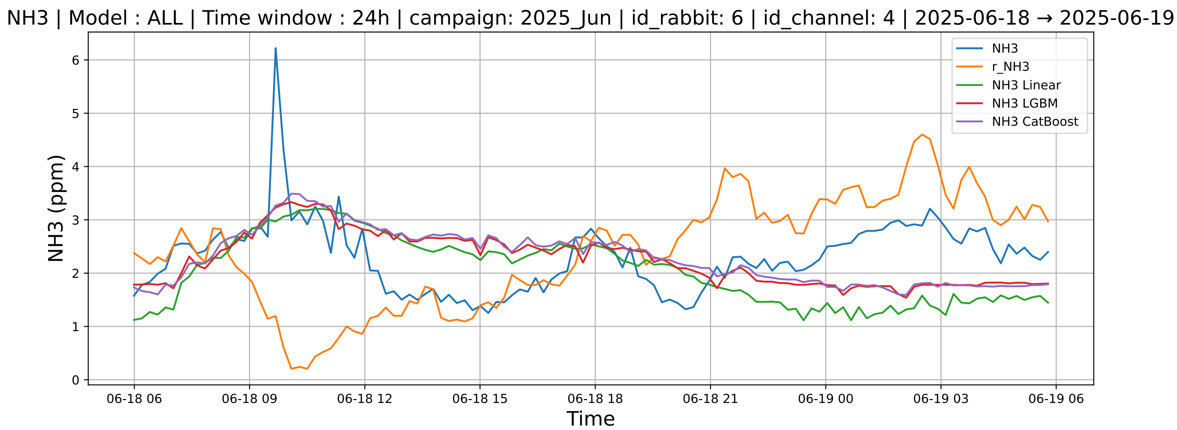


4.2.3.3 Time window : ALL

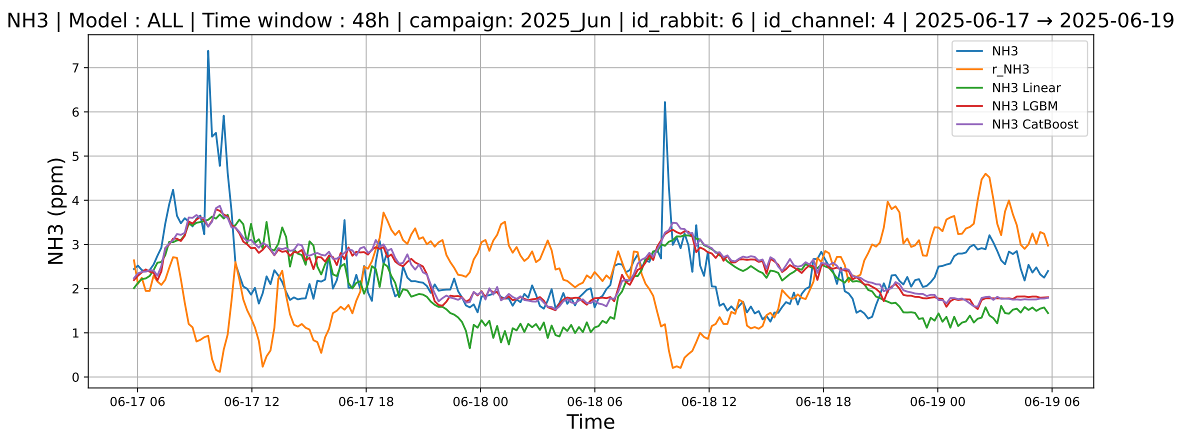


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

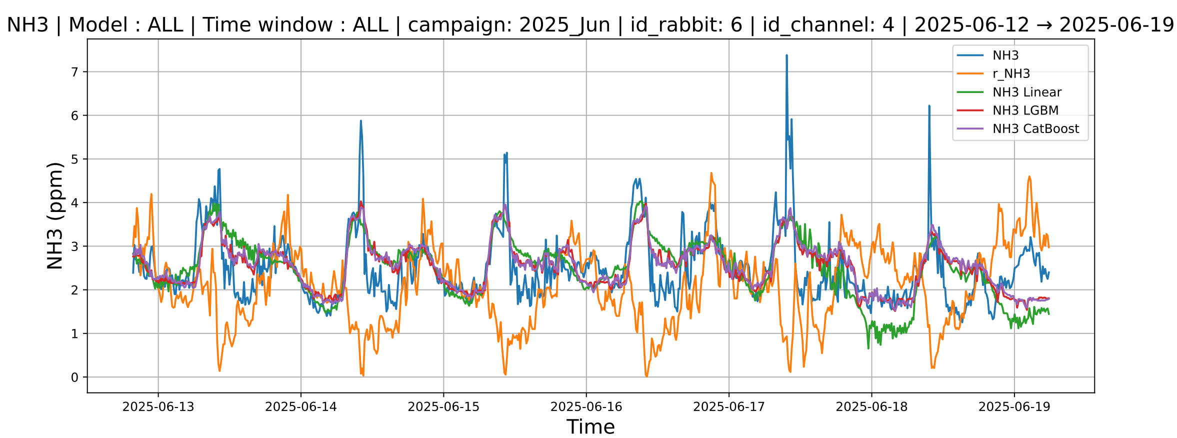
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



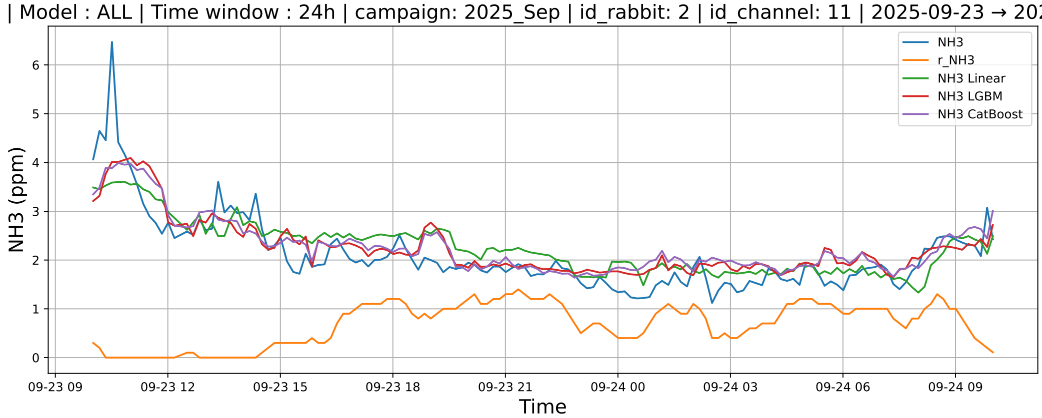
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

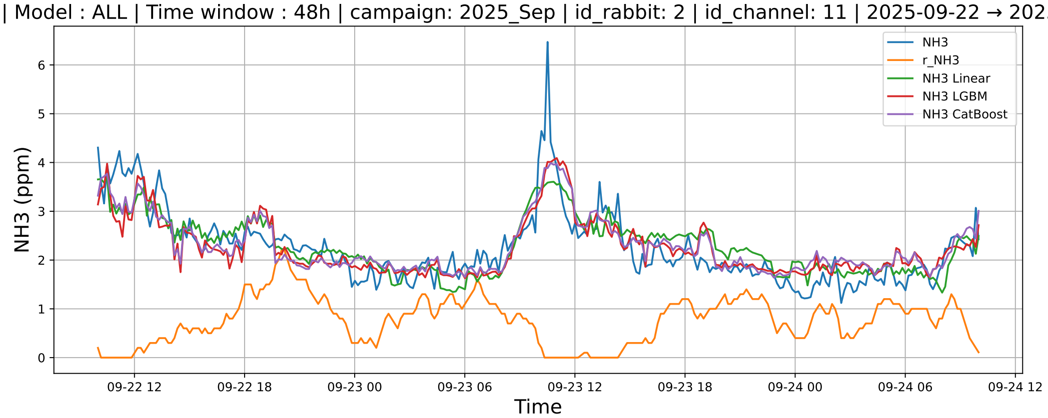
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

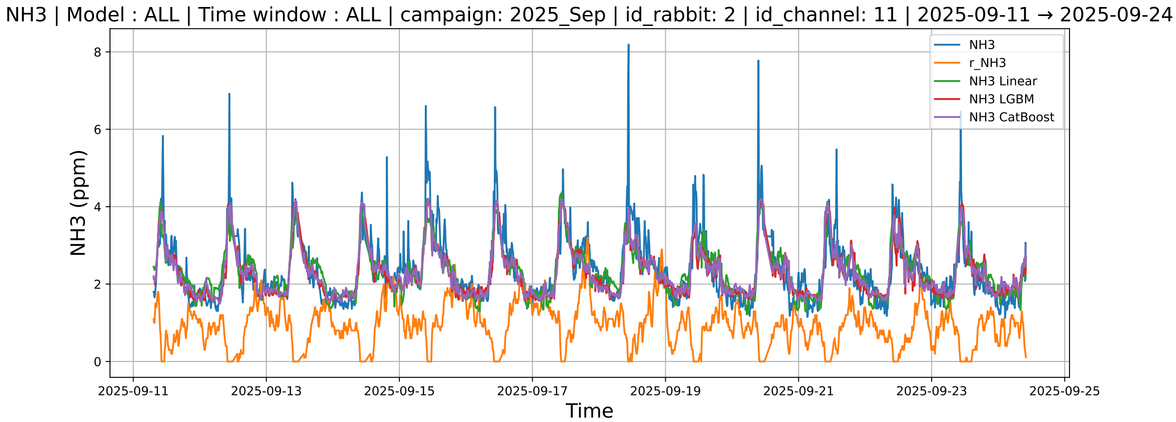


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

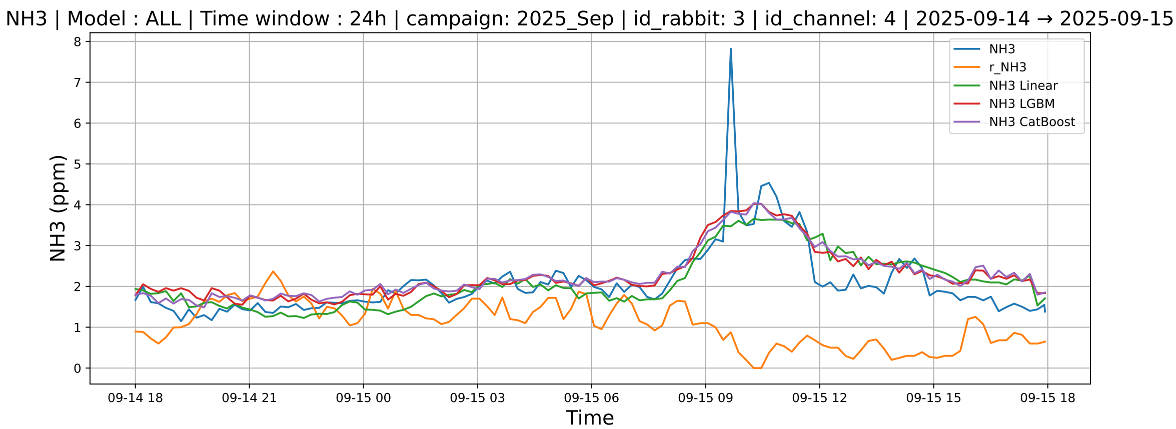


4.2.5.3 Time window : ALL



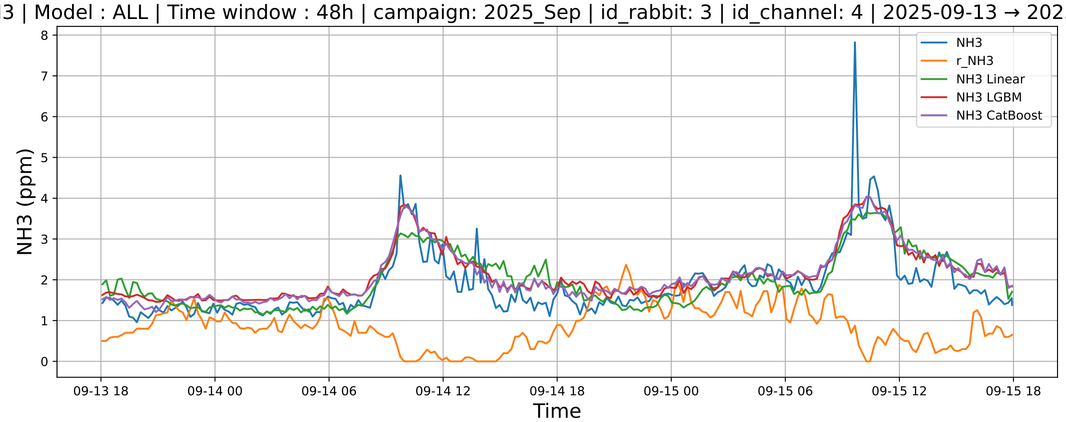
4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.2.6.1 Time window : 24



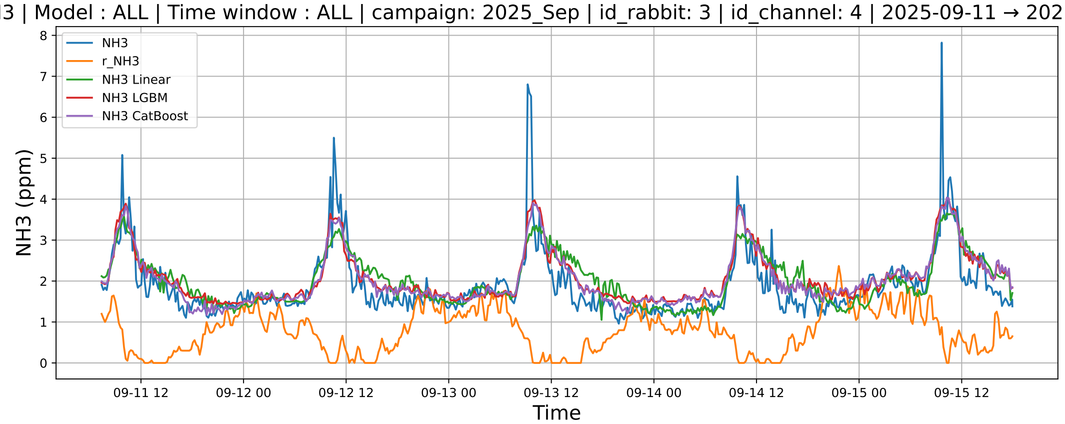
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



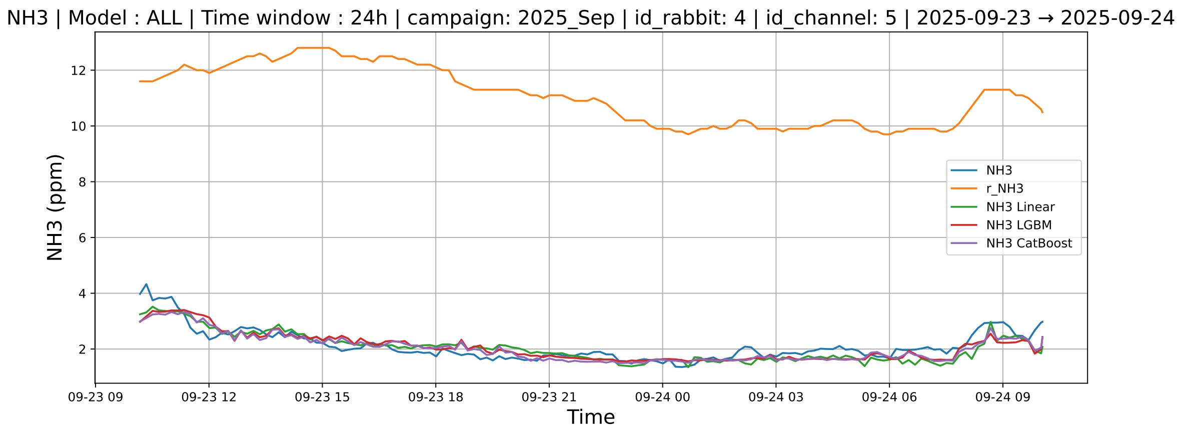
4.2.6.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

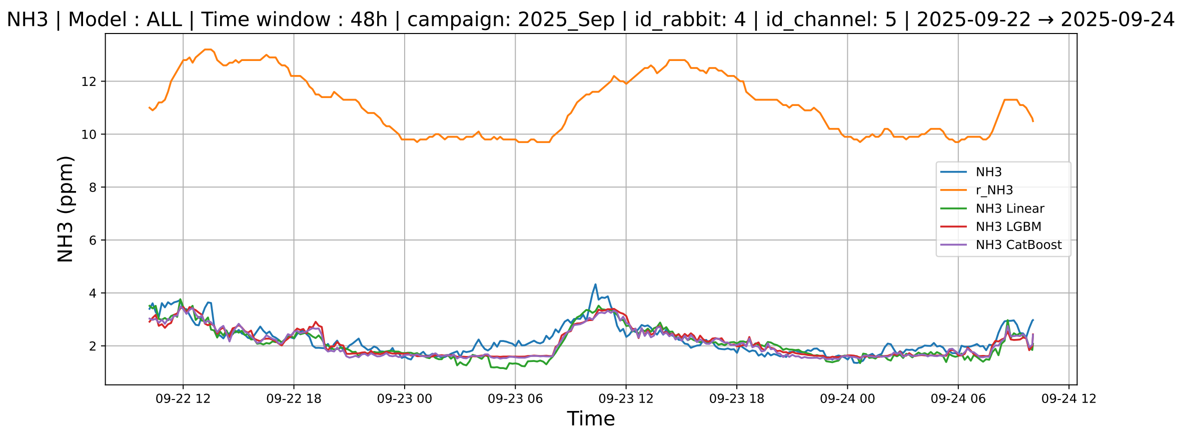


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

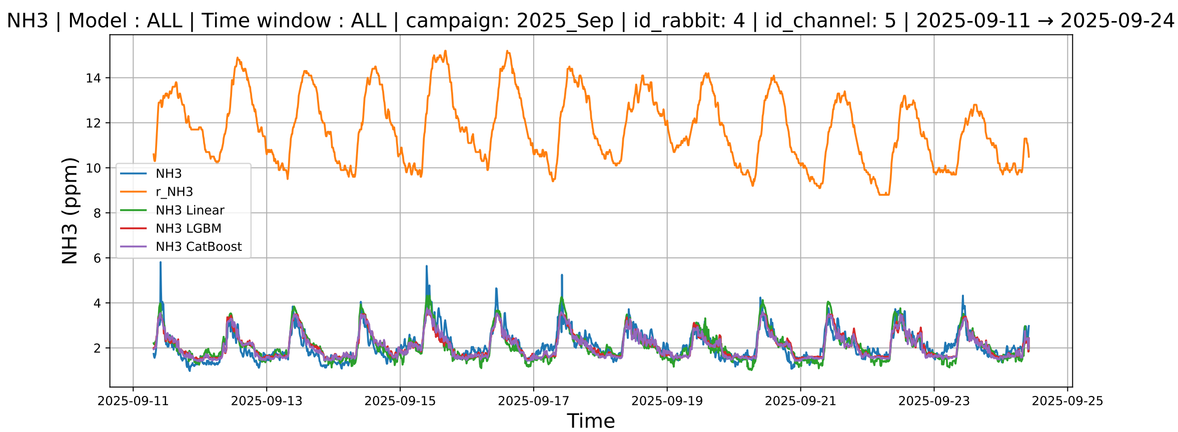
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48



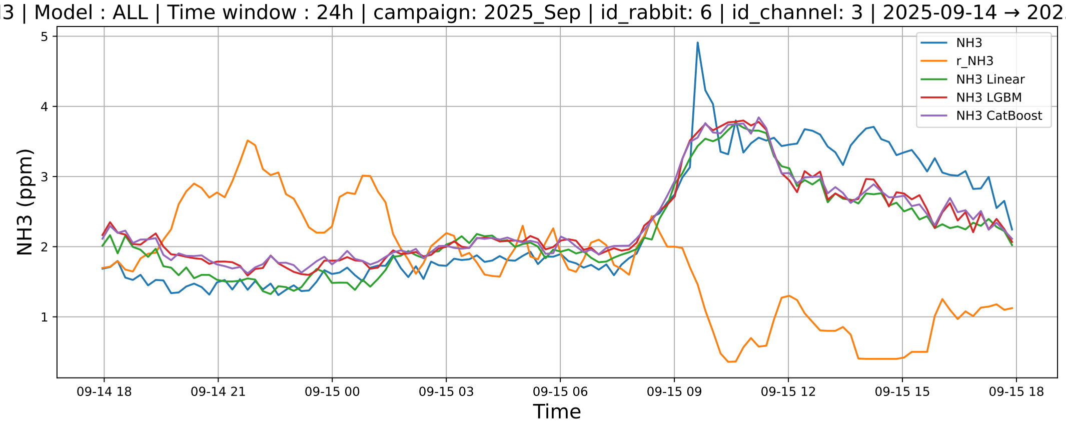
4.2.7.3 Time window : ALL



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

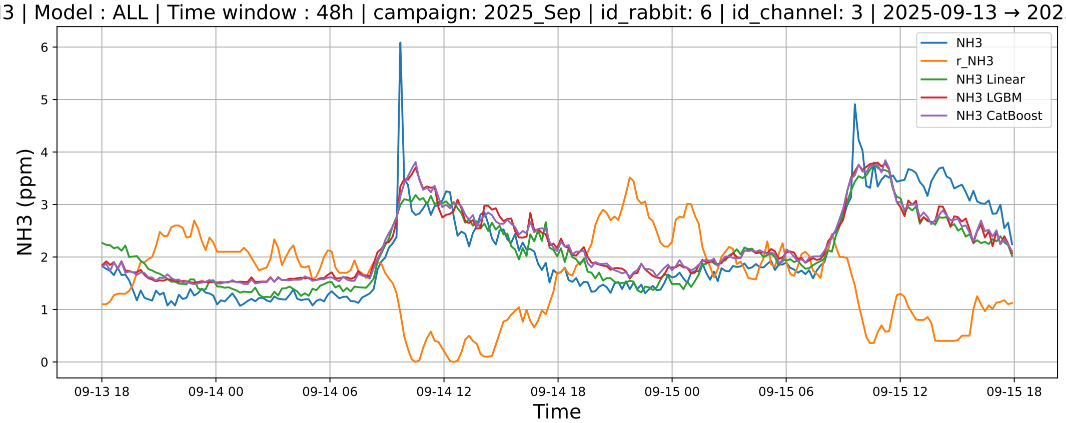
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

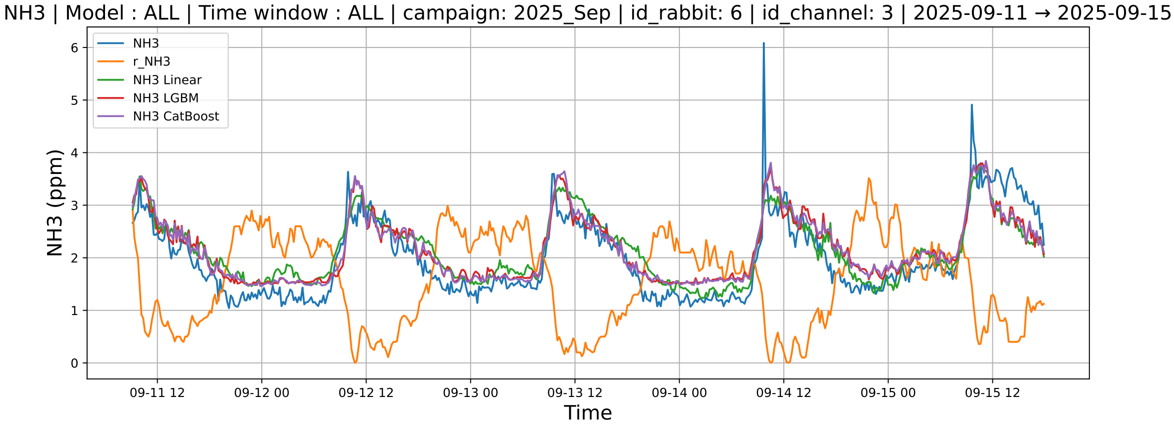


4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL



NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
09-11 12
09-12 00
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